

RESULT OF SEA TRIAL

“DUBAI ANGEL”

This sea trial was carried out on south sea off Busan during Aug. 27th to 30th, 2010 / Sep. 3rd to 5th, 2010

This report has been composed of as below.

- 1) General Particulars**
- 2) Hull & Hull Outfit Part**
 1. Hull Part
 2. Hull Outfit Part
- 3) Machinery Part**
- 4) Electric Part**
- 5) Noise & Vibration Part**

**Prepared by Basic Design Part.
on Sep. 07th, 2010.**

1) General Particulars

1) Ship's particulars

Ship's name	DUBAI ANGEL
Kind of Ship / Navigation area	Crude Oil Tanker / World-Wide
IMO Number	9 4 2 2 5 2 4
Call Sign	V 7 T T 6
Official No.	3 8 8 2
Nationality	Marshall Island
Port of Registry	Majuro
Owner	ANGEL MARITIME INC.
Builder's name & address	Hanjin Heavy Ind. & Construction Co., Ltd. 29, 5-Ga, Pongnae-Dong, Yongdo-Ku, Busan, Korea
Hull No.	N208
Date of delivery	2010. 09. 15
Classification	LR +100A1, Double Hull Oil Tanker, CSR, ESP, +IWS, LI, Shipright (CM, SCM, PCWBT, SERS) with notes Pt. Ht., +LMC, UMS, IGS, COW, SBT, MPMS, BWMP(S+F).

2) Principal Dimensions

Length Overall (LOA)	250.000	m
Length Between Perpen.(LBP)	239.000	m
Breadth (Mld)	44.000	m
Depth (Mld)	21.350	m
Draft, design (Mld)	13.500	m
Draft, summer (Ext)	14.986	m

3) Main Engine

MAN B&W 6S 60MC-C
MCR : 13,560 kW x 105.0 rpm
NCR : 12,204 kW x 101.4 rpm (90% MCR)

4) Propeller

Type	: Fixed Pitch Propeller
No. of Blade	: 4
Diameter	: 7.200 m
Pitch at 0.7R	: 5.001 m

5) Rudder

Type	: Semi-Balanced Spade
Area(Movable Part)	: 65.39 m ²

6) Steering Gear

Type	: 2-Rams, 4-Cylinders Rapson-Slide type
Max. Torque	: 2,030 Kn-m at 35 deg.
Max. Working Pressure	: 240 kg/cm ²
Main motor	: NO.1 : 55 kW x 1,800 rpm NO.2 : 55 KW / 7.5KW x 1,800 / 900 rpm

2) Hull & Hull Outfit Part

1. Hull Part

C O N T E N T S

1)	Draft Measurement	1
2)	Progressive Speed Trial	2~5
3)	Turning Circle Test	6~8
4)	Crash Stop Astern Test	9~10
5)	Z-Maneuvering Test	11~16
6)	Steering Gear Test w/ Emergency Steering	17

● NOTE

As this vessel(N 208, "DUBAI ANGEL") is the subsequent sister vessel of "DUBAI PRINCESS" (Hull No. N 198) the following tests were not carried out during this sea trial.

Therefore, the results for this vessel were obtained from the sister vessel.

- Turning Circle test
- Crash Stop Astern test at half ahead speed
- Z-Maneuvering test

[illegible]

	Test No. H-				PAGE	1	
	Sea Trial Results for 1) Draft Measurement				Ship No.	N208	
					Date	10. 09. 03	
Condition : Sea Trial Condition					Measured Time AM 09:30		
Item		Method	Measured Draft				
			Aft	Midship	Fore		
(1)	Draft at Draft Marks	Port Side	13.670	13.620	13.660		
		Starboard Side	13.670	13.630	13.660		
(2)		Mean	13.670	13.625	13.660		
(3)	Apparent Trim	Fore mark - Aft mark	-0.010				
(4)	Draft Correction Factor		-0.0746	-0.0032	0.0394		
(5)	Draft Mark Correction	(3) x (4)	0.001	0.000	0.000		
(6)	Draft at (AP)(MIDSHIP)(FP)	(2) + (5)	13.671	13.625	13.660		
(7)	Factor		1	6	1		
(8)	Deflection Correction	(6) X (7)	13.671	81.750	13.660		
(9)	Mean Draft	Sum (8) / 8	13.635				
(10)	Actual Trim	Draft at (FP) - (AP)	-0.011				
(11)	Displacement & LCB(LCG) from the Trimmed Hydrostatic Table						
	Item	Displacement			LCB (=LCG)		
	Trim(m) Draft(m)	-0.500	-0.011	0.000	-0.500	-0.011	0.000
	13.600	121235.1	121215.4	121215.0	7.385	8.063	8.078
	13.700	122217.8	122196.5	122196.0	7.317	7.990	8.005
	13.635	121558.8 ton			8.037 m		
(12)	Actual S.G. of Sea Water	(As measured)			1.01900		
(13)	Displacement Correction due to S.G.	(11) x {(12) - 1.025/ 1.025}			-711.6		
(14)	Actual Disp't after All Corrections	(11) + (13)			120847.2		

	Test No. H-						Page	2			
	Sea Trial Results for 2) Progressive Speed Trial						Ship No.	N208			
							Date	10. 09 .04			
Trial date : 2010.09.04					Trial Place : Off Busan						
Ship Condition : Trial Ballast Condition											
Draft at FP				13.660 m		Trim by Stern				-0.011	
Draft at AP				13.671 m		Displacement				120847.2	
Draft, mean				13.635 m		Propeller Immersion				184%	
Weather & Sea Condition :											
Weather :				Fine		Sea Condition :				Calm	
Depth of Sea Water :				128		Temperature of Atmosphere				28	
Temperature of Sea Water :				26		Specific Gravity of Sea Water				1.019	
Trial Result											
M/E Load	Run No.	Course of Run	Time in	Speed (knot)	M/E rpm	M/E BHP(kW)	Wind		Wave (Average)		
							Vel.(m/s)	dir.(deg)	hgt.(m)	dir.(deg)	
Abt. 90%	1	200	13:07	15.146	106.4	12565.0	9.5	S 25.5	1.0	60	
	2	20	14:16	15.929	106.4	12533.0	5.8	P 23.2	1.0	60	
	Mean				15.538	106.4	12549.0				
Abt. 100%	1	200	15:37	15.121	109.3	13911.0	9.8	S 22.5	1.0	60	
	2	20	16:45	16.831	109.8	14068.0	7.4	P 25.6	1.0	60	
	Mean				15.976	109.6	13989.5				
* The M/E power used in this report was measured by HHIC's Torsion-meter.											

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SPEED TRIAL ANALYSIS

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BY NSMB METHOD

SHIP NAME: N208 EMARAT TRIAL DATE: 2010.09.03
 * LBP = 239.000 M * BREADTH = 44.000 M * TA/TF = 13.671/13.660 M
 * DISPL = 120847. TONS * T MEAN = 13.665 M

SPEED TRIAL RESULTS

LOAD (%)	RUNNING DIRECT. (DEG)	WIND DIR. (DEG)	WIND SPEED (M/S)	SHIP SPEED (KTS)	RATE REVS. (RPM)	SHAFT POWER (KW)	CURRENT SPEED (KTS)	INNING TIME
90.	200.	S 25.5	9.5	15.146	106.40	12565.	0.000	13: 7
90.	20.	P 23.2	5.8	15.929	106.40	12533.	0.000	14:17
				15.538	106.40	12549.		
100.	200.	S 22.5	9.8	15.121	109.30	13911.	0.000	15:37
100.	20.	P 25.6	7.4	16.831	109.80	14068.	0.000	16:45
				15.976	109.55	13990.		

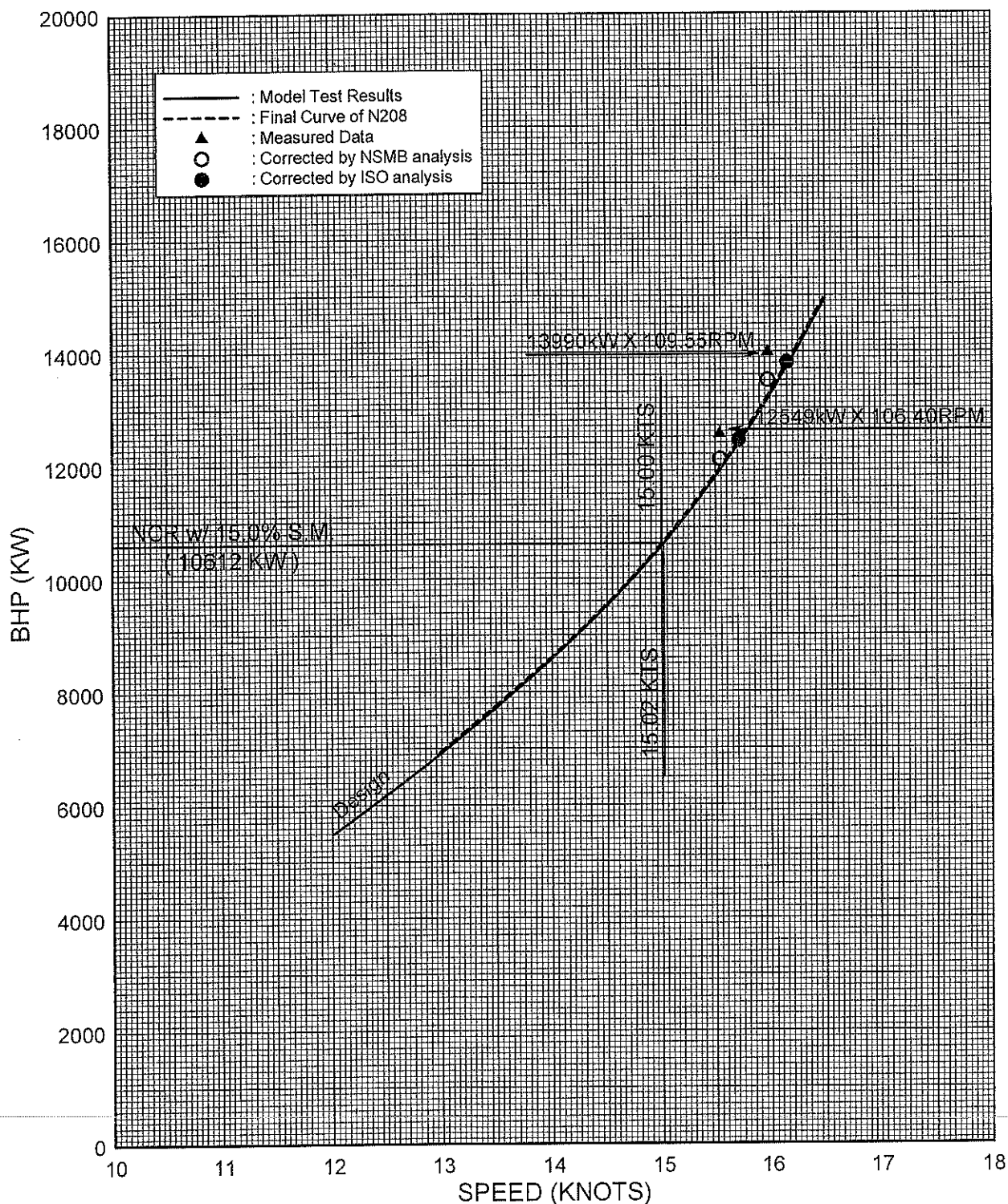
CORRECTION TO BEAUFORT NO. 0 (CALM SEA)

LOAD (%)	RUNNING DIRECT. (DEG)	TRUE W.DIR (DEG)	TRUE W.S (KTS)	WIND SCALE (B.N)	CORR. FACTOR	SHIP SPEED (KTS)	CORR. RPM	CORR. BHP (KW)	DISPL. CORR. RPM	CORR. BHP
90.	200.	279.2	8.09	3.4	1.036	15.146	105.16	12129.	105.06	12095
90.	20.	238.6	7.12	3.2	1.032	15.929	105.30	12150.	105.21	12116
				3.3	1.034	15.538	105.23	12139.	105.13	12106
100.	200.	271.2	7.70	3.3	1.034	15.121	108.08	13451.	107.98	13414
100.	20.	258.2	7.32	3.2	1.032	16.831	108.64	13626.	108.54	13588
				3.3	1.033	15.976	108.36	13539.	108.26	13501

* DISPLACEMENT CORRECTION : BY ADMIRALTY COEFFICIENT
 MODEL TEST DISPLACEMENT : 120343. TONS

SPEED-POWER CURVE

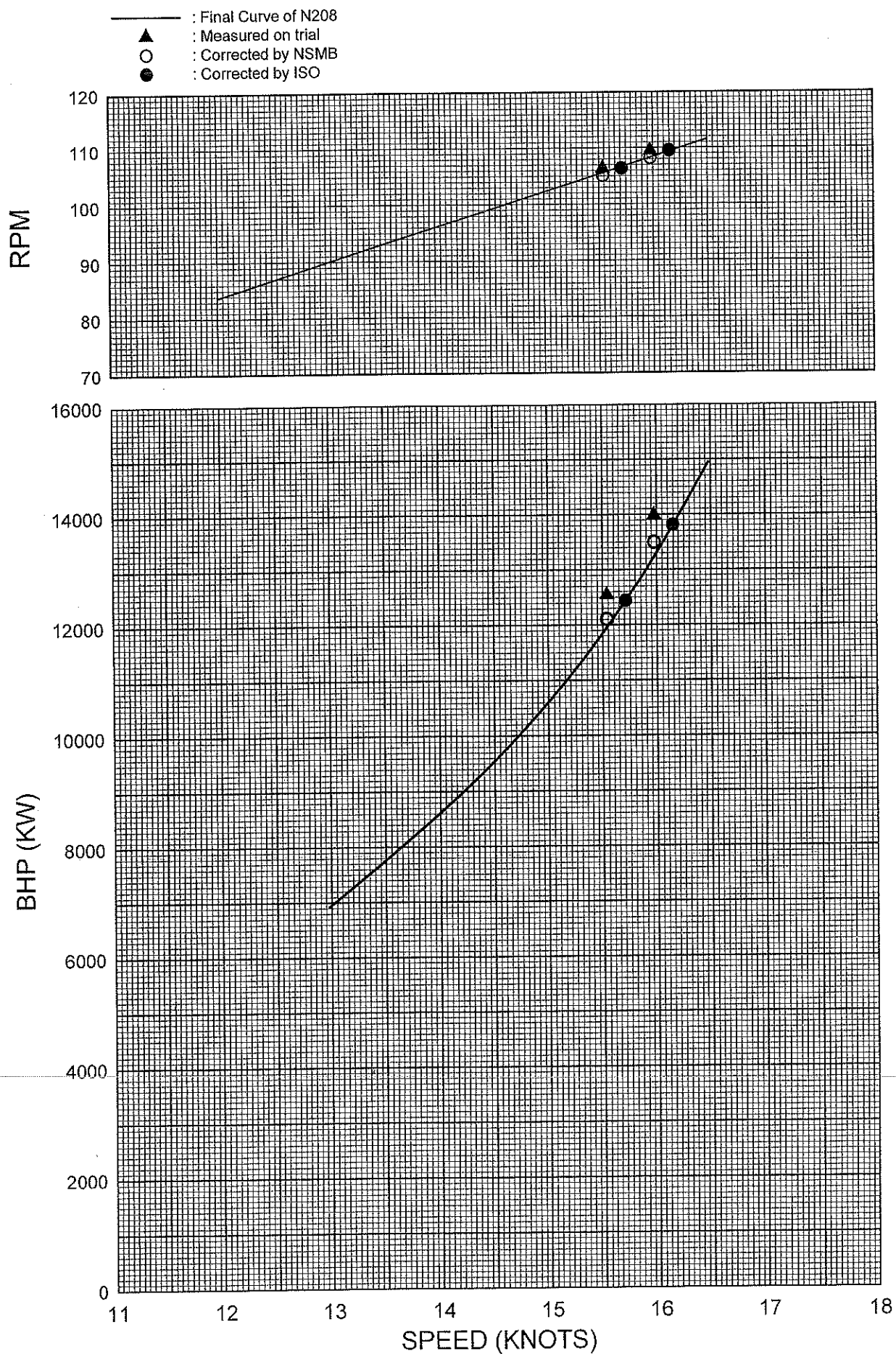
N-208



DESIGN LOAD

Service Speed at NCR w/ 15% S.M. : 15.00 KTS

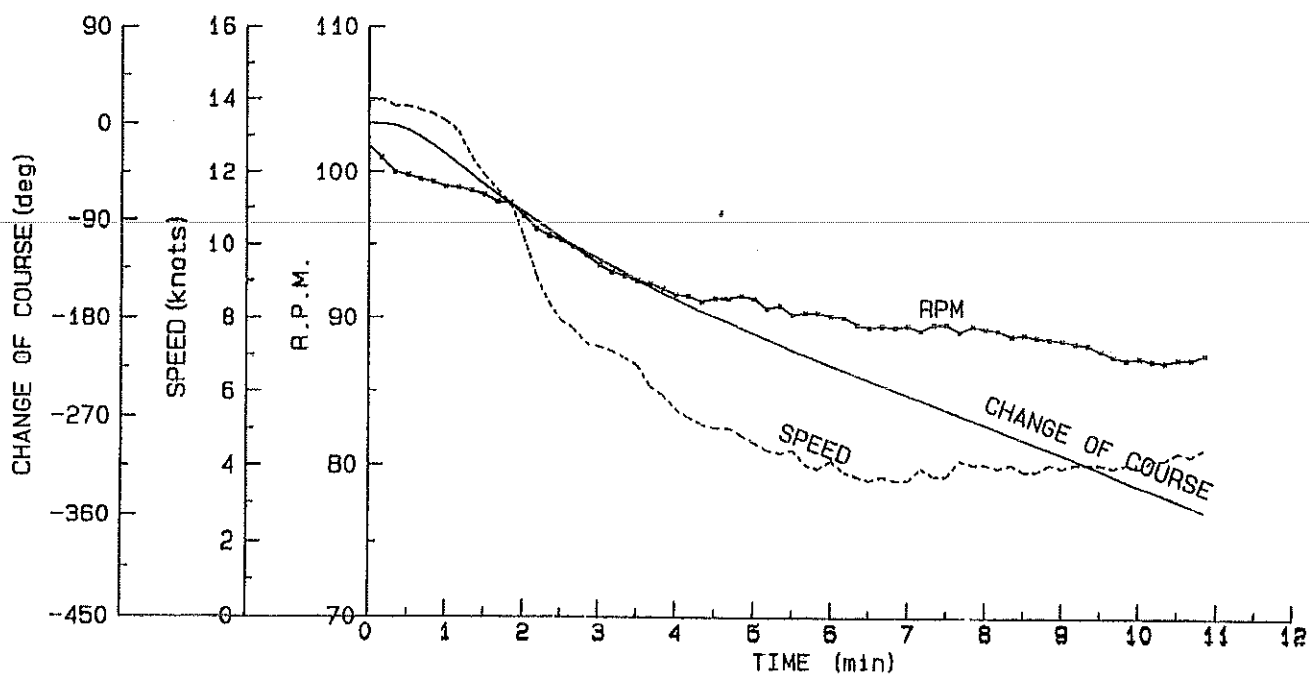
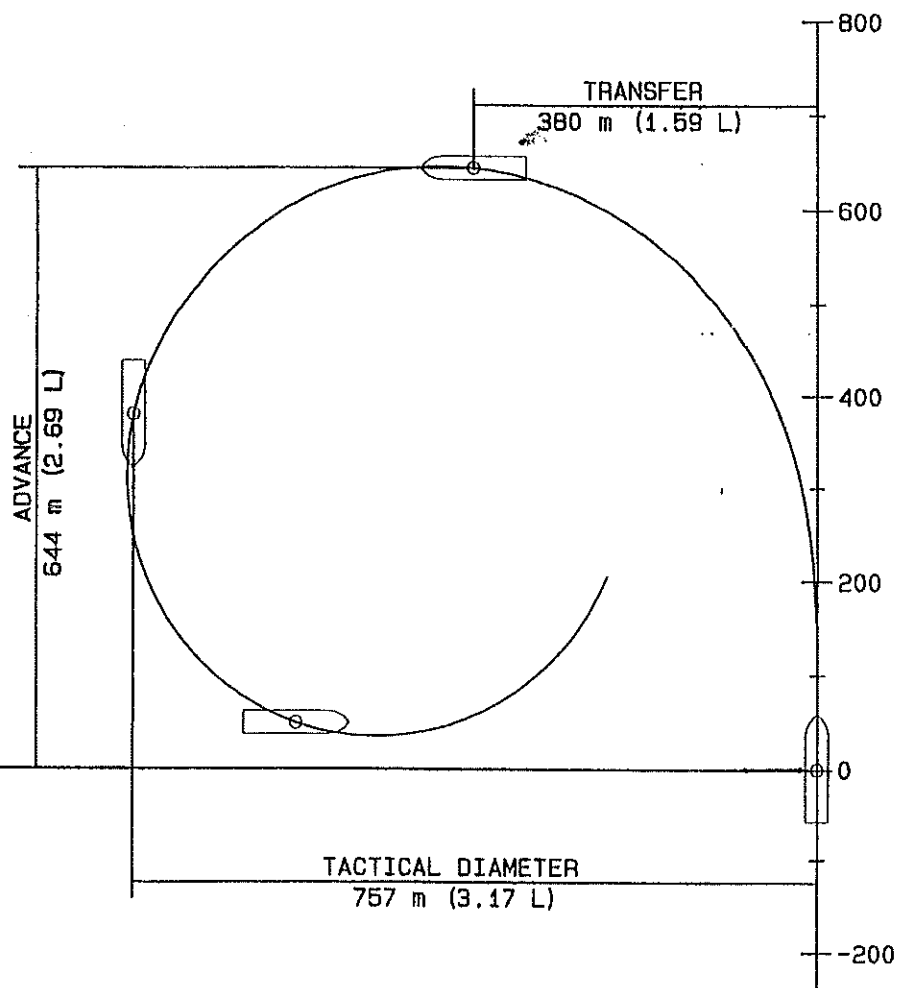
SPEED-POWER CURVE



			Page	6
	Sea Trial Results for 3) Turning Circle Test		Ship No.	N198
			Date	09.01.12
Trial date : 2009. 01. 12		Trial Place : Off Busan		
Ship Condition : Trial Ballast Condition		Sea Condition		
Draft, Fore	13.589 m	Weather	Windy	
Draft, Aft	13.756 m	Sea Condition	Rough	
Displacement	122069.3 ton	Wind Dir. & Velocity	S60°, 9.0 m/s	
		Depth of Sea Water	121 m	
Trial Result at Full Ahead Speed (Ful Speed)				
		Port Turn	Starboard Turn	
Clock Time		19:35	19: 00	
Base Course		200°	200°	
Rudder Angle		S - 60°	S - 55°	
Shaft rpm		100 rpm	100 rpm	
Ship speed		13.9 knots	14.0 knots	
Turning Time To (sec)	090 deg	138	124	
	180 deg	282	278	
	270 deg	471	465	
	360 deg	665	659	
	450 deg	855	860	
	540 deg	1045	1078	
	630 deg	1245	1277	
	720 deg	1440	1480	

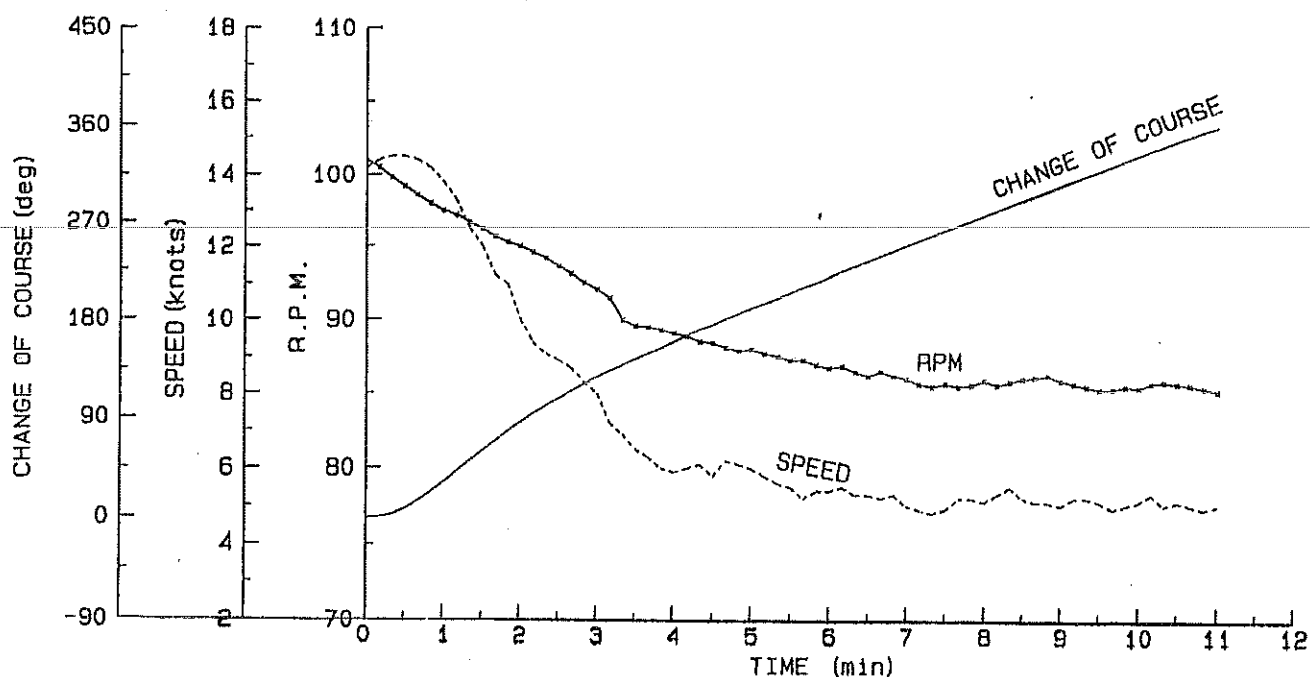
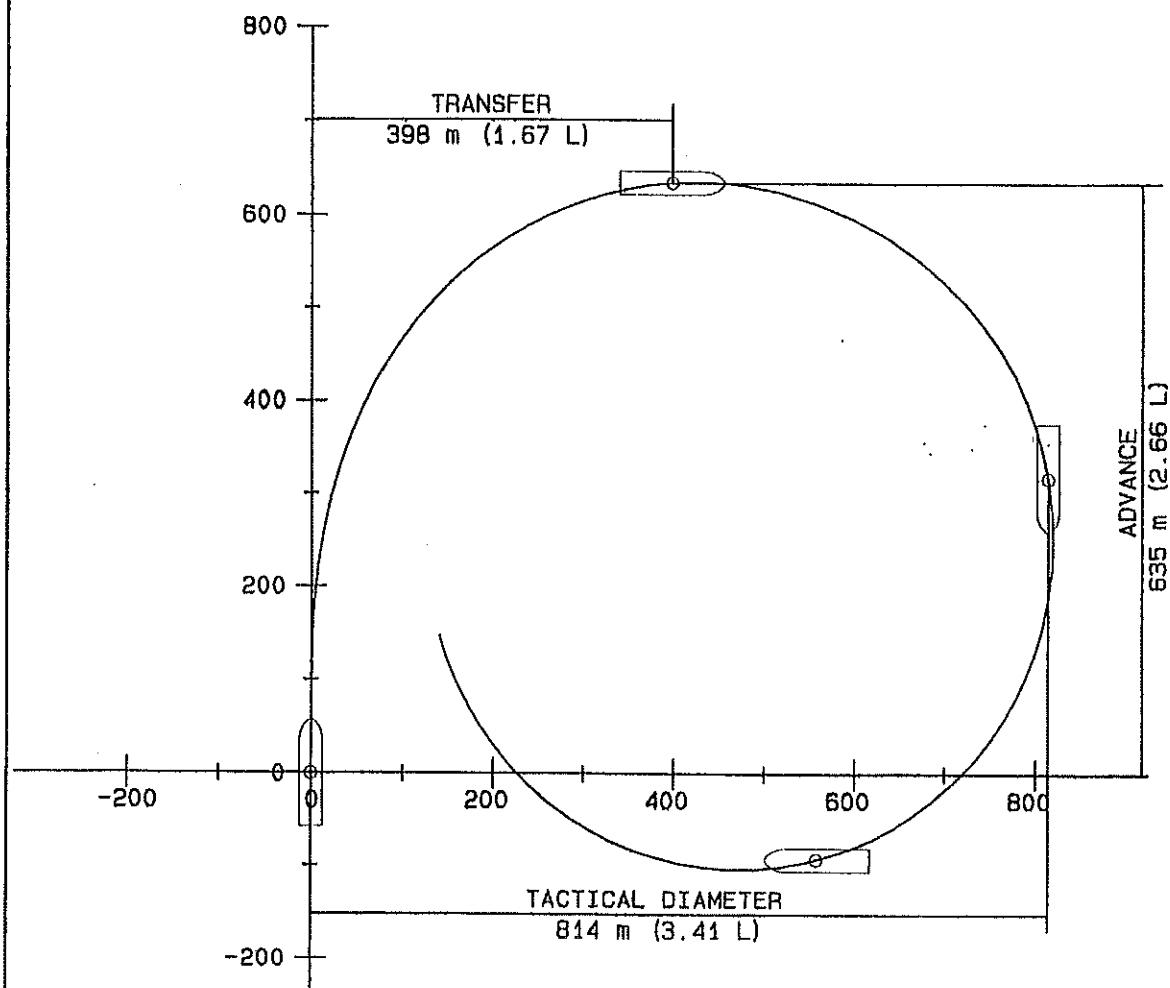
TURNING CIRCLE TO PORT

(BALLASTED CONDITION at IMO SPEED)



TURNING CIRCLE TO STARBOARD

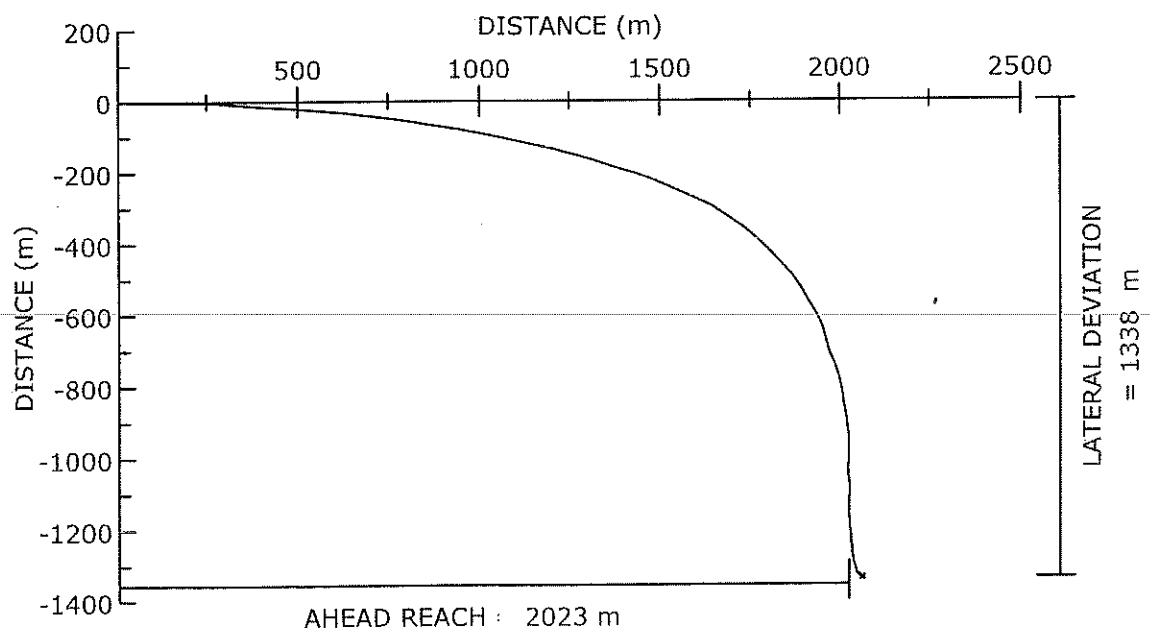
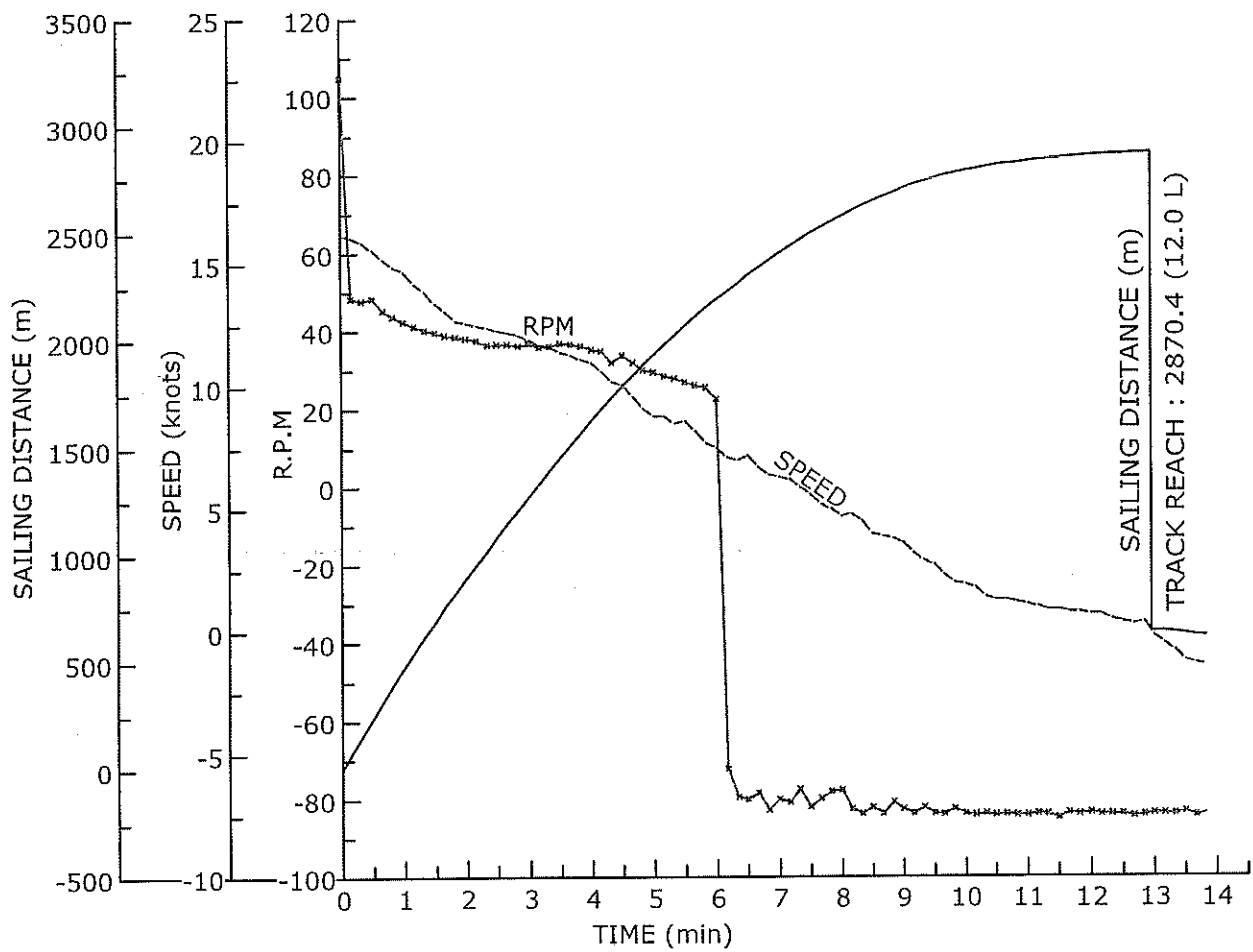
(BALLASTED CONDITION at IMO SPEED)



 HHIC	Test No. H-		Page	9
	Sea Trial Results for 4) Crash Stop Astern Test		Ship No.	N208
			Date	10.09.05
Crash Stop Astern Test				
Trial date : 2010. 09. 05		Trial Place : OFF PUSAN		
Ship Condition : <i>Trial Ballast Condition</i>		Sea Condition :		
Draft, Fore	13.660 m	Weather	Fine	
Draft, Aft	13.671 m	Sea Condition	Calm	
Displacement	120847.2	Wind Dir. & Velocity	S-10°, 12.9 m/s	
		Depth of Sea Water	129	
Trial Result				
	IMO speed to Full Astern			
Clock Time	00 : 25			
Base Course	1.9			
Shaft rpm	105			
Ship speed	16.2 kts			
Time to Start Shaft Astern	6min. 02sec.			
Time to Stop Ship	12 min. 54sec.			
Time to Reach Astern Steady rpm	10 min. 09sec.			
Astern Steady rpm	-84			

CRASH STOP ASTERN

DESIGN LOADED CONDITION at FULL SPEED

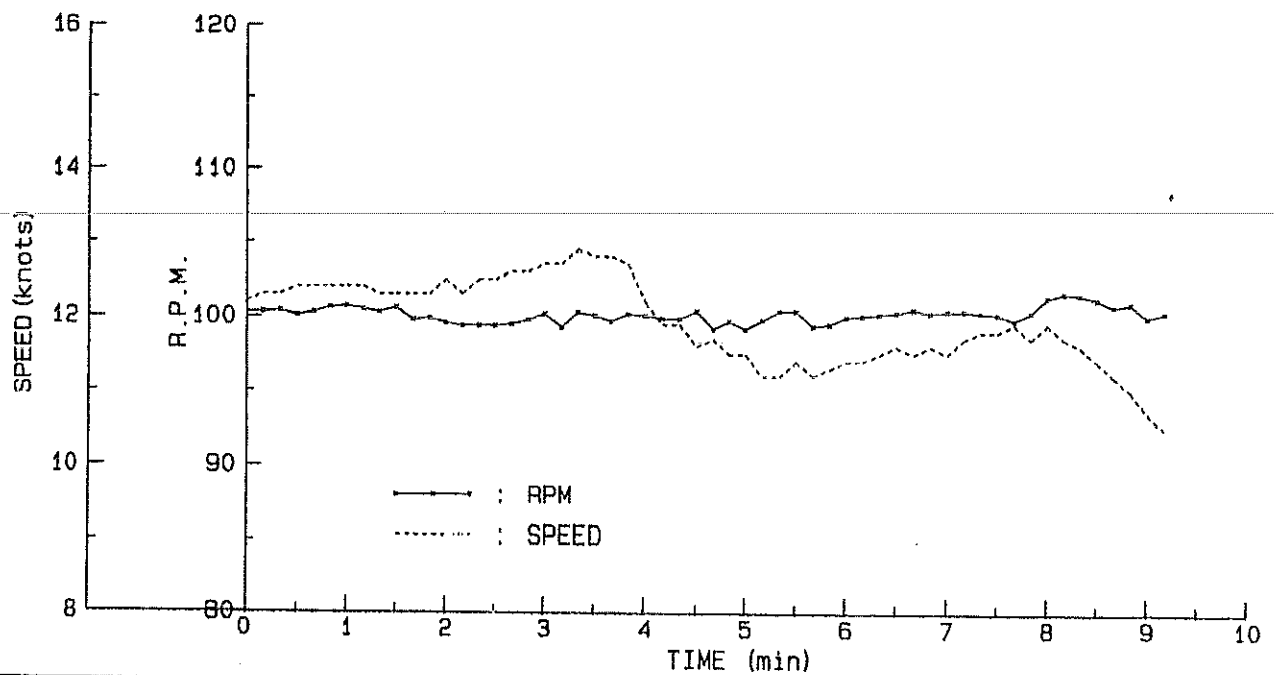
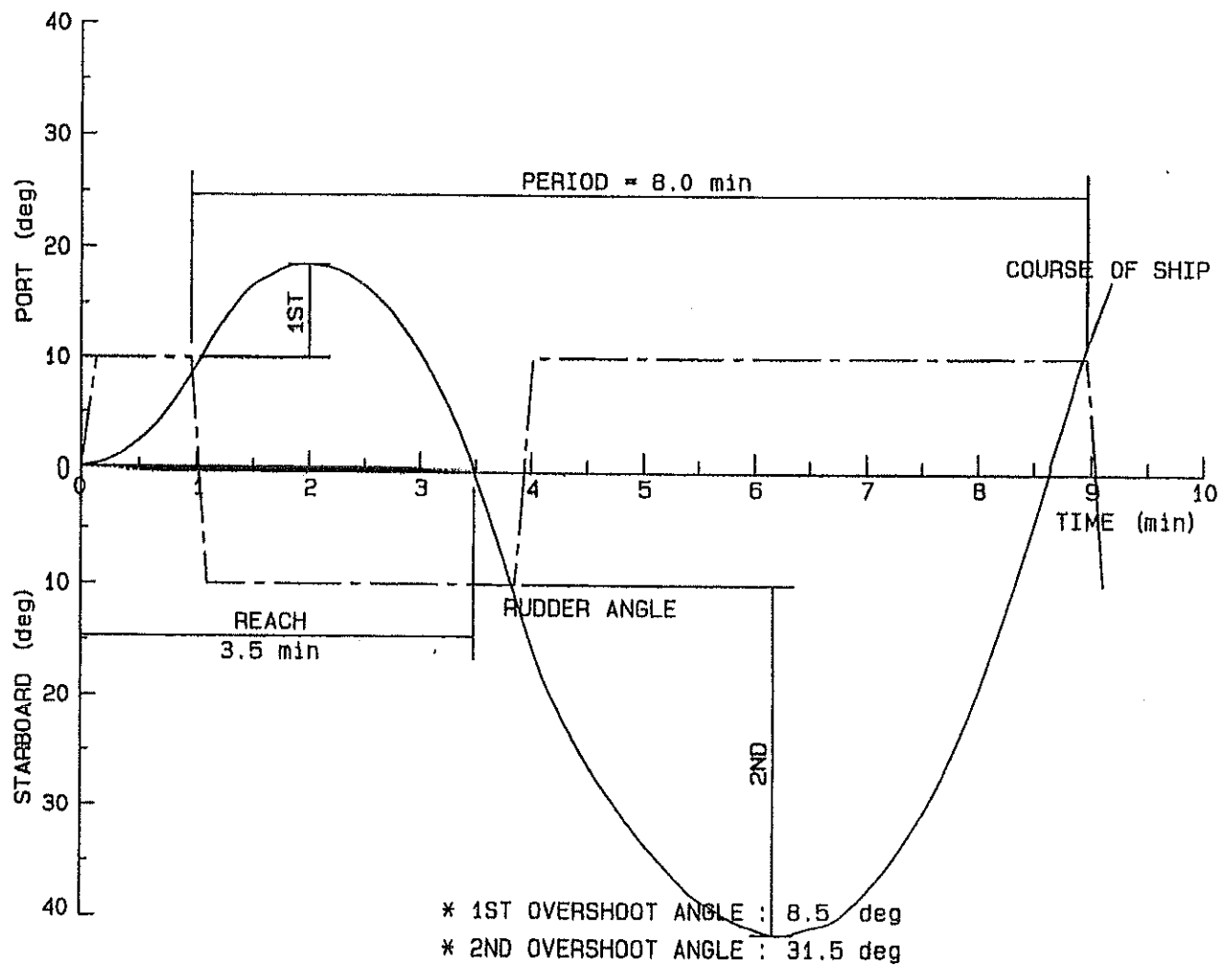


			Page	11
	Sea Trial Results for 5) "Z" Maneuver Test		Ship No.	N198
			Date	09.01.12
Trial date : 2009. 01. 12		Trial Place : Off Busan		
Ship Condition : Trial Ballast Condition		Sea Condition		
Draft, Fore	13.589 m	Weather	Windy	
Draft, Aft	13.756 m	Sea Condition	Rough	
Displacement	122069.3 ton	Wind Dir. & Velocity	S-20°, 10 m/s	
		Depth of Sea Water	129 m	
Trial Result (Rudder angle/Course= 10/10 deg)				
		Port Rudder First	Starboard Rudder First	
Clock Time		20 : 30	20 : 00	
Base Course		240°	240°	
Shaft rpm		100.0 rpm	100.0 rpm	
Ship speed		13.80 knot	13.80 knot	
Overshoot Angle	1st	16.9°	8.5°	
	2nd	27.5°	31.5°	
Trial Result (Rudder angle/Course= 20/20 deg)				
		Port Rudder First	Starboard Rudder First	
Clock Time		21 : 30	21 : 0105	
Base Course		240°	240°	
Shaft rpm		100.0 rpm	100.0 rpm	
Ship speed		13.80 knots	13.80 knots	
Overshoot Angle	1st	18.0°	19.6°	
	2nd	18.1°	19.2°	

"Z" MANEUVER

(BALLASTED CONDITION at IMO SPEED)

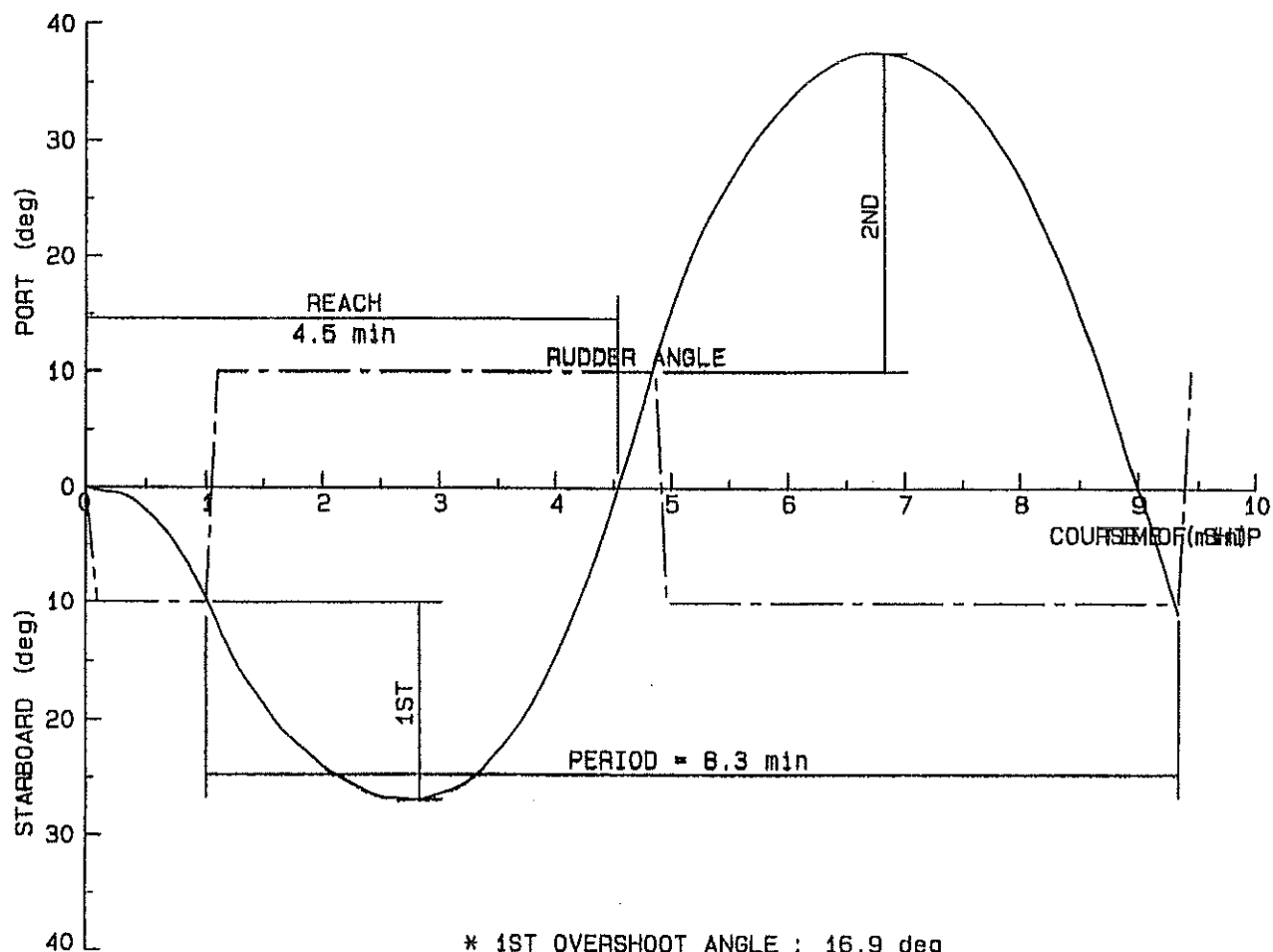
(P-10/10 deg)



"Z" MANEUVER

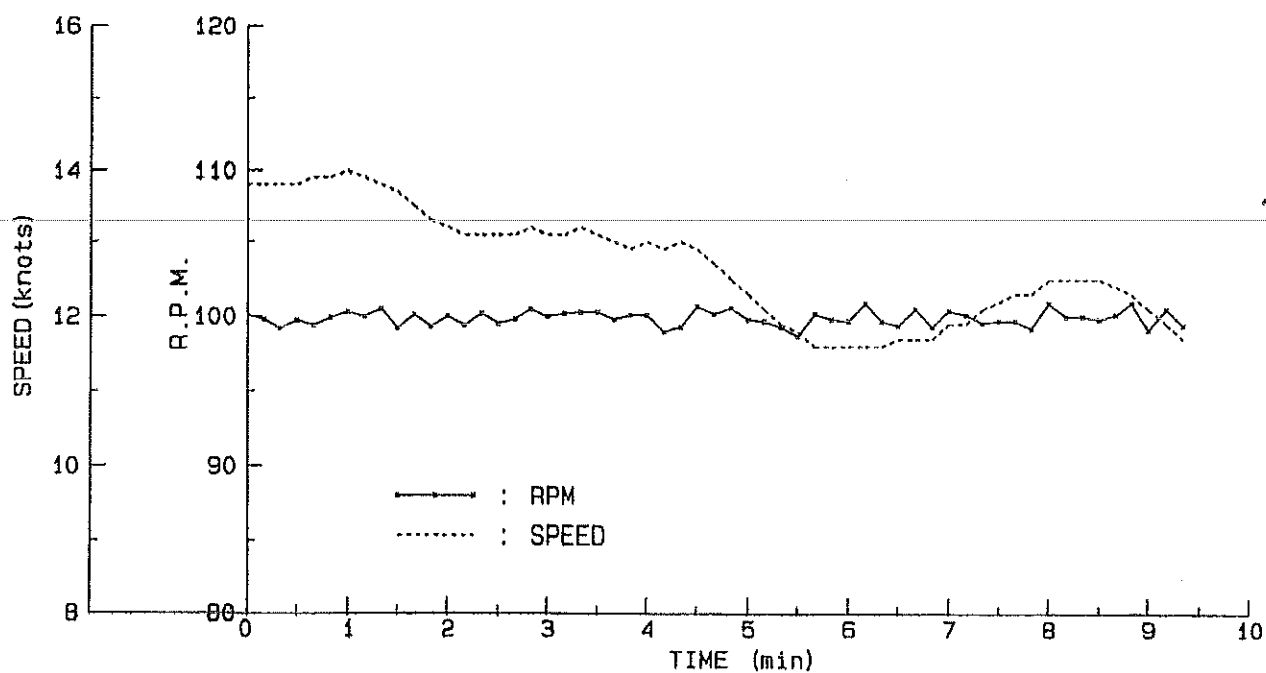
(BALLASTED CONDITION at IMO SPEED)

(S-10/10 deg)



* 1ST OVERSHOOT ANGLE : 16.9 deg

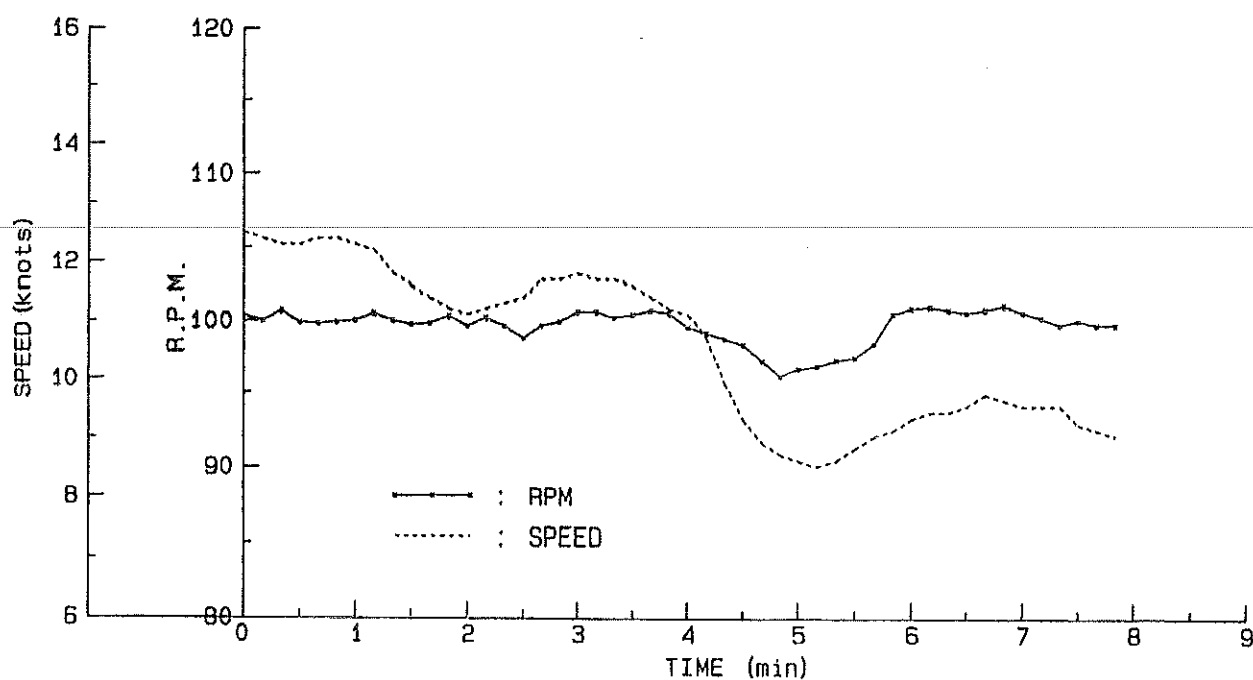
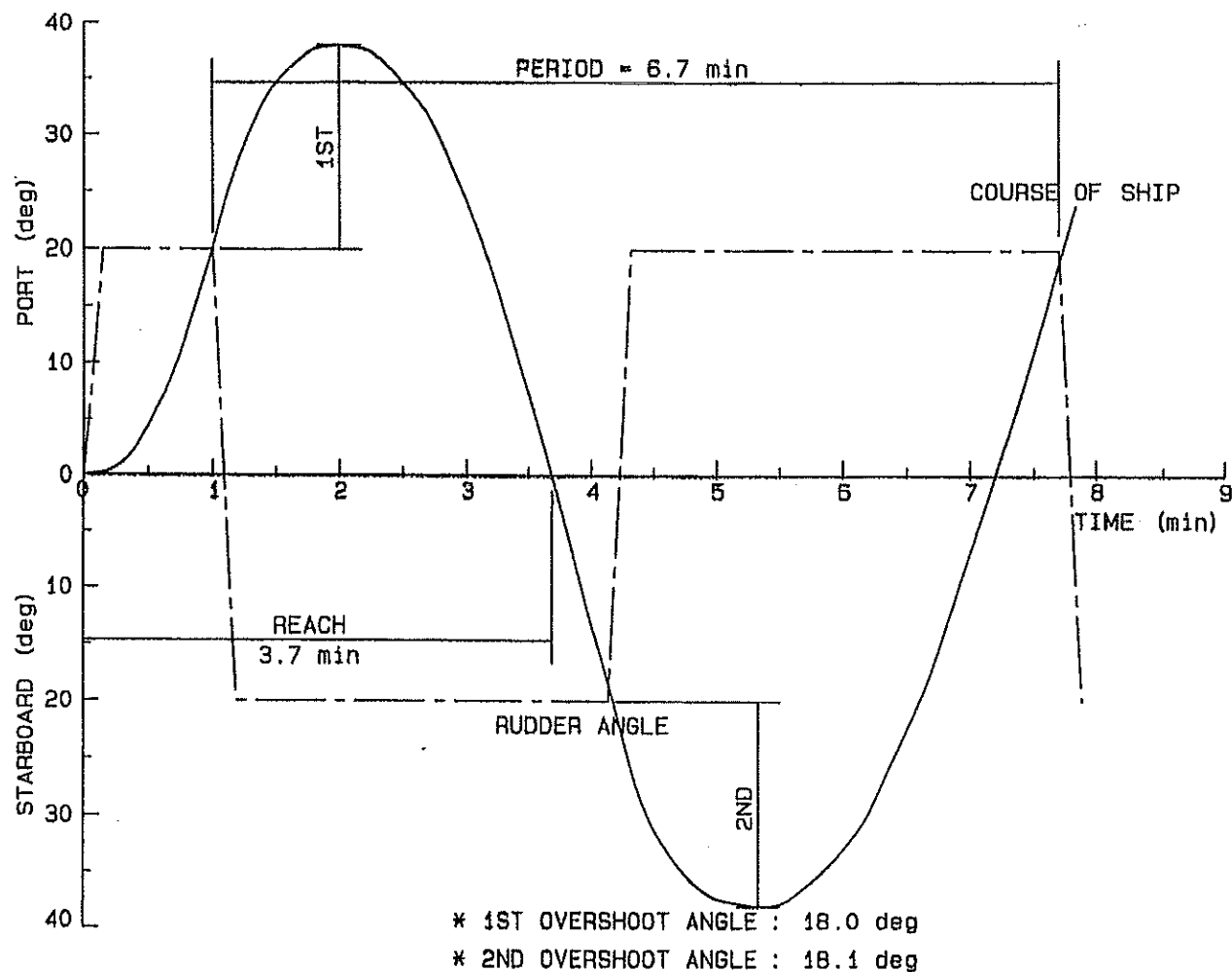
* 2ND OVERSHOOT ANGLE : 27.5 deg



"Z" MANEUVER

(BALLASTED CONDITION at IMO SPEED)

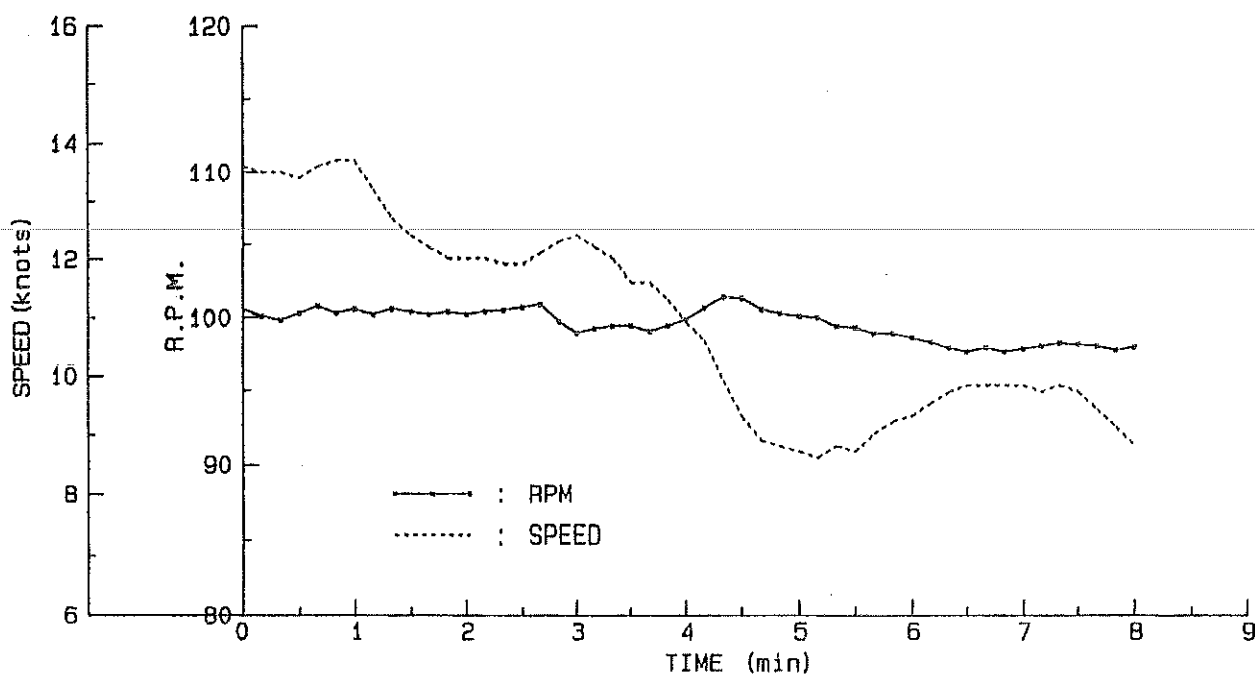
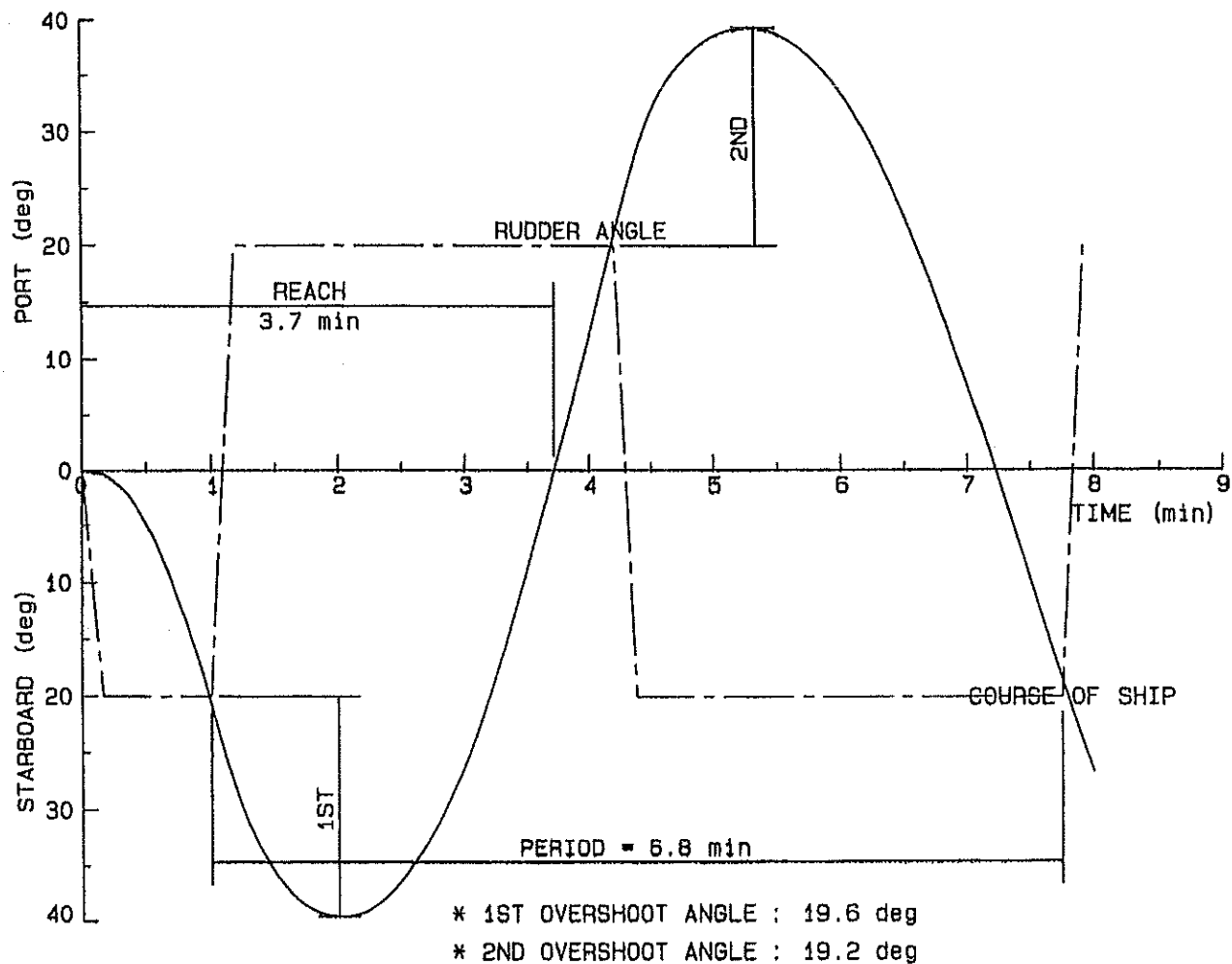
(P-20/20 deg)



"Z" MANEUVER

(BALLASTED CONDITION at IMO SPEED)

(S-20/20 deg)



INITIAL TURNING ABILITY

at 10/10 ZIGZAG TEST

(unit : meter)

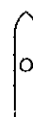
PORT

STARBOARD

Initial Speed
: 13.7 kts

Initial Speed
: 13.8 kts

* REMARK :



: 10 deg changed
position of course

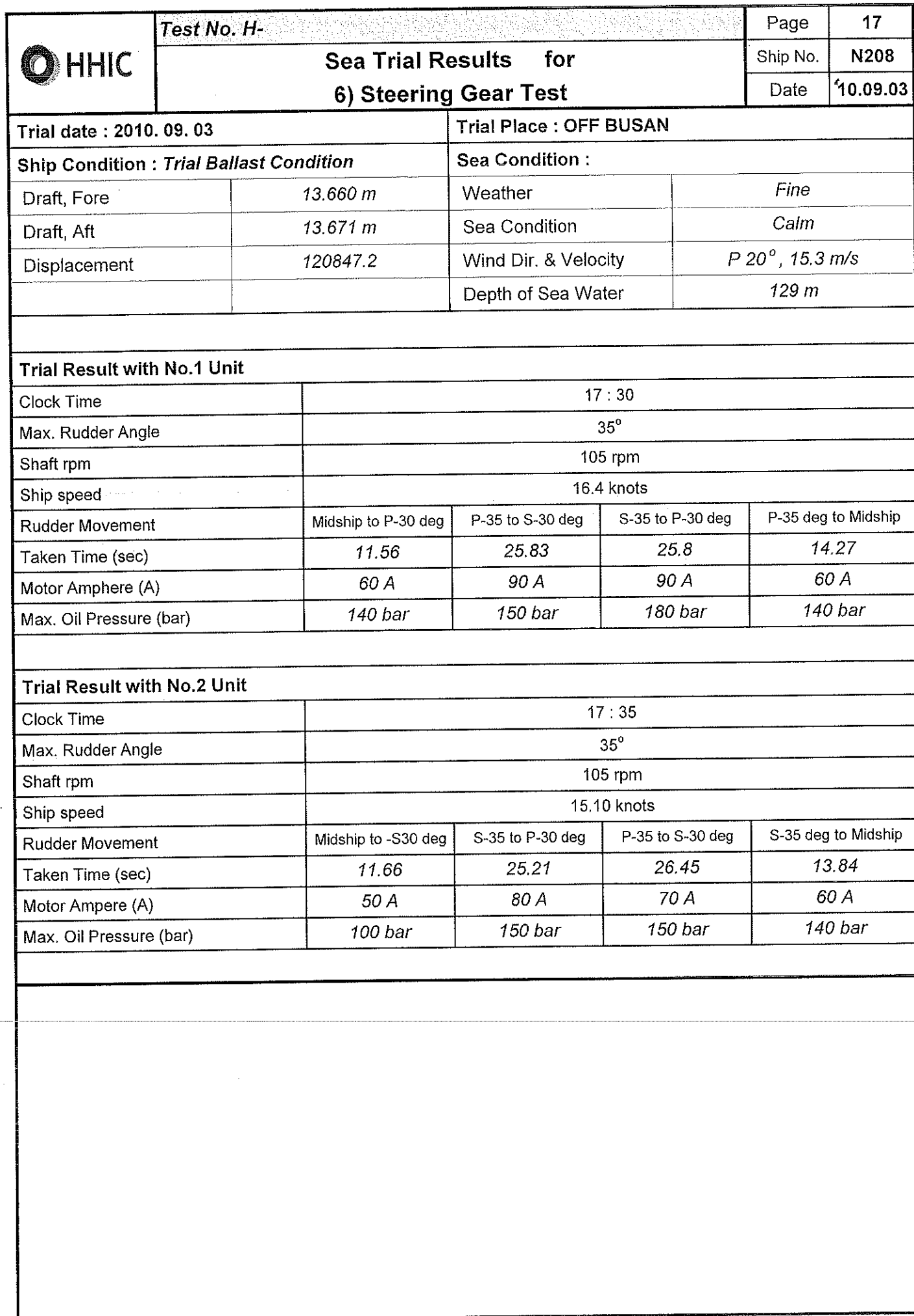
-100

0

100

TEST RESULT

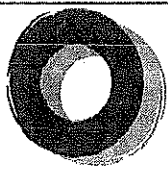
	PORT TURN	STB' D TURN
TRACK REACH	437.4 m (1.83 L)	435.2 m (1.82 L)
ELAPSED TIME	/ min 03 sec	/ min 0/ sec



2. Hull Outfit Part

C O N T E N T S

1)	Anchor Windlass Test	1
2)	Fire Line Shooting Test	2
3)	Window Wiper & Heater Test	3
4)	Life & Rescue Boat / Davit Test	4

	체크시이트 번호 CHECK SHEET NO.	페이지 Page of
	검사/시험 성적서 RESULT OF INSPECTION / TEST	
	발송일자 Date Issued	

선 번 Project No.	N-208	검사항목 Inspection Item	WINDLASS / MOORING WINCH OPERATION	검사종류 Type of Inspection
검사관련기관 Class/Authority	LR	부서 Dept.	QA	장소 Place
			ON BOARD	날씨/온도 Weather/Temp.
				℃

단위

Unit: _____

1) ANCHOR WINDLASS TEST**1.0 GENERAL**

1.1 TEST DATE : 29-Aug-10

1.2 ATTENDENTS : LR, SHIP OWNER

2.0 TEST RECORD**2.1 TRIAL RESULT**

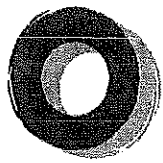
	PORT ANCHOR / WINDLASS	STB'D ANCHOR / WINDLASS
CLOCK TIME	8:20	8:40
TOTAL CHAIN LENGTH	375.5	375.5
LENGTH OF CHAIN DROPPED	128.5	128.5
TIME TO HOIST 30 FATHOMS (TWO SHACKLES)	4' 31" (271 sec)	4' 32" (272 sec)
AVERAGE HOIST SPEED	12 m / min	12 m / min
WINDLASS MOTOR AMPERE	90	90

2.2 FUNCTION CHECK

CHAIN LOWERING CONDITION	OK	OK
CHAIN CABLE COUNTER	OK	OK
BRAKE CONDITION	OK	OK
CHAIN COMPRESSOR	OK	OK
HOISTING CONDITION (ONE ANCHOR RESPECTIVELY)	OK	OK
SLOW DOWN FUNCTION	OK	OK
ANCHOR STOWING CONDITION	OK	OK
CHAIN SECURING CONDITION	OK	OK
MISCELLANEOUS	OK	OK

	Dept. QA		Witnessed / Approved by			비고 Remarks :
	검사자 Checker	Part 장 Apporoved by	선 주 Owner	선 급 Class	해당기관 Authority	
일 자 Date	29-Aug-10					
확인자 BY	DH.KIN					

 (HHIC-QC-004
90.1.1.1 명목)

	체크시이트 번호 CHECK SHEET NO.		페이지 Page of	
	검사/시험 성적서 RESULT OF INSPECTION / TEST		발송일자 Date Issued	
선 번 Project No.	N208	검사항목 Inspection Item	FIRE LINE SHOOTING TEST	검사종류 Type of Inspection
검사관련기관 Class/Authority	LR	부서 Dept.	QA	장소 Place
			ON BOARD	날씨/온도 Weather/Temp. °C

단위

Unit: _____

2) FIRE LINE SHOOTING & WATER SPRAY TEST

1.0 GENERAL

1.1 TEST DATE : 27-Aug-10

1.2 ATTENDENTS : LR, SHIP OWNER

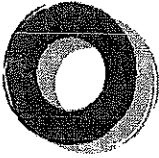
2.0 FIRE LINE SHOOTING

ITEM		PUMP SIDE	FIRE & G/S PUMP	FIRE & BILGE PUMP	EM'CY FIRE PUMP
PRESSURE	PUMP SIDE (SUCTION/DISCH.)		0.2 / 11.5	0.25 / 12.3	0 / 12.5
	NAV.BRIDGE DECK (kg/cm ²)		8.0	8.8	10.0
	F'CLE DECK (kg/cm ²)		9.4	10.0	12.0
PUMP ON/OFF CONDITION	NAV.BRIDGE DECK		OK	OK	OK
	FIRE CONT. STATION		OK	OK	OK
	PUMP SIDE		OK	OK	OK
	EM'CY GEN. ROOM				OK
	ECR		OK	OK	OK
WORKING CONDITION OF HYDRANT			OK		

3.0 WATER SPRAY

NO.	LOCATION	VALVE NO.	RESULT
1	PAINT & LAMP STORE	WD 68	OK
2	CHEMICAL STORE	WD 67	OK

	Dept. QA		Witnessed / Approved by			비 고 Remarks :
	검사자 Checker	Part 장 Approved by	선 주 Owner	선 급 Class	해당기관 Authority	
일 자 Date	27-Aug-10					
확인자 BY	DH.KIM					

	체크시이트 번호 CHECK SHEET NO.	페이지 Page of
	검사/시험 성적서 RESULT OF INSPECTION / TEST	발송일자 Date Issued

선 번 Project No.	N208	검사항목 Inspection Item	WINDOW WIPER OPERATION TEST	검사종류 Type of Inspection	
검사관련기관 Class/Authority	부서 Dept.	QA	장소 Place	ON BOARD	날씨/온도 Weather/Temp. °C

단위
Unit: cycle/min

3) WIPER & C.V.S OPERATION TEST

1.0 GENERAL

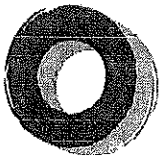
1.1 TEST DATE : 27-Aug-10

1.2 ATTENDENTS : SHIP OWNER

2.0 TEST RECORD

NO.	RESULT		LAMP CONDITION	HEATER LAMP CONDITION
	LOW	HIGH		
W1	OK	OK	OK	OK
W2	OK	OK	OK	OK
W3	OK	OK	OK	OK
C.V.S TEST				
PORT		OK		
STB'D		OK		
<u>REMARKS :</u>				

	Dept. QA		Witnessed / Approved by			비 고 Remarks :
	검사자 Checker	Part 장 Approved by	선 주 Owner	선 급 Class	해당기관 Authority	
일 자 Date	27-Aug-10					
확인자 BY	DH.KIM					

	체크시이트 번호 CHECK SHEET NO.	페이지 Page of
	검사/시험 성적서 RESULT OF INSPECTION / TEST	
	발송일자 Date Issued	

선 번 Project No.	N208	검사항목 Inspection Item	LIFE & RESCUE BOAT / DAVIT	검사종류 Type of Inspection
검사관련기관 Class/Authority	LR	부서 Dept.	QA	장소 Place
		ON BOARD		날씨/온도 Weather/Temp. °C

단위

Unit: _____

4) LIFE & RESCUE BOAT / DAVIT

1.0 GENERAL

1.1 TEST DATE : 4-Sep-10

1.2 ATTENDENTS : LR, SHIP OWNER

*Wb= BOAT + EQUIP'T WEIGHT : 3,262kg
(include floating block weight)

2.0 TEST RECORD

*Wp=WEIGHT OF 1 PEREON : 75kg

NO	TEST ITEM	RESULT	
		STB'D	PORT
DAVIT OPERATION TEST WITHOUT LOAD			
1	OUT-REACH FROM SHIP SIDE	1260 mm	1200 mm
2	CONDITION OF LIMIT SWITCH	OK	OK
3	CONDITION OF DAVIT LOCKING DEVICE	OK	OK
4	OPERATION OF 20° HEELING CONDITION	OK	OK
5	CONDITION OF MANUAL OPERATION	OK	OK
DAVIT OPERATION TEST (TEST LOAD : 3712 kg=Wb + 6 Wp)			
6	RECOVERY DEMONSTRATION (RESCUE BOAT ONLY)		OK
7	HOISTING SPEED (RESCUE BOAT ONLY) : Min 0.3m/s		0.32 m/s
LOAD TEST (TEST LOAD : 6475.7 kg= 1.1 × (Wb + 35 Wp)			
8	LOWERING SPEED (0.72m/s ~ 1.3m/s)	1.18 m/s	1.21 m/s
9	BRAKE CONDITION	OK	OK
10	HOOK RELEASING CONDITION	OK	OK
11	LENGTH OF FALLS (WIRE ROPE REMAINING CHECK)	20 LAYER	20 LAYER
12	LIFE BOAT START/RUN CONDITION	OK	OK
13	VISUAL INSPECTION AFTER TEST	OK	OK
14	BOAT LAUNCHING AT 5 KNOTS (DURING SEA TRIAL)		OK
REMARKS:			

	Dept. QA		Witnessed / Approved by			비고 Remarks :
	검사자 Checker	Part 장 Approved by	선 주 Owner	선 급 Class	해당기관 Authority	
일 자 Date	04-Sep-10					
확인자 BY	DH.KIM					

3) Machinery Part

C O N T E N T S

1 Operating Characteristics

1-1)	M/E Performance Curve	3
1-2)	M/E Characteristic Curve(I)	4
1-3)	M/E Characteristic Curve(II)	5
1-4)	Fuel Limit Diagram of Electronic Governor	6
1-5)	Record for M/E Condition	7~11
1-6)	M/E Indicator Diagram	12~21

2 F.O. Consumption Measurement 22~23

3 M/E Min. Revolution Test 24

4 M/E Starting Test 25

5 M/E Crankshaft Deflection Test 26

[illegible]

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REMARK

DESIGNED BY MACH. OUTFITTING DESIGN TEAM

TOTAL (24) SHEETS WITH A COVER

SHIP NO.

N - 208

DWG NO.

"DUBAI ANGEL"

8200-300

C

Z103016/기장기본설계PART/2010-09-06 13:35



Sea Trial Reports for
MACHINERY PART

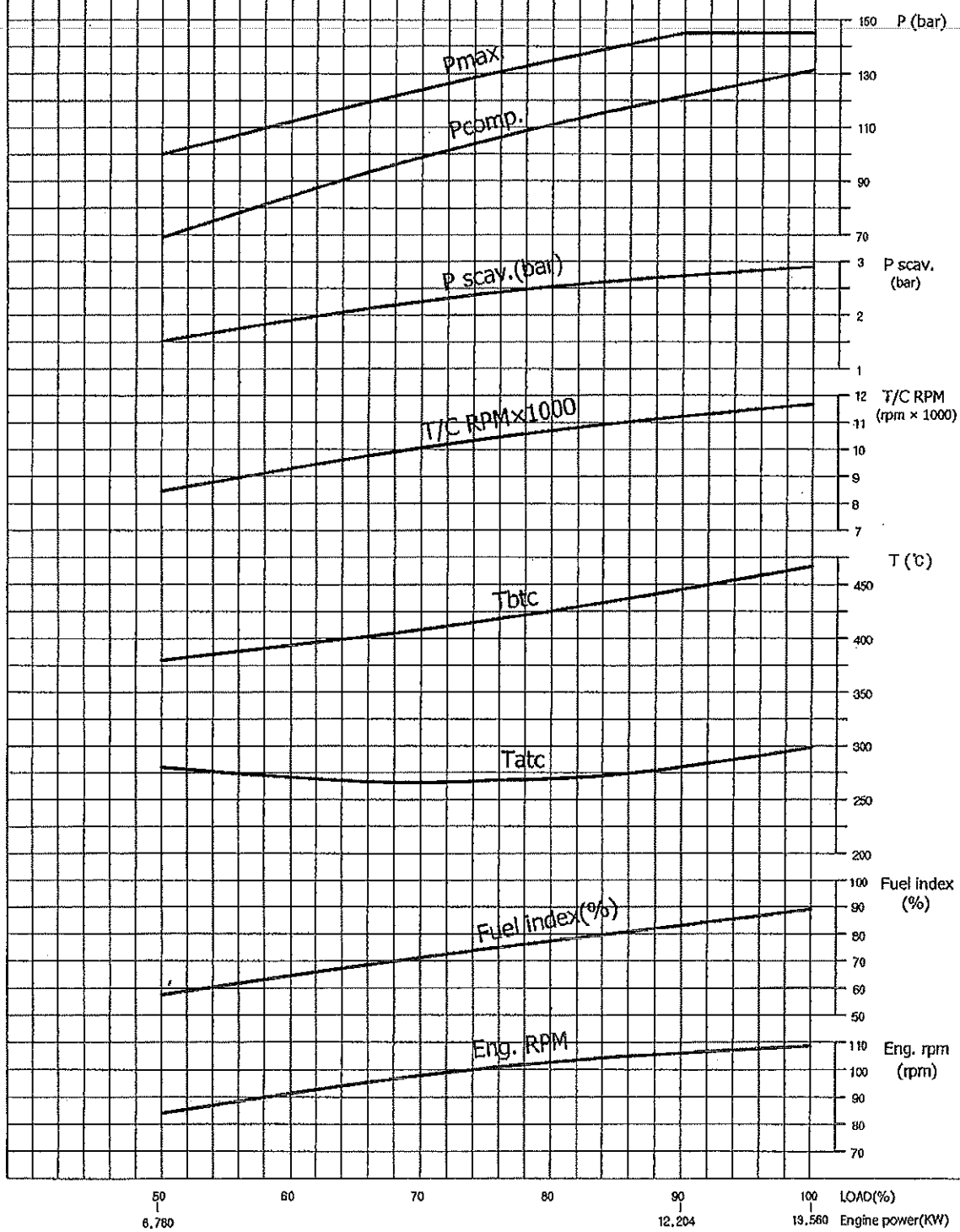
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Date	2010.09.06

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<u>NO.</u>	<u>DESCRIPTION</u>	<u>PAGE</u>
1	OPERATING CHARACTERISTICS	
	1-1) M/E PERFORMANCE CURVE	3
	1-2) M/E CHARACTERISTIC CURVE (I)	4
	1-3) M/E CHARACTERISTIC CURVE (II)	5
	1-4) Fuel Limit Diagram of Electronic Governor	6
	1-5) RECORD FOR M/E CONDITION	7 ~ 11
	1-6) M/E INDICATOR DIAGRAM	12 ~ 21
2	F.O. CONSUMPTION MEASUREMENT	22 ~ 23

1-1) M/E PERFORMANCE CURVE

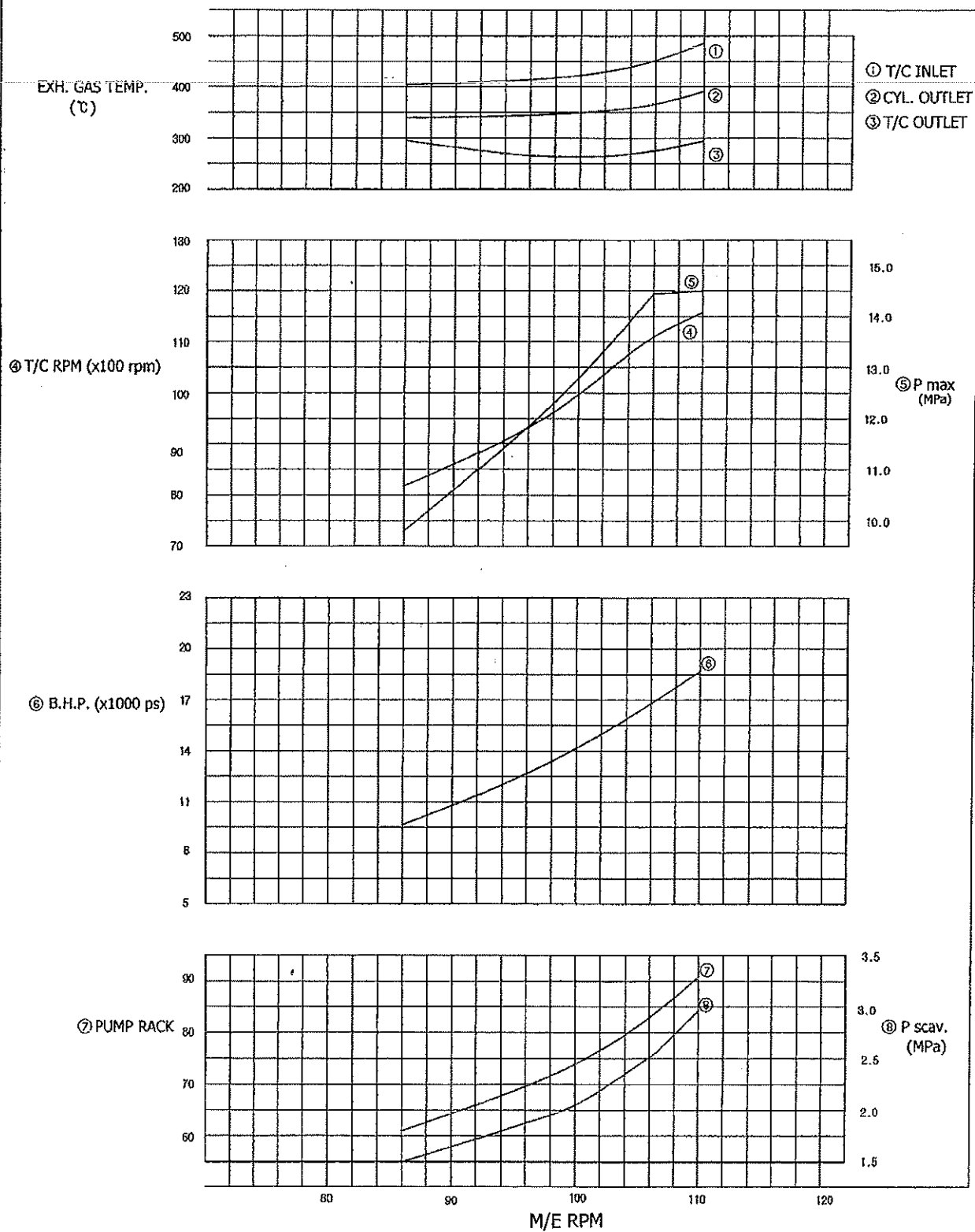
HULL NO. : N - 208



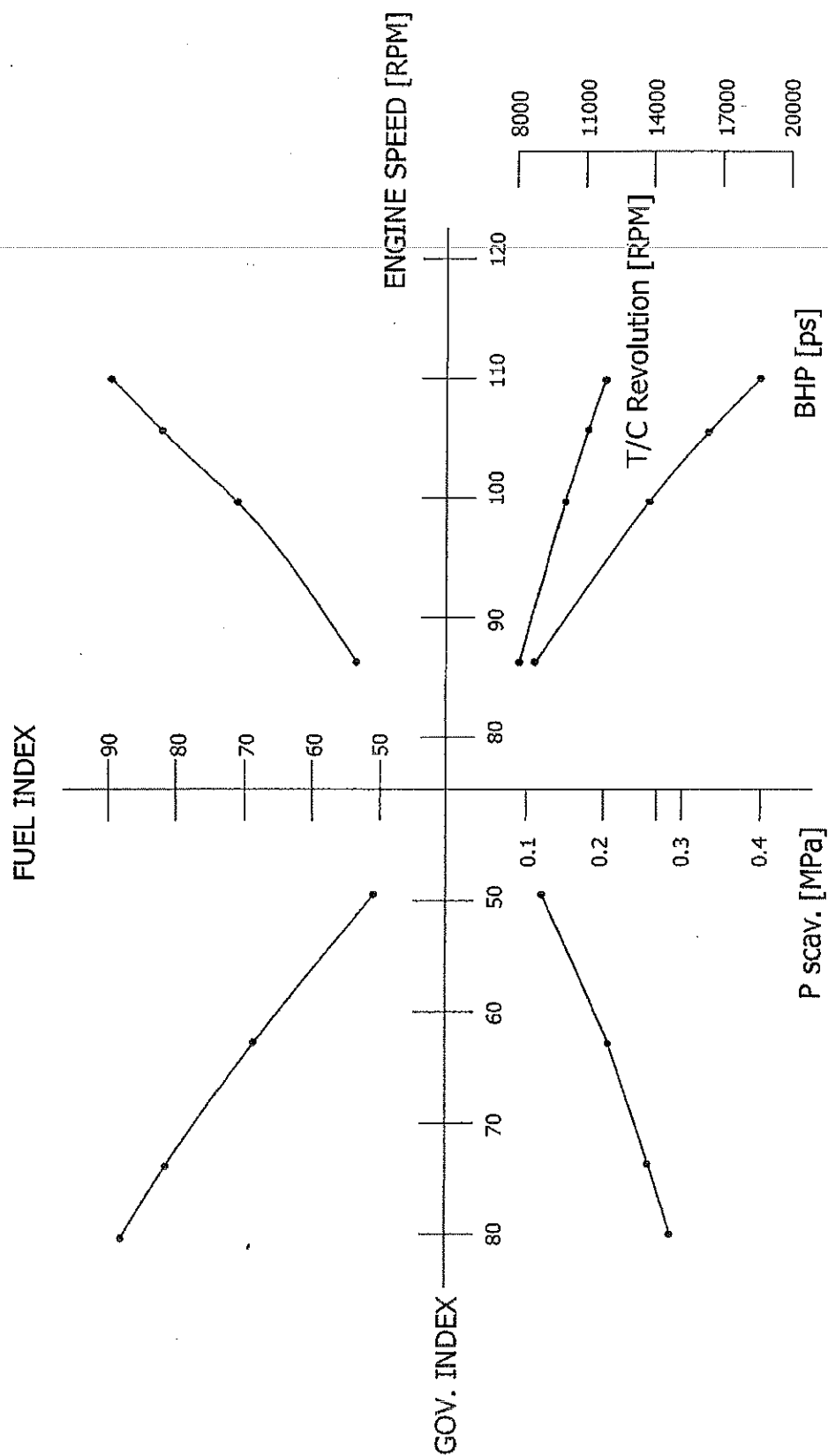
1-2) M/E CHARACTERISTIC CURVE (I)

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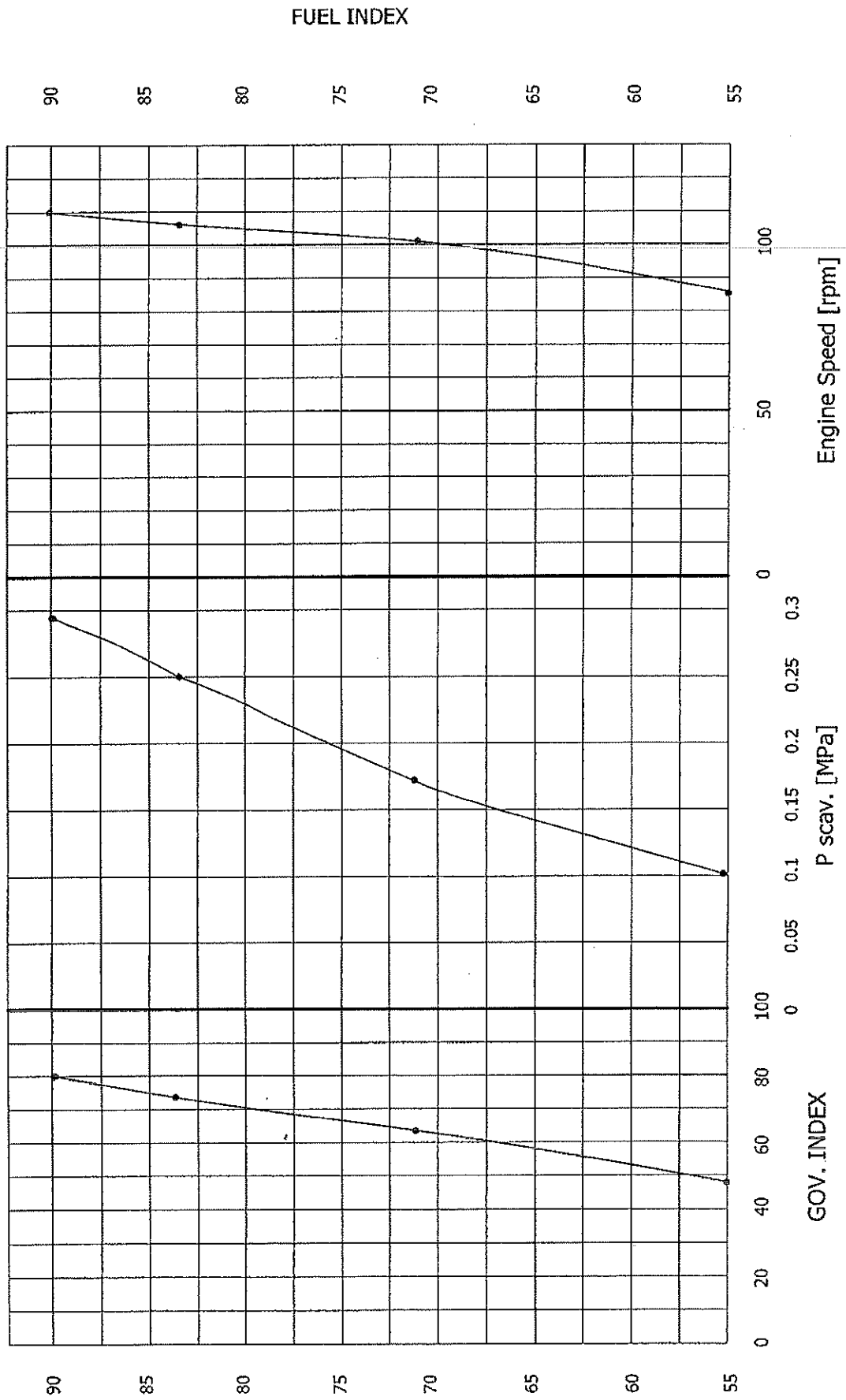
HULL NO. : N - 208



1-3) M/E CHARACTERISTIC CURVE (II)



1-4) Fuel Limit Diagram of Electronic Governor



	Test No. 1-5) Sea Trial Records for						Page		7 / 23				
	M/E PERFORMANCE						Ship No.		N-208				
							Date		2010.09.03				
REMARK		SPEED TEST				MEAS. TIME		13:10					
LOAD (%)		90% R1		LOAD. INDEX		92.7%		VIT AIR (bar)		-			
ENG. SPEED (rpm)		106.4		HANDLE NOTCH		-		HUMIDITY (%)		-			
ENG. POWER	(kW)	12,565		GOV. INDEX (mm)	75	PRESS. (mmbar)	E/R	1011					
	(HP)	16,843					ATM	1010					
F.O. SERV TK TEMP. (°C)		92		ROOM TEMP. (°C)		36		ATM TEMP. (°C)		34			
GAUGE	PRESSURE (kg/cm ²)						TEMP. (°C)						
	MAIN L.O.	F.O.	P.C.O.	J.C.W (IN/OUT)	STARTING AIR	MAIN L.O.	CAM L.O. (OUTLET)	F.O.	THRUST PAD				
	1.80	7.5	-	3.3 / 1.5	29	48	53	132	62				
CYL. No.	Pmax	Pcomp	Texh	FUEL INDEX	VIT INDEX	F.W.	P.C.O.	AIR COOLER					
	bar	°C	mm	OUT(°C)	OUT(°C)	IN	77	IN	-	1	2	3	AVG
AVG	144.0	122.8	371.7	83.5	4.87	84.0	61.3	PRESS	PRESS DROP mmAq	220			
1	143	122	380	83	4.9	84	62		F.W. IN	2.2			
2	145	124	380	84	4.8	84	64		F.W. OUT	-			
3	145	122	350	83	4.9	84	60	TEMP °C	AIR IN	193			
4	145	122	370	83	4.8	84	60		AIR OUT	42			
5	142	122	380	84	4.9	84	61		F.W. IN	35			
6	144	125	370	84	4.9	84	61		F.W. OUT	52			
SCAV. AIR													
								PRESS.	bar	2.5			
									mmHg	-			
								TEMP.	°C	42			
COMP. AIR PRESS. (kg/cm ²)													
								SPRING	CONTROL	SAFETY			
								6.5	6.5	6.5			
T/C	SPEED	BLOWER INLET		BEF. T	AFTER T/C		LUB. OIL						
	rpm	°C	mmAq	°C	°C	mmAq	L.O IN (bar)	L.O IN (°C)	L.O OUT(°C)				
	NO.1	11,218	39 / 47	25	450	280	70	0.9	47	71			
	NO.2												
	NO.3												
AVG													
* VIT-ACTUATOR CONTROL PRESS. : 3.3													

 HANJIN HEAVY INDUSTRIES & CONSTRUCTION CO., LTD

	Test No. 2010-09-03						Page		8 / 23					
	1-5) Sea Trial Records for M/E PERFORMANCE						Ship No.		N-208					
							Date		2010.09.03					
REMARK		SPEED TEST						MEAS. TIME		14:20				
LOAD (%)		90% R2		LOAD. INDEX		92.4%		VIT AIR (bar)		-				
ENG. SPEED (rpm)		106.4		HANDLE NOTCH		-		HUMIDITY (%)		-				
ENG. POWER	(kW)	12,533		GOV. INDEX (mm)		74		PRESS. (mmbar)		E/R	1005			
	(HP)	16,800								ATM	1004			
F.O.SERV TK TEMP. (°C)		92		ROOM TEMP. (°C)		37		ATM TEMP. (°C)		33				
GAUGE	PRESSURE (kg/cm ²)						TEMP. (°C)							
	MAIN L.O.	F.O.	P.C.O	J.C.W (IN/OUT)	STARTING AIR	MAIN L.O.	CAM L.O. (OUTLET)	F.O.	THRUST PAD					
	1.95	7.5	-	3.3 / 1.5	28	47	53	132	61					
CYL. No.	Pmax	Pcomp	Texh	FUEL INDEX	VIT INDEX	F.W. IN	76	P.C.O. IN	-	AIR COOLER				
	bar		°C	mm		OUT(°C)		OUT(°C)		1	2	3	AVG	
AVG	145.8	120.5	368.3	82.8	5.33	84.0	60.8	PRESS	PRESS DROP mmAq	215				
1	146	121	370	83	5.4	84	61		F.W. IN	2.2				
2	147	121	370	83	5.3	84	64		F.W OUT	-				
3	147	120	350	82	5.3	84	60		TEMP °C	AIR IN	192			
4	145	120	360	82	5.3	84	60			AIR OUT	42			
5	145	120	370	83	5.3	84	60			F.W. IN	35			
6	145	121	390	84	5.4	84	60	F.W. OUT	51					
						SCAV. AIR								
						PRESS.		bar		2.5				
								mmHg		-				
						TEMP.		°C		42				
						COMP. AIR PRESS. (kg/cm ²)								
						SPRING		CONTROL		SAFETY				
						6.8		6.8		6.8				
T/C	SPEED	BLOWER INLET		BEF. T	AFTER T/C		LUB. OIL							
	rpm	°C		mmAq	°C	°C	mmAq	L.O IN (bar)	L.O IN (°C)	L.O OUT(°C)				
	NO.1	11,157	39 / 47	25	450	280	70	1.0	47	70				
	NO.2													
	NO.3													
AVG														
* VIT-ACTUATOR CONTROL PRESS. : 3.4														

 HANJIN HEAVY INDUSTRIES & CONSTRUCTION CO., LTD

HHIC	Test No. 2010-09-03						Page		9 / 23					
	1-5) Sea Trial Records for M/E PERFORMANCE						Ship No.		N-208					
							Date		2010.09.03					
REMARK		SPEED TEST				MEAS. TIME		15:40						
LOAD (%)		100% R1		LOAD. INDEX		102.6%		VIT AIR (bar)		-				
ENG. SPEED (rpm)		109.3		HANDLE NOTCH		-		HUMIDITY (%)		-				
ENG. POWER	(kW)	13,911		GOV. INDEX (mm)	81	PRESS. (mmbar)	E/R	1004						
	(HP)	18,647					ATM	1003						
F.O.SERV TK TEMP.(°C)		92		ROOM TEMP. (°C)		37		ATM TEMP. (°C)		34				
GAUGE	PRESSURE (kg/cm2)					TEMP. (°C)								
	MAIN L.O.	F.O.	P.C.O	J.C.W (IN/OUT)	STARTING AIR	MAIN L.O.	CAM L.O. (OUTLET)	F.O.	THRUST PAD					
	1.90	7.4	-	3.3 / 1.6	29	48	53	133	62					
CYL. No.	Pmax	Pcomp	Texh	FUEL INDEX	VIT INDEX	F.W. IN	74	P.C.O. IN	-	AIR COOLER				
	bar		°C	mm		OUT(°C)		OUT(°C)		1	2	3	AVG	
AVG	145.0	131.7	395.0	89.0	3.38	84.0	61.5	PRESS	PRESS DROP mmAq	230				
1	145	131	380	90	3.3	84	62		F.W. IN	2.1				
2	146	132	400	89	3.5	84	64		F.W OUT	-				
3	145	131	400	88	3.4	84	60		TEMP °C	AIR IN	208			
4	145	131	400	88	3.3	84	60			AIR OUT	44			
5	144	131	390	89	3.4	84	61			F.W. IN	35			
6	145	134	400	90	3.4	84	62			F.W. OUT	54			
						SCAV. AIR								
						PRESS.		bar		2.8				
								mmHg		-				
						TEMP.		°C		44				
						COMP. AIR PRESS. (kg/cm2)								
						SPRING		CONTROL		SAFETY				
						6.9		6.9		6.9				
T/C		SPEED	BLOWER INLET	BEF. T	AFTER T/C		LUB. OIL							
		rpm	°C	mmAq	°C	°C	mmAq	L.O IN (bar)	L.O IN (°C)	L.O OUT(°C)				
	NO.1	11,736	40 / 48	25	480	290	70	0.9	47	72				
	NO.2													
	NO.3													
AVG														
* VIT-ACTUATOR CONTROL PRESS. : 2.3														

HANJIN HEAVY INDUSTRIES & CONSTRUCTION CO., LTD

	Test No. 1-5) Sea Trial Records for						Page		10 / 23				
	M/E PERFORMANCE						Ship No.		N-208				
							Date		2010.09.03				
REMARK		SPEED TEST				MEAS. TIME		16:45					
LOAD (%)		100% R2		LOAD. INDEX		103.7%		VIT AIR (bar)		-			
ENG. SPEED (rpm)		109.8		HANDLE NOTCH		-		HUMIDITY (%)		-			
ENG. POWER	(kW)	14,068		GOV. INDEX (mm)	81		PRESS. (mmbar)	E/R	1004				
	(HP)	18,858						ATM	1003				
F.O.SERV TK TEMP.(°C)		92		ROOM TEMP. (°C)		37		ATM TEMP. (°C)		32			
GAUGE	PRESSURE (kg/cm2)						TEMP. (°C)						
	MAIN L.O.	F.O.	P.C.O	J.C.W (IN/OUT)	STARTING AIR		MAIN L.O.	CAM L.O. (OUTLET)	F.O.	THRUST PAD			
	1.90	7.3	-	3.2 / 1.5	28		49	53	132	62			
CYL. No.	Pmax	Pcomp	Texh	FUEL INDEX	VIT INDEX	F.W. IN	P.C.O. IN	AIR COOLER					
	bar		°C	mm		OUT(°C)	OUT(°C)		1	2	3	AVG	
AVG	145.0	133.2	395.0	89.2	3.10	81.7	61.3	PRESS	PRESS DROP mmAq	235			
1	144	132	380	90	3.1	82	62		F.W. IN	2.1			
2	145	133	400	90	3.2	82	64		F.W OUT	-			
3	146	133	400	86	3.1	80	60	TEMP °C	AIR IN	211			
4	145	134	400	89	3.0	82	60		AIR OUT	44			
5	145	133	400	90	3.1	82	61		F.W. IN	35			
6	145	134	390	90	3.1	82	61		F.W. OUT	55			
								SCAV. AIR					
								PRESS.	bar	2.9			
									mmHg	-			
								TEMP.	°C	44			
								COMP. AIR PRESS. (kg/cm2)					
								SPRING	CONTROL	SAFETY			
								6.5	6.5	6.5			
T/C		SPEED	BLOWER INLET	BEF. T	AFTER T/C		LUB. OIL						
		rpm	°C	mmAq	°C	°C	mmAq	L.O IN (bar)	L.O IN (°C)	L.O OUT(°C)			
	NO.1	11,828	39 / 48	25	485	295	80	0.9	47	72			
	NO.2												
	NO.3												
AVG													
* VIT-ACTUATOR CONTROL PRESS. : 2.2													

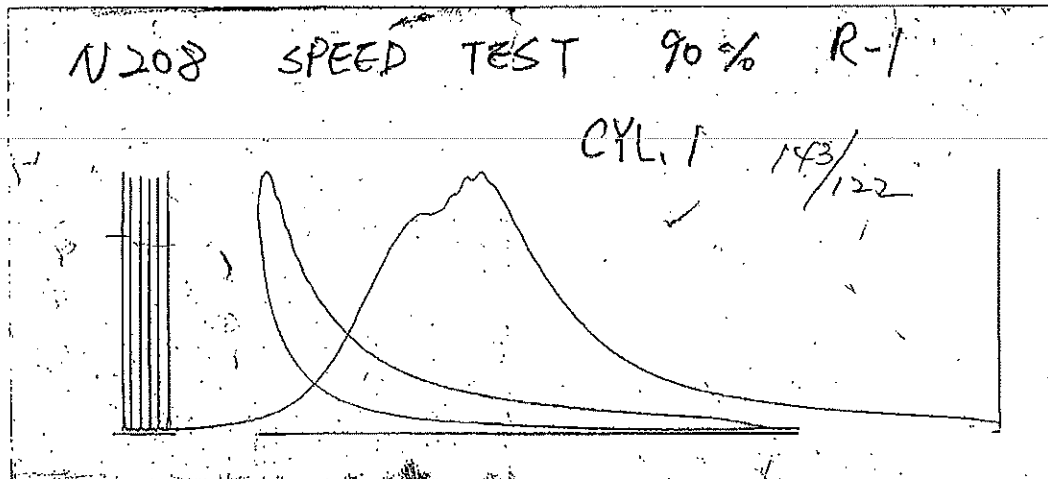
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	Test No. 1-5) Sea Trial Records for						Page 11 / 23								
	M/E PERFORMANCE						Ship No. N-208								
							Date 2010.09.03								
REMARK		ENDURANCE TEST				MEAS. TIME		17:50							
LOAD (%)		END. (90%)	LOAD. INDEX		95.0%		VIT AIR (bar)		-						
ENG. SPEED (rpm)		106.2	HANDLE NOTCH		-		HUMIDITY (%)		-						
ENG. POWER	(kW)	12,882	GOV. INDEX (mm)		76		PRESS. (mmbar)	E/R	1004						
	(HP)	17,268					ATM	1003							
F.O.SERV TK TEMP. (°C)		92	ROOM TEMP. (°C)		37		ATM TEMP. (°C)		31						
GAUGE	PRESSURE (kg/cm ²)						TEMP. (°C)								
	MAIN L.O.	F.O.	P.C.O	J.C.W (IN/OUT)	STARTING AIR		MAIN L.O.	CAM L.O. (OUTLET)	F.O.	THRUST PAD					
	1.90	7.5	-	3.2 / 1.4	30		47	53	132	61					
CYL. No.	Pmax	Pcomp	Texh	FUEL INDEX	VIT INDEX	F.W.		P.C.O.		AIR COOLER					
	bar		°C	mm		OUT(°C)		OUT(°C)		1 2 3 AVG					
AVG	144.7	122.8	373.3	84.5	4.63	80.7		60.5		PRESS DROP mmAq	225				
1	145	122	360	84	4.3	80		61		F.W. IN	2.2				
2	145	124	380	85	4.7	82		62		F.W OUT	-				
3	146	124	360	84	4.8	80		60		TEMP °C	AIR	IN	199		
4	144	123	380	84	4.6	82		60			OUT	43			
5	144	122	370	85	4.7	80		60			F.W.	IN	35		
6	144	122	390	85	4.7	80		60			OUT	52			
										SCAV. AIR					
										PRESS.	bar		2.9		
											mmHg		-		
										TEMP.	°C		43		
										COMP. AIR PRESS. (kg/cm ²)					
										SPRING	CONTROL	SAFETY			
										6.9	6.9	6.9			
T/C	SPEED	BLOWER INLET		BEF. T	AFTER T/C		LUB. OIL								
		rpm	°C	mmAq	°C	°C	mmAq	L.O IN (bar)	L.O IN (°C)	L.O OUT(°C)					
	NO.1	11,308	39 / 47	25	460	280	70	0.9	47	71					
	NO.2														
	NO.3														
AVG															
* VIT-ACTUATOR CONTROL PRESS. : 3.2															

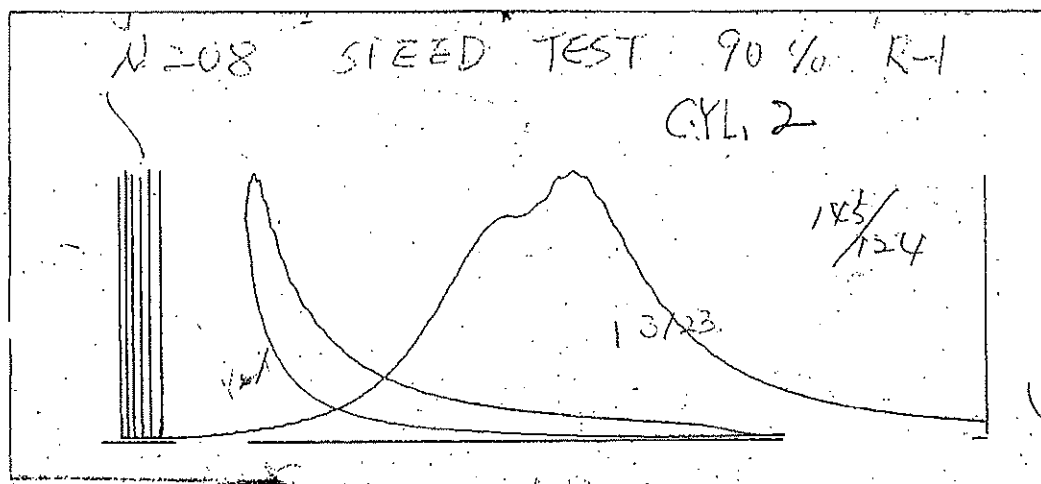
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HHIC	SPEED TEST	90% R1	PAGE : 12/28.
			SHIP NO : N - 208
			DATE : 10.09.06

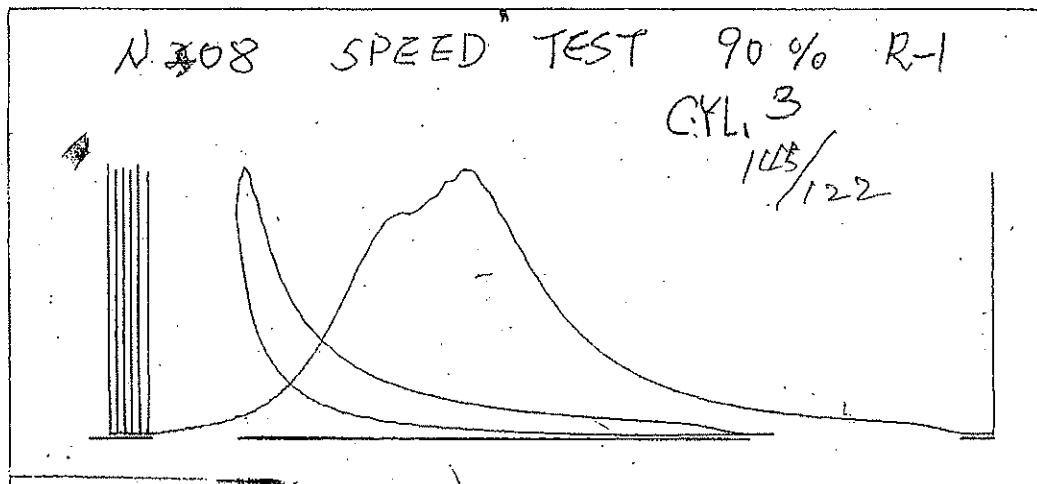
NO. 1 CYLINDER



NO. 2 CYLINDER



NO. 3 CYLINDER



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HHIC

SPEED TEST

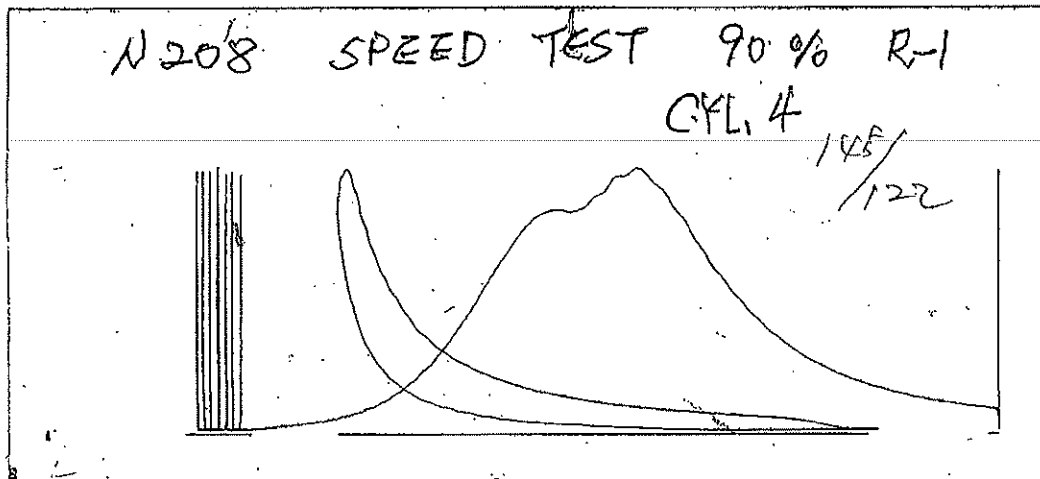
90% R1

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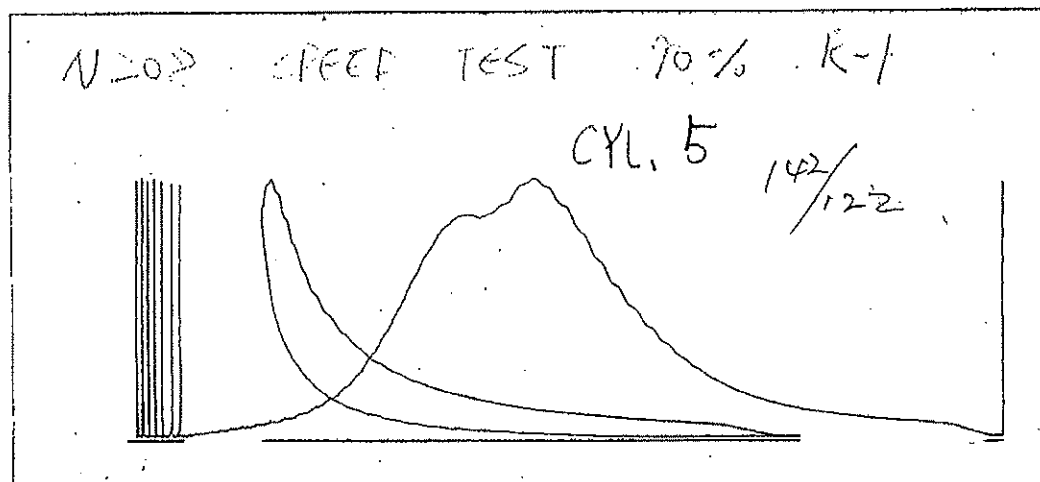
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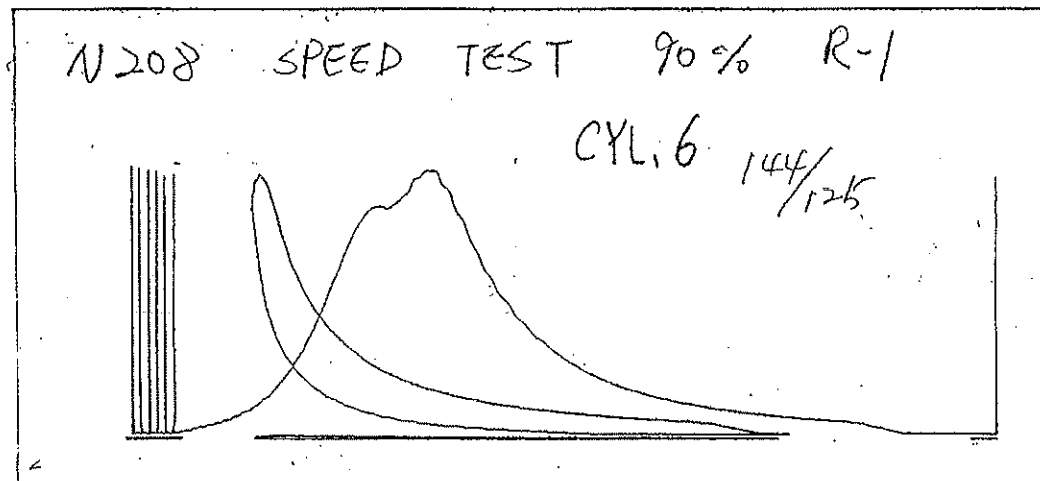
NO. 4 CYLINDER



NO. 5 CYLINDER



NO. 6 CYLINDER



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HHIC

SPEED TEST

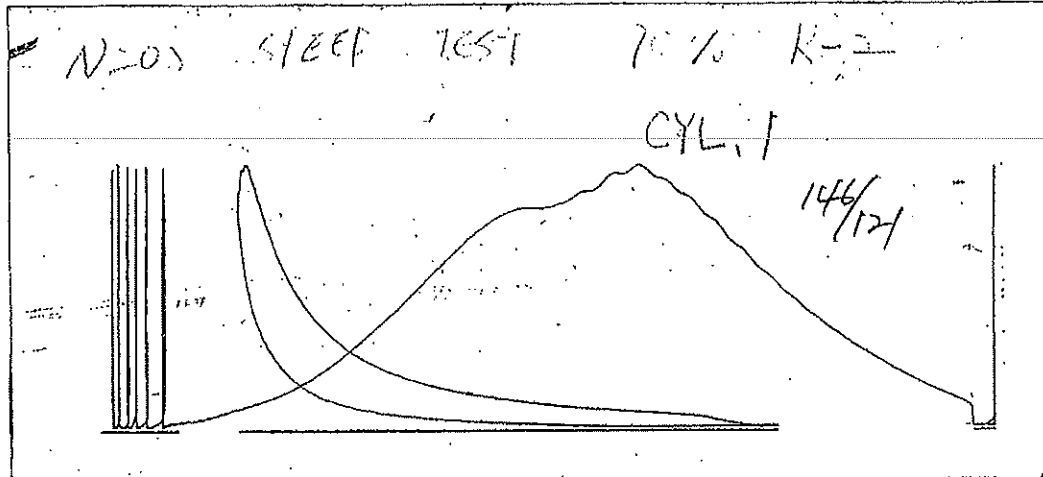
90% R 2

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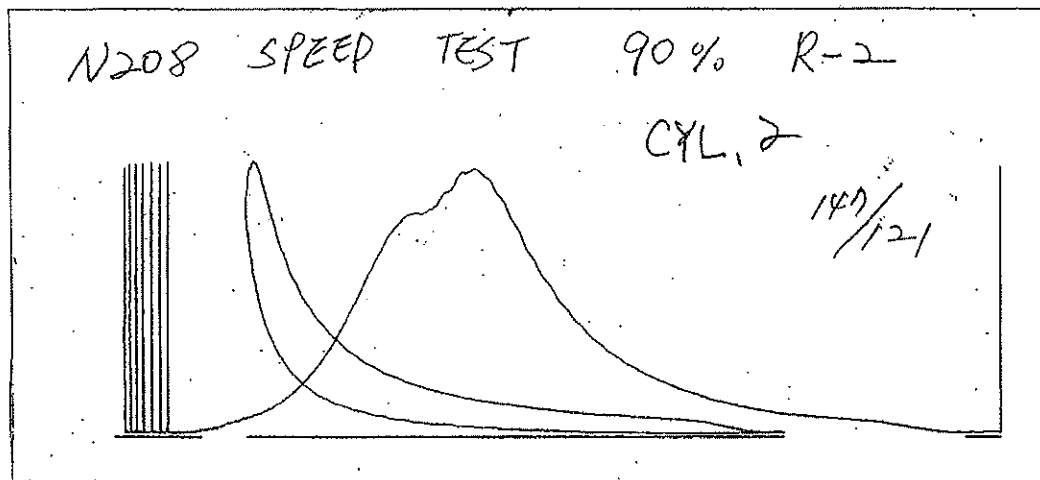
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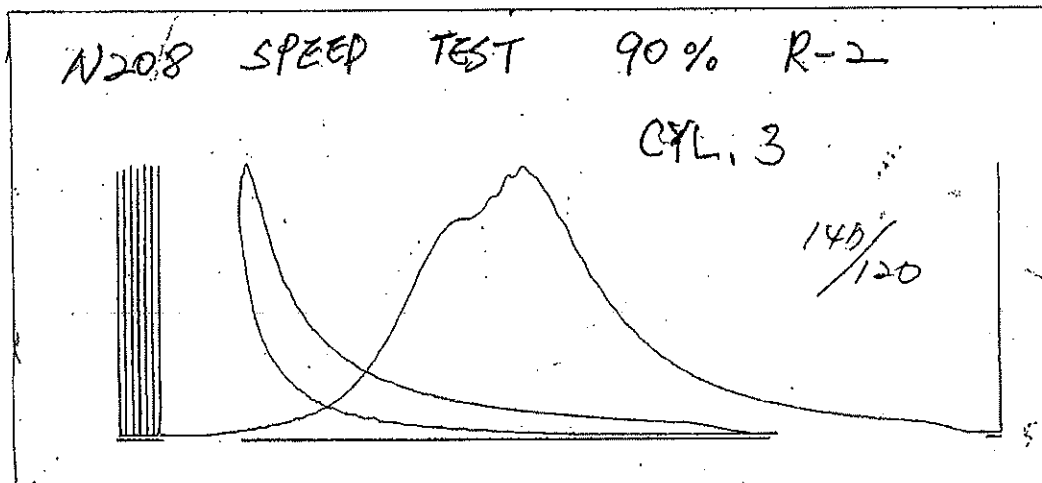
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NO. 2 CYLINDER



NO. 3 CYLINDER



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HHIC

SPEED TEST

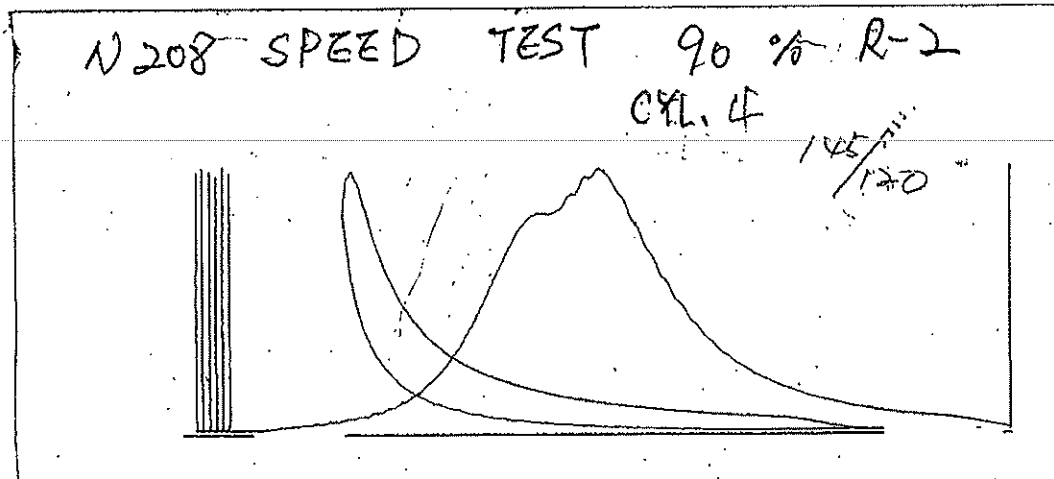
90% R 2

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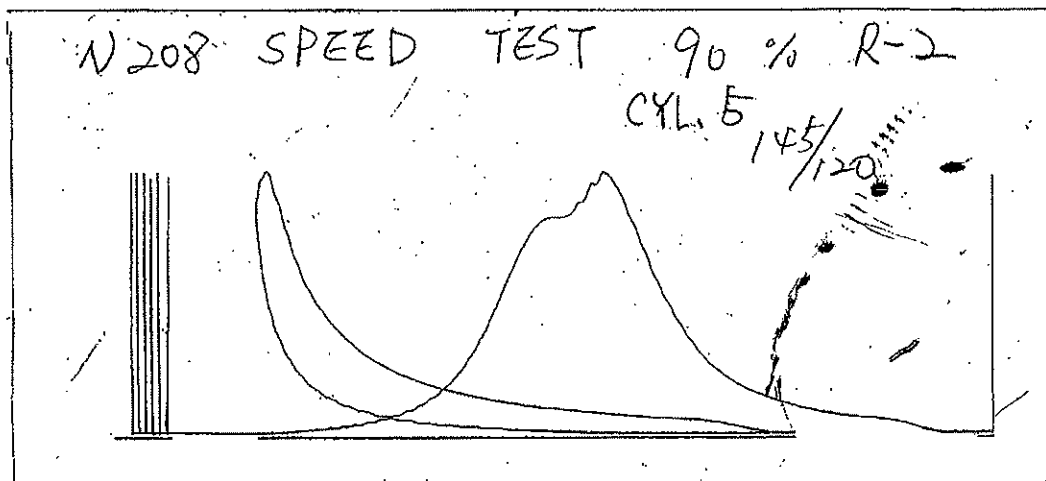
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DATE : 10.09.06

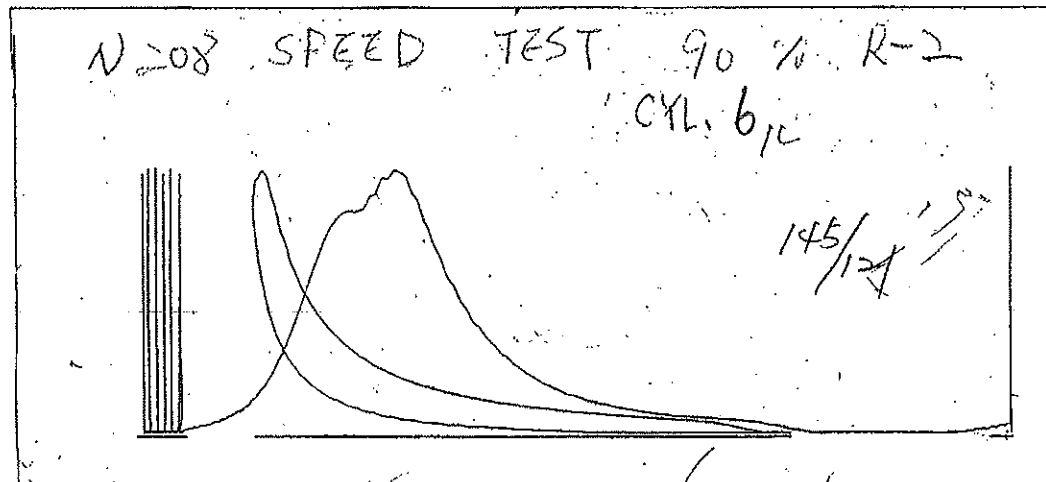
NO. 4 CYLINDER



NO. 5 CYLINDER



NO. 6 CYLINDER



HANJIN HEAVY INDUSTRIES CO., LTD

HHIC

SPEED TEST

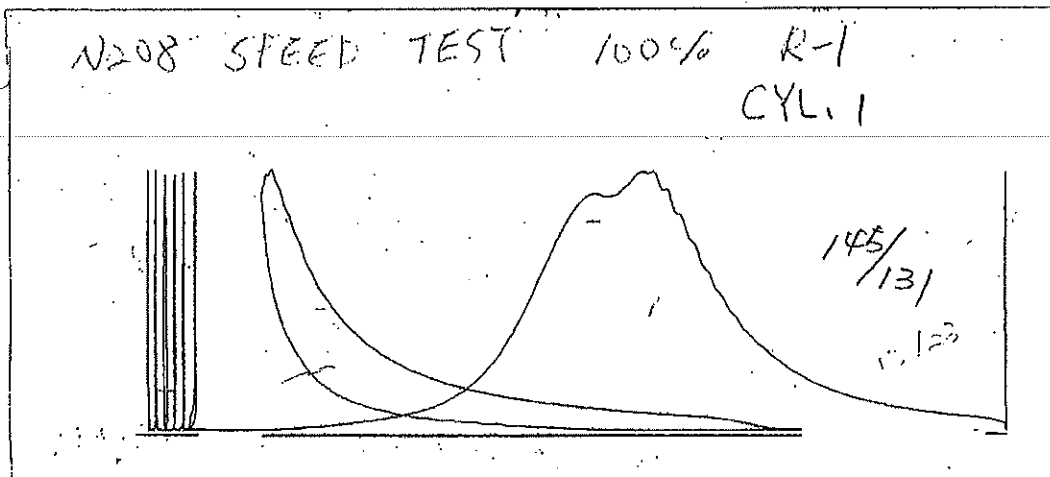
100% R1

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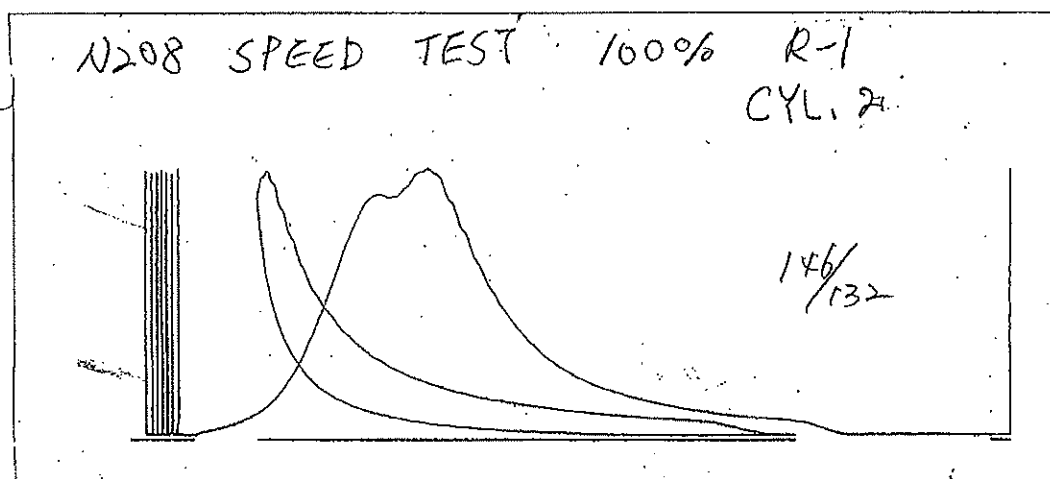
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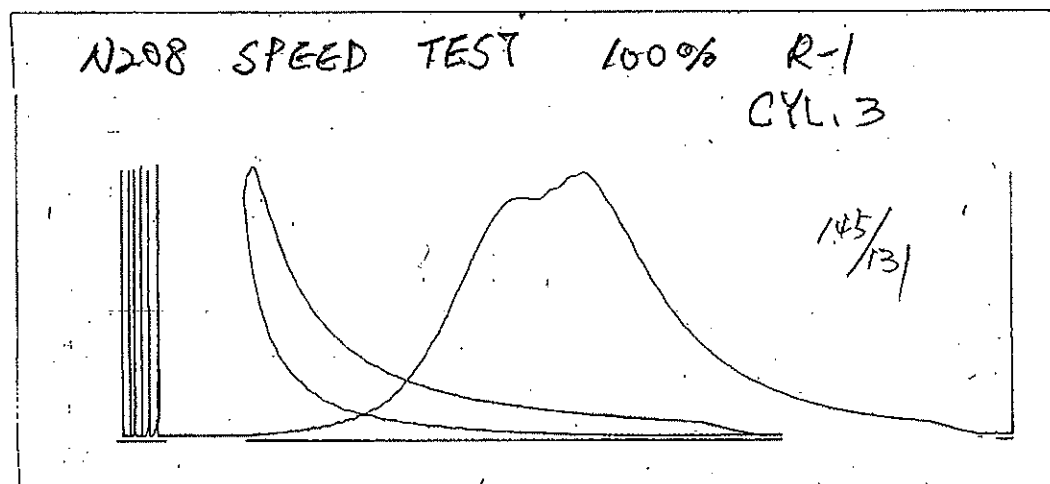
NO. 1 CYLINDER



NO. 2 CYLINDER



NO. 3 CYLINDER



HANJIN HEAVY INDUSTRIES CO., LTD

HHIC

SPEED TEST

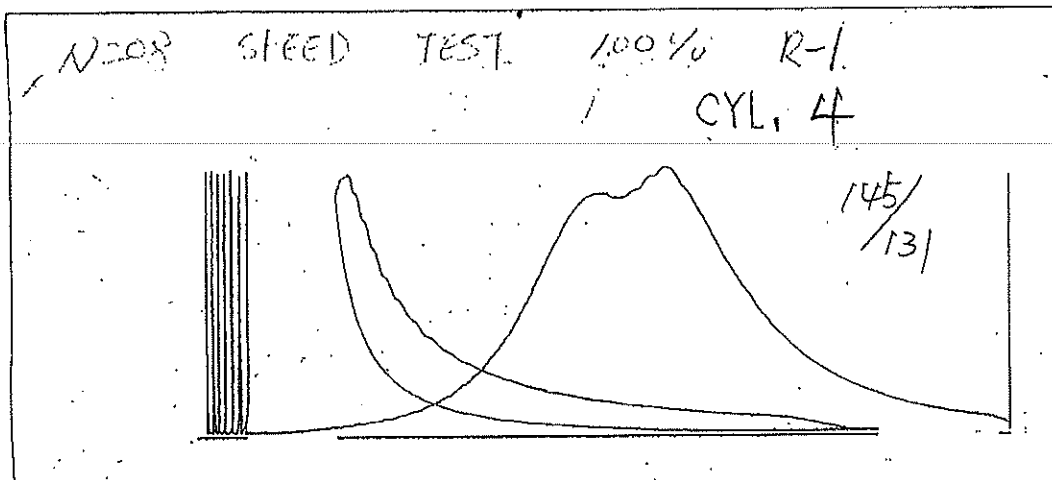
100% R1

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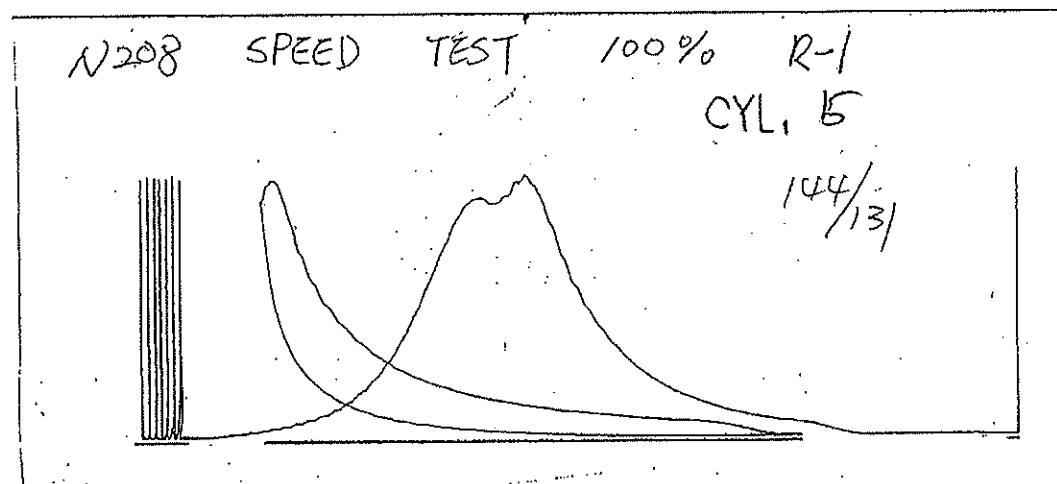
SHIP NO : N-208

DATE : 10.09.06

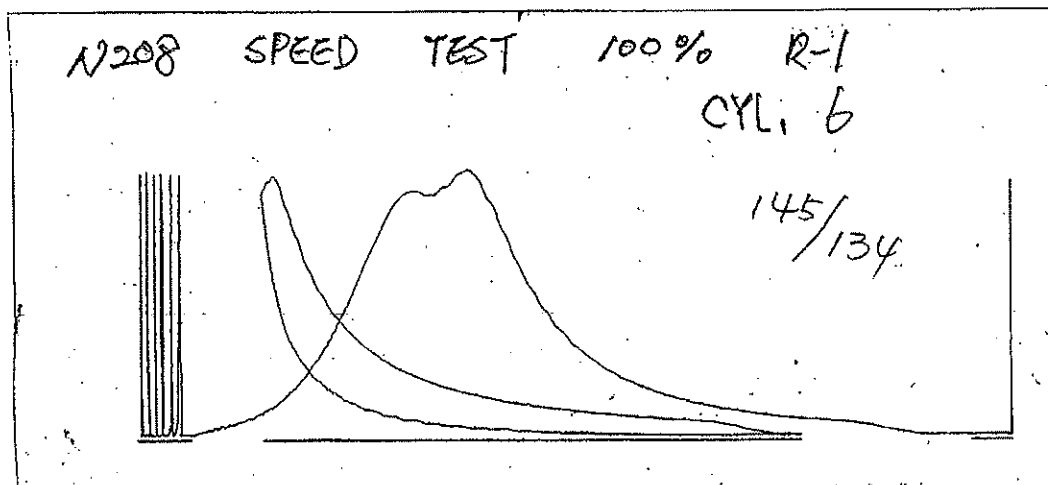
NO. 4 CYLINDER



NO. 5 CYLINDER



NO. 6 CYLINDER



HANJIN HEAVY INDUSTRIES CO., LTD

HHIC

SPEED TEST

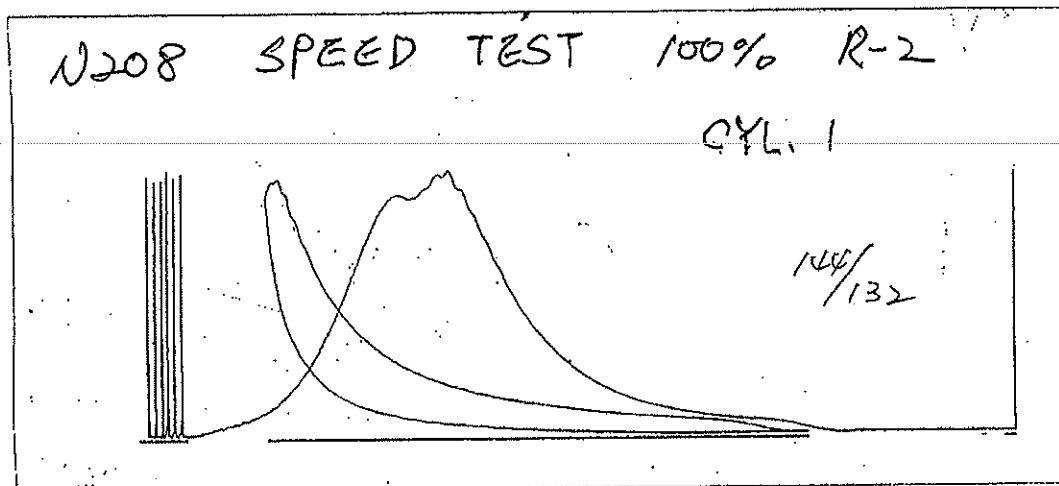
100% R 2

PAGE : 18/23

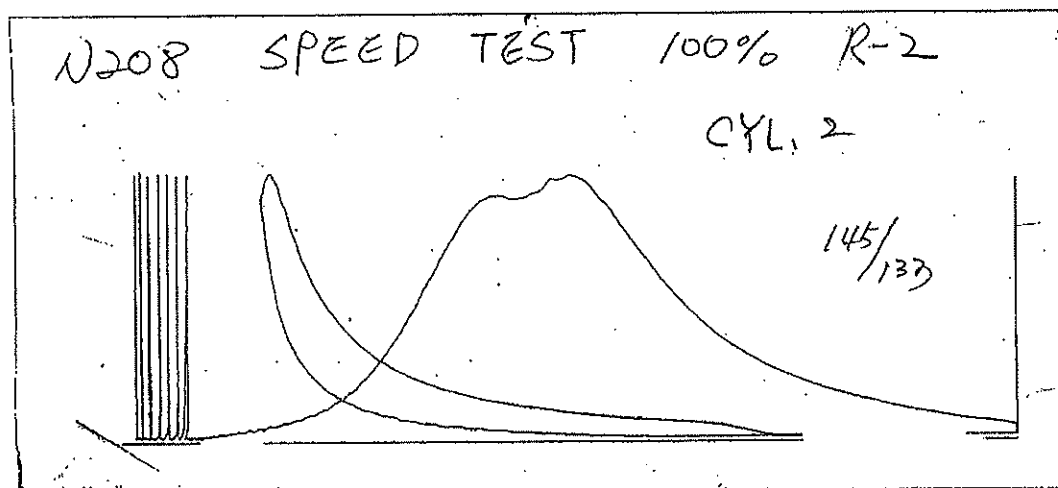
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DATE : 10.09.06

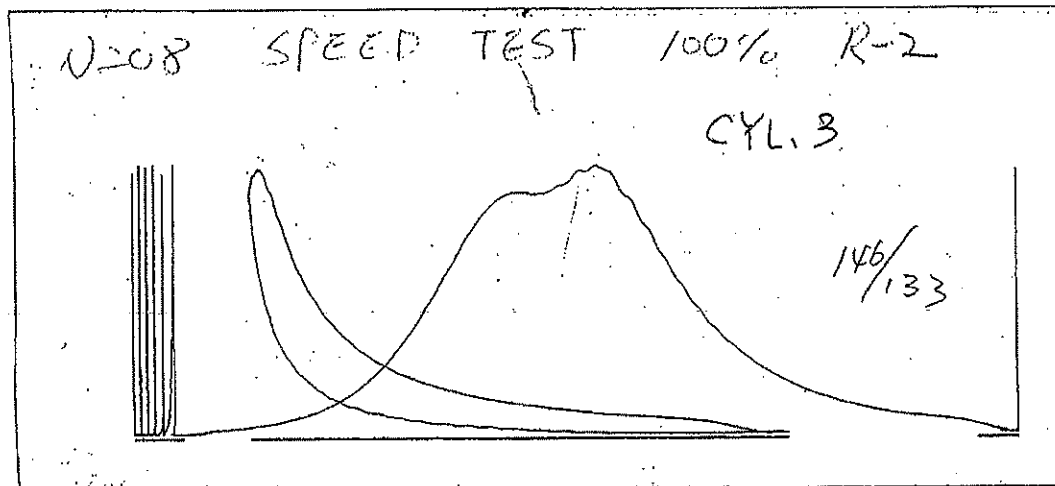
NO. 1 CYLINDER



NO. 2 CYLINDER



NO. 3 CYLINDER



HANJIN HEAVY INDUSTRIES CO., LTD

HHIC

SPEED TEST

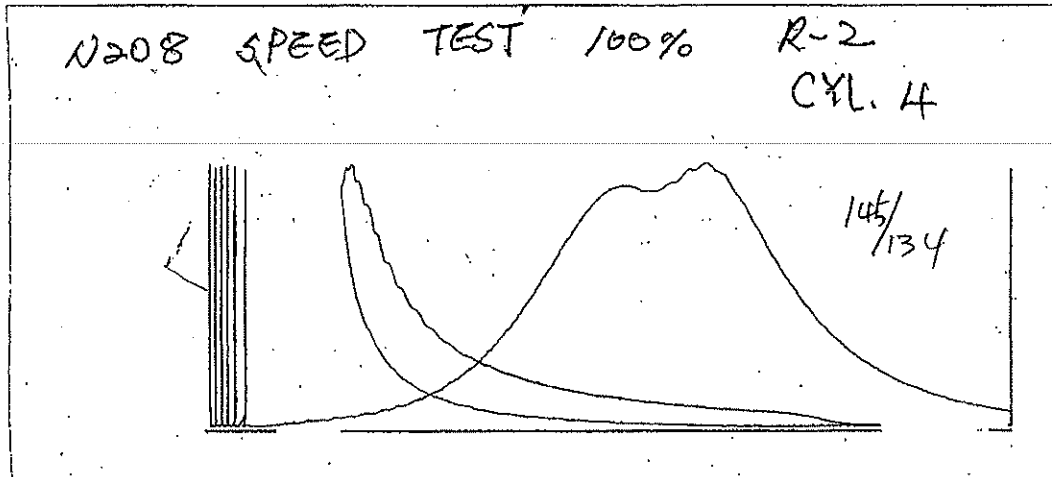
100% R 2

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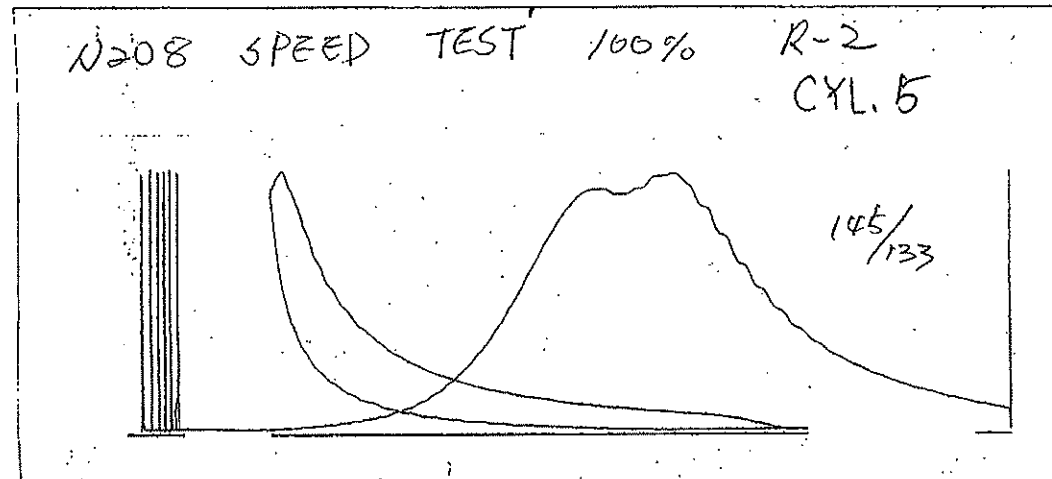
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DATE : 10.09.06

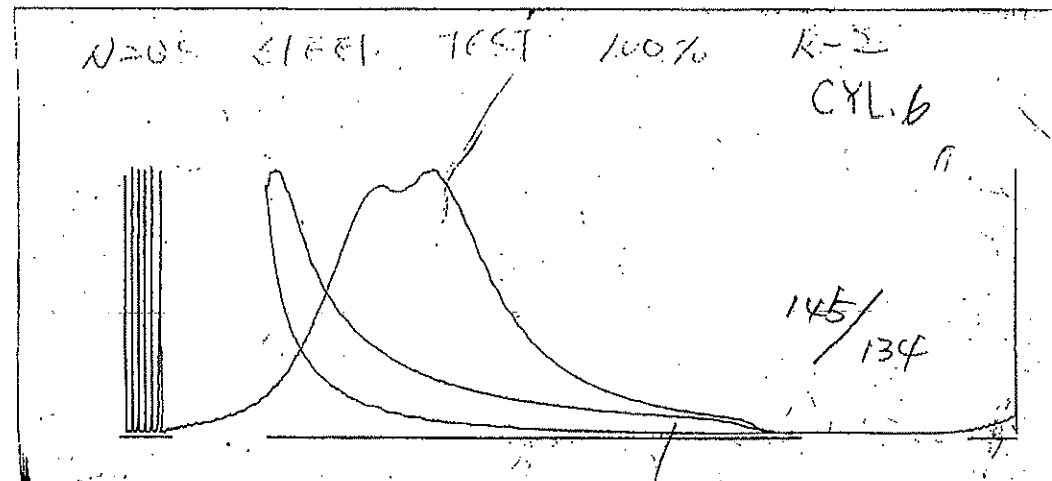
NO. 4 CYLINDER



NO. 5 CYLINDER



NO. 6 CYLINDER



HANJIN HEAVY INDUSTRIES CO., LTD

HHIC

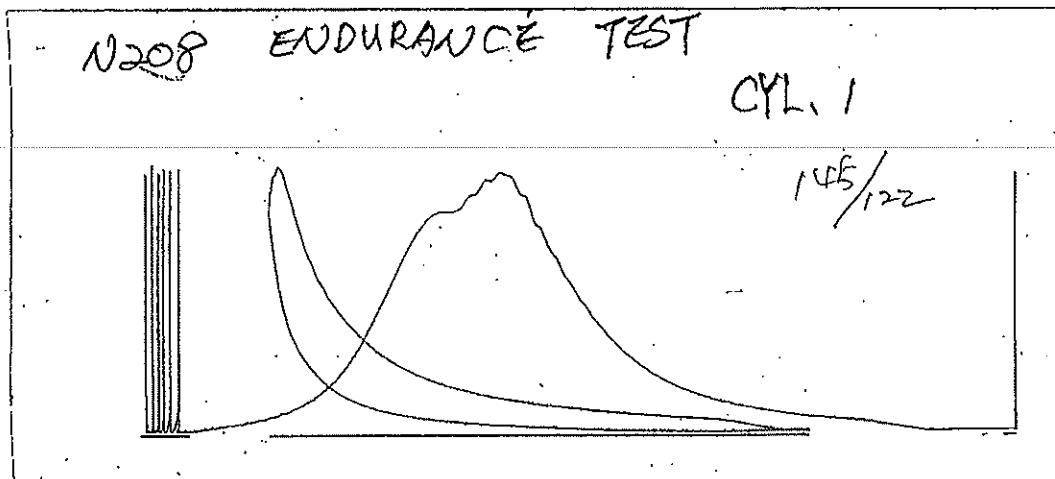
ENDURANCE TEST

PAGE : 20/23

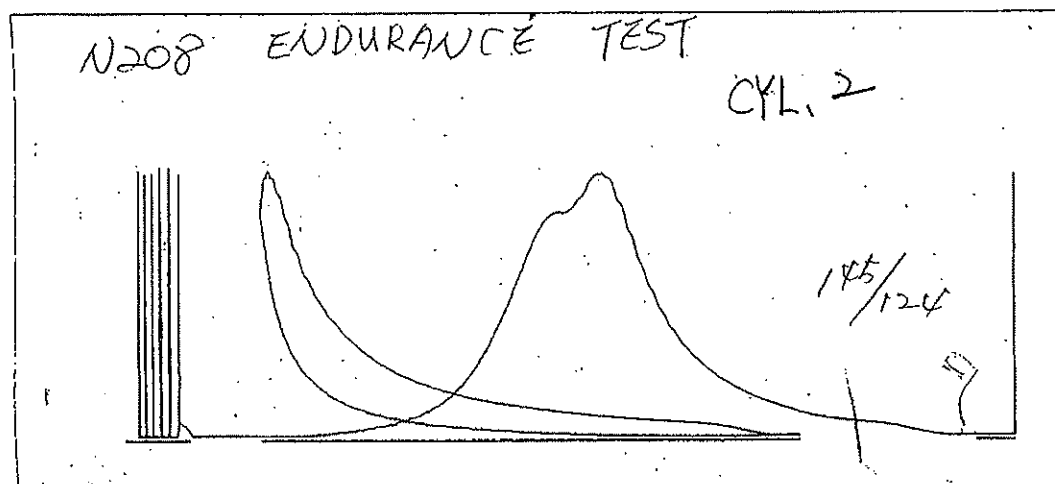
SHIP NO : N - 208

DATE : 10.09.06

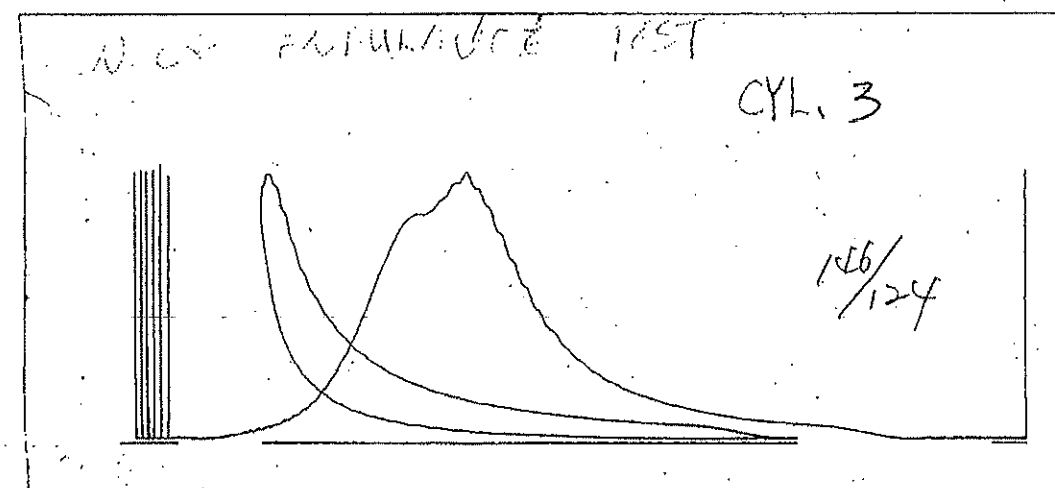
NO. 1 CYLINDER



NO. 2 CYLINDER



NO. 3 CYLINDER



HANJIN HEAVY INDUSTRIES CO., LTD

HHIC

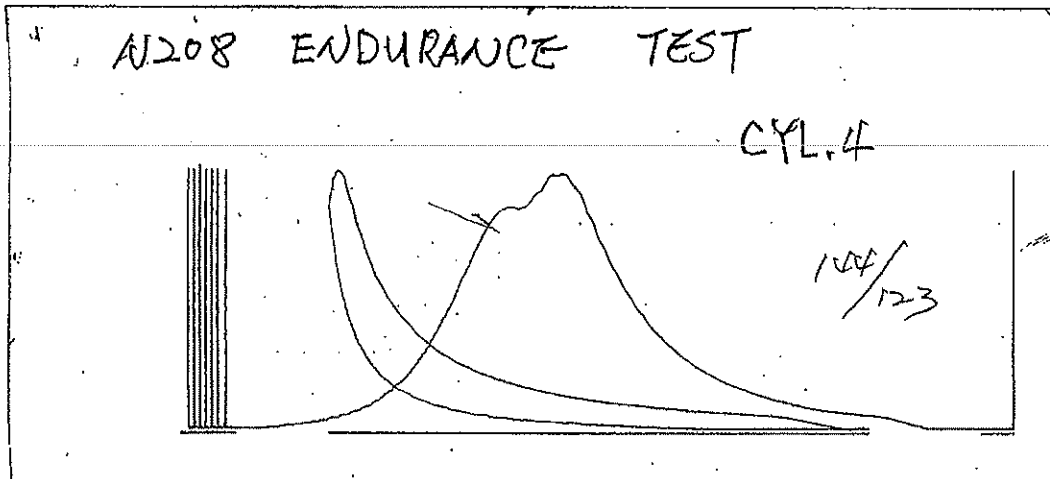
ENDURANCE TEST

PAGE : 21/23

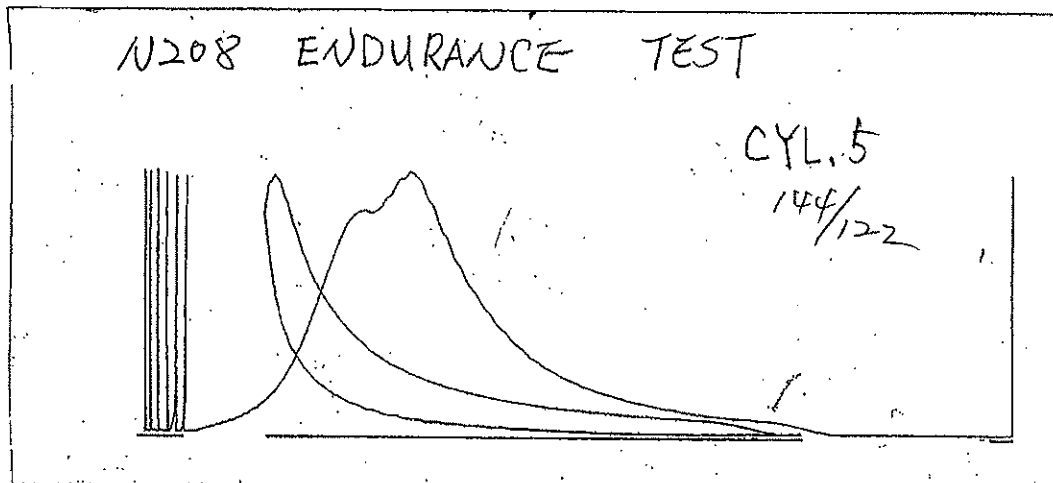
SHIP NO : N - 208

DATE : 10.09.06

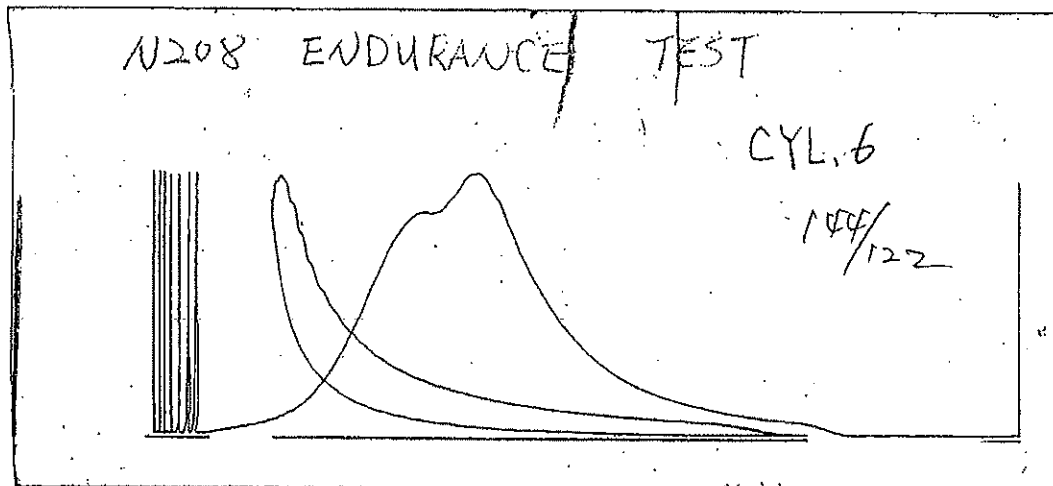
NO. 4 CYLINDER



NO. 5 CYLINDER



NO. 6 CYLINDER



HANJIN HEAVY INDUSTRIES CO., LTD

	Test No.		22 / 23
	Sea Trial Results for 2. M/E Fuel Oil consumption Test	Ship No.	N-208
		Date	2010.09.03

1. PARTICULARS

1) ENGINE MODEL : B&W 6S60MC-C7
 2) OUTPUT x RPM :
 3) DATE / TIME OF TEST : 2010. 09. 03.
 (During Endurance Test - 106RPM)

2. MEASURING METHOD

1) F.O. SPEC. USED

-	KIND		BUNKER - C
-	API GRAVITY 60F		11.6
SG	SPECIFIC GRAVITY	15/4 C	0.9882
-	VISCOSITY KIN 50 C (cSt)	mmf/s	304
-	REDWOOD #1 37.8 C (Sec)		-
LCV	LOWER CAL. VALUE	Kcal/kg	9519
-	FLASH POINT	°C (MIN. 60 C)	90
-	CARBON	W%	-
-	WATER & SEDIMENT	W% (MAX. 1.0)	0.05%
-	ASH	W%	-
-	SULFUR	W% (MAX. 4.0)	3.46
-	ASPHALT	W%	-

2) CALCULATION TABLE FOR F.O. CONSUMPTION

		UNIT	M/E	G/E IN	G/E OUT	REMARK
MEASURE TIME		HOUR	1H 36'	1H 17'	1H 41'	
LOAD		%				
POWER	PS	KW	12882.06			
START	SEC	T1	0	0	0	
	L	ID1	175,420	849,820	783,140	
FINISH	SEC	T2	3636	3617	3641	
	L	ID2	178,240	854,400	787,580	
DEVIAT.	SEC	Ta = T2 - T1	3636	3617	3,641	
	L	IDa = ID2 - ID1	2820	4580	4440	
FO FLOWRATE (L/H)		A = IDx3600/T	2792.1	4558.5	4390.0	
F.O. TEMP. IN		°C	88	125	125	
VOL. FACTOR		VF	0.9487	0.9326	0.9326	BY TABLE
WEIGHT OF F.O. (Kg/H)		B1 = VF x A x SG	2617.6	4201.1	4045.8	
WEIGHT OF M/E F.O. (Kg/H)		B=B1-(B2-B3)	2462.3			
F.O. CONSUMP. (g / KW.H)		B x 1000 / PS x (LCV / 10200)	178.9			REFERENCE

 HANJIN HEAVY INDUSTRIES & CONSTRUCTION CO., LTD

23/23



CERTIFICATE OF ANALYSIS

Product	MF 380CST	Date Sampled	2010.08.31 08:00
Sampling Point	부산물류센터 #03	Date Analyzed	2010.09.01 14:23
Sample No.	1	Lot No.	00000025583
VEHICLE	77양우호	Quantity	
Customer	한진중공업 (N-208호)		

Test Item	Unit	Result	Specification	Test Method
API Gravity, 60°F		11.6		ASTM D 1298
Specific Gravity, 15/4°C		0.9882	Max 0.991	ASTM D 1298
Flash Point, PMCC	Celsius	90.0	Max 60	ASTM D 93
Kinematic Viscosity, 50°C	mm ² /s	304.000	Max 380	ASTM D 445
Sulfur	wt %	0.1000	Max 4.5	ASTM D 1291
Water & Sediment	%(V)	0.050	Max 1.0	ASTM D 1748
Net Heat of Combustion	Kcal/Kg	9519		ASTM D 2382
Gross Heat of Combustion	Kcal/Kg	10063		ASTM D 1740

I CERTIFY THE ABOVE OF QUALITY TO BE TRUE AND CORRECT

Print Date : 2010.09.01

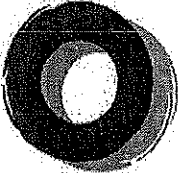
SK Energy. Busan Logistics Center



The Test Results May Be Used As The Specified Purpose

330-1 Cheonghak-dong Yeongdo-gu Busan 606-070 Korea TEL: (051) 413-1643 FAX: (051) 413-1647

			Page	24
	Sea Trial Results for Minimum Revolution Test		Ship No.	N208
			Date	10. 08. 29
Minimum Revolution Test				
Trial date : 2010. 08. 29.		Trial Place : Off Busan		
Ship Condition : Trial Ballast Condition		Sea Condition		
Draft, Fore	13.660 m	Weather	Windy	
Draft, Aft	13.671 m	Sea Condition	Moderate	
Displacement	120847.2 ton	Wind Dir. & Velocity	S-14°, 9 m/s	
		Depth of Sea Water	98.3 m	
Trial Result				
Clock Time		04 : 15		
Base Course		57.8 °		
Min. Lowest running rpm		26 rpm		
Ship's Speed, corresponding		abt. 3.5 knots		

	체크시이트 번호 Check Sheet No. _____	페이지 Page of 25
	검사/시험성적서 RESULT OF INSPECTION/TEST	발송일자 Date Issued

선 번 Project No	N-208	검사항목 Inspection Item	M/E STARTING	검사종류 Type of Inspection	1/1
검사관련기관 Class/Authority	LR	부서 Dept.	QA	장소 Place	ON BOARD
				날씨/온도 Weather/Temp.	℃
				단위 Unit	

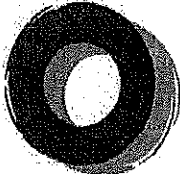
STARTING TEST FOR MAIN ENGINE

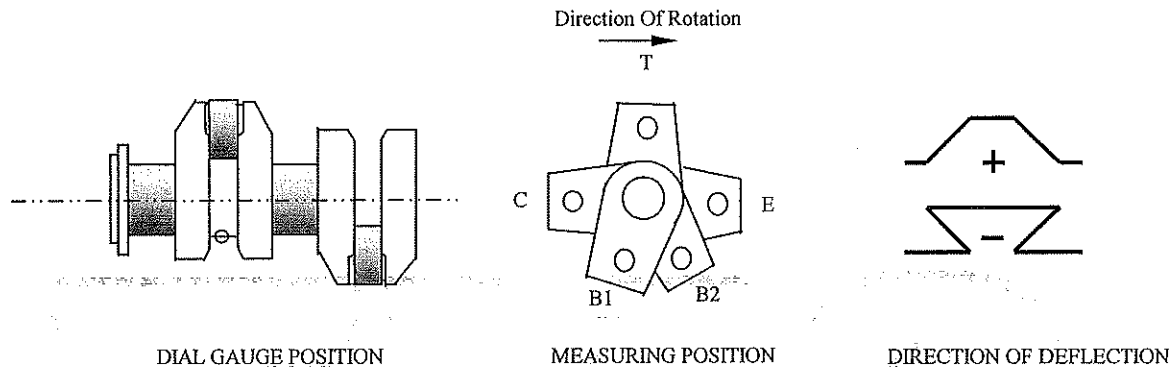
NO.	AHEAD		CONTROL POSITION	NO.	ASTERN		CONTROL POSITION
	BEFORE	AFTER			BEFORE	AFTER	
1	29.99	26.28	WHEEL HOUSE	2	26.28	23.29	WHEEL HOUSE
3	23.29	20.95	WHEEL HOUSE	4	20.95	18.83	WHEEL HOUSE
5	18.83	17.05	WHEEL HOUSE	6	17.05	15.08	WHEEL HOUSE
7	15.08	13.85	WHEEL HOUSE	8	13.85	12.33	WHEEL HOUSE
9	12.33	11.30	WHEEL HOUSE	10	11.30	9.10	ENGINE CONTROL RM
11	9.10	7.50	ENGINE CONTROL RM	12			
13				14			
15				16			
17				18			

REMARK

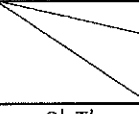
- 1) MANOEUVRING POSITION BY REMOTE CONTROL : BRIDGE, E/C/R, EM'CY STAND
- 2) NUMBER AND CONTENTS OF AIR RESERVOIR
 - * FIRST VESSEL : WITH TWO(2) AIR RESERVOIRS
 - * SISTER VESSEL : WITH ONE(1) AIR RESERVOIR
- 3) INITIAL PRESSURE ON COMMENCEMENT OF TEST : 30 kg/cm²

	QA Team		Witnessed/Approved by			비고 Remarks :
	검사자 Inspector	부서장 Team Manager	선주 Owner	선급 Class	해당기관 Authority	
	일자 Date	28.AUG.2010				
확인자 By	J. H. KIM					

	체크시이트 번호 Check sheet No. _____		페이지 Page of 26	
	검사/시험성적서 RESULT OF INSPECTION/TEST		발송일자 Date Issued	
			1/1	
선 번 Project No. N-208	검사항목 Inspection Item M/E CRANK SHAFT		검사종류 Type of Inspection DEFLECTION CHECK	
검사관련기관 Class/Authority LR	부서 Dept. QA	장소 Place ON BOARD	날씨/온도 Weather/Temp. 37 °C	단위 Unit 1/100 mm

M/E C/SHAFT DEFLECTION

ENGINE CONDITION	CYL. NO POSITION	1	2	3	4	5	6	REMARK
☐ COLD CONDITION ☒ HOT CONDITION	B1	0	0	0	0	0	0	HYUNDAI-MAN B&W 6S-60MC TEMP : 37°C DRAFT FWD : 13.5 M AFT : 13.5 M
	C	-12	-9	-4	-4	-1	0	
	T	-20	-9	0	-1	6	9	
	E	-11	-11	-6	-4	2	-2	
	B2	0	-2	-5	0	2	-4	
B=(B1+B2)/2		0.0	-1.0	-2.5	0.0	1.0	-2.0	
T-B		-20.0	-8.0	-2.5	-1.0	5.0	11.0	VERTICAL DEFLECTION
E-C		1	-2	-2	0	3	-2	HORIZONTAL DEFLECTION

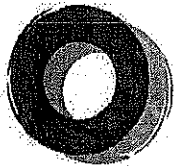
	QA Team		Witnessed / Approved by			비고 Remarks :
	검사자 Inspector	부서장 Team Manager	선 주 Owner	선 급 Class	해당기관 Authority	
	일 자 Date	28.AUG.2010				
확인자 By	J. H. KIM					

 (HHIC-QC-004)
 90.1.1 등록

4) Electric Part

C O N T E N T S

1)	Bridge Maneuvering Test	1
2)	Blackout Test	2
3)	Voltage Drop Check	3



체크시트 번호
CHECK SHEET NO.

페이지
Page of 1

검사/시험성적서 RESULT OF INSPECTION/TEST

발송일자
Date Issued

선 번 N-208
Project No.

검사항목 Bridge Maneuvering Sys.
Inspection Item

검사종류 Function Test
Type of Inspection

검사관련기관 LR
Class/Authority

부서 QA
Dept.

장소 ON BOARD
Place

날씨/온도
Weather/Temp. °C

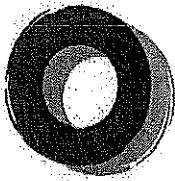
** MANEUVERING TEST TABLE **

		RPM	KNOT	
A H E A D	FULL	87		
	HALF	70		
	SLOW	53		
	DEAD SLOW	35		
A S T E R N	DEAD SLOW	35		
	SLOW	53		
	HALF	70		
	FULL	84		

- *a : WHEN RPM IS STABILIZED
- *b : APPROXIMATELY 2/3 OF FULL SPEED
- *c : APPROXIMATELY 5 MINUTES KEEPING
- *d : DEAD IN THE WATER(0, KNOT)
- *e : APPROXIMATELY HALF SPEED AHEAD
- *f : WITHIN 1 SECOND

	QA Dept.		Witnessed / Approved by			비 고 Remarks :
	검사자 Checker	부서장 Senior manager	선 주 Owner	선 급 Class	해당기관 Authority	
일 자 Date	2010.8.29					
확인자 BY	G.I.LEE					

(HHIC-QC-004
90.1.1.1 등 록)

체크시이트 번호
CHECK SHEET NO. _____페이지
Page of 2검사/시험성적서
RESULT OF INSPECTION/TEST발송일자
Date Issued _____선 번 N-208
Project No. _____검사항목 BLACK OUT
Inspection Item _____검사종류 FUNCTION TEST
Type of Inspection _____검사관련기관
Class/Authority LR부서
Dept. QA장소
Place ON BOARD날씨/온도
Weather/Temp. °CBLACK OUT TEST

1. PREPARATION

1-1. SHIP'S SPEED	6.5	KNOTS
1-2. M/E RPM	106	RPM
1-3. GEN. LOAD	560	KW
1-4. SINGLE GEN. POWER SUPPLYING		
1-5. EM'CY GEN. IS IN AUTO. POSITION		

2. PERFORMANCE : 1ST STEP

RESULT : O K

- 2-1. MAKE BLACK OUT BY ACB TRIP
- 2-2. 1ST ST'BY GEN/ENG WILL START & ACB CLOSED AUTO
- 2-3. SEQUENTIAL START OF VITAL PUMPS

3. PERFORMANCE : 2ND STEP

RESULT : O K

- 3-1. SELECT G/E CONTROL MODE "MANUAL"
- 3-2. MAKE BLACK OUT AGAIN
- 3-3. E/G WILL START & ACB CLOSED AUTO
- 3-4. CHECK THE SHIP'S EM'CY SOURCE

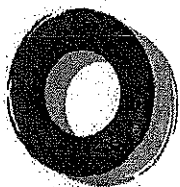
41 SEC

4. PERFORMANCE : 3RD STEP

RESULT : O K

- 4-1. START & CLOSE THE ACB OF ST'BY GEN.
- 4-2. OPEN THE ACB OF E/G & CLOSE THE BUS TIE.
- 4-3. E/G STOP

	Q.A Team		Witnessed / Approved by			비 고 Remarks :
	검사자 Checker	파트장 Part manager	선 주 Owner	선 급 Class	해당기관 Authority	
일 자 Date						
확인자 BY						



체크시이트 번호
CHECK SHEET NO.

페이지
Page of 3

검사/시험성적서
RESULT OF INSPECTION/TEST

발송일자
Date Issued

선 번 N-208
Project No.

검사항목 VOLTAGE DROP
Inspection Item

검사종류 MEASURING TEST
Type of Inspection

검사관련기관
Class/Authority LR

부서
Dept. QA

장소
Place ON BOARD

날씨/온도
Weather/Temp. °C

VOLTAGE DROP TEST

UNDER THE NORMAL SERVICE CONDITIONS THE VOLTAGE DROP BETWEEN THE BUS BAR
AND THE FAREST CIRCUIT POINT IN THE SYSTEM CONCERNED NOT EXCEED 5%.

VOLTAGE DROP CHECK SHEET

VOLTAGE	CIRCUIT NO.	NAME OF EQUIP.	SOURCE SIDE	TERMINAL SIDE
440 V	MSB P1207	P-1(S), FWD DK MACH. D/B	454 V	448 V
220 V	MSB P1605	L13003B, L-3 UPP. DECK LIGHT D/B	233 V	232 V
220 V	ESB P2004/04A	L23014, EL-3 BOSUN STORE	235 V	233 V

	QA Dept.		Witnessed / Approved by			비 고 Remarks :
	검사자 Checker	파트장 Part manager	선 주 Owner	선 급 Class	해당기관 Authority	
일 자 Date	2010.8.29		2010.8.10			
확인자 BY	G.I.LEE		E10			

(HHNC-QC-004
90.1.1 등록)

5) Noise & Vibration Part

1. Report of Noise Measurement

2. Report of Local Vibration Measurement

DISTRIBUTION		Q'TY	BIBLIOGRAPHY			PAGE	1/7					
Basic R&D Team			ALT. NO.	DATE	DESCRIPTION	REMARK						
General Design Part.		1	0	2010.09. 06.	BASIC R&D TEAM							
					VIBRATION & NOISE R&D PART							
TOTAL		1										
APP. BY	J.U.Park	DATE: 2010. 09. 06.						SCALE: NONE	SHIP. NO.			
APP. BY	M.S.Kim	TITLE : 114K CRUDE OIL TANKER THE REPORT OF NOISE MEASUREMENT M.V. "EMARAT DUBAI ANGEL"				N-208						
CHK. BY	J.H.Kwon					DWG. NO.						
DWN. BY	S.H.Lee					TRIMV00NS001R						
TEL. NO	7355					ALT NO.	0					
 HANJIN HEAVY INDUSTRIES & CONSTRUCTION CO., LTD.												

CONTENTS

1. Preface	2
2. Noise measurement	2
2. 1 Instruments	2
2.2 Test conditions	2
2.3 Measured spaces and results	2

Tables and Figures

Table 1	Particulars of test ship
Table 2	Conditions during noise measurement
Table 3 ~ 4	Noise levels at measured spaces

1. Preface

The noise measurement on board N-208 (M.V. "EMARAT DUBAI ANGEL") was carried out during endurance test (106.0 rpm) at sea trial (2010. 09. 03.)

The particulars of test ship are tabulated in Table 1.

2. Noise measurement

2.1 Instruments

The noise measurement was performed by portable noise meter PA101 (SV)

whose major components are listed below;

* Modular precision sound analyzer	:	REALWAVE POCKET ANALYZER PA101(SV)
* Measurement microphone	:	BSWA TECH SV22
* Calibrator	:	Type 4231(B&K)
* Microphone windscreen	:	Type UA 0237(B&K)

2.2 Test conditions

The actual test condition during measurement and sea trial condition are tabulated in Table 2.

2.3 Measured spaces and results

The measured spaces and noise levels are listed in Table 3 ~ 4.

The measured levels of all spaces are satisfied with the criteria of specification.

Table 1 Particulars of test ship

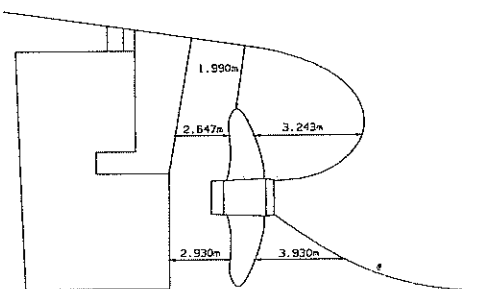
HULL			MAIN ENGINE		
Kind and type		TANKER	Kind and type		MAN B&W 6S60MC-C
Class		LR	Bore (MM)		600
Length, LBP (M)		239.00	Stroke (MM)		2,400
Breadth, B (M)		44.00	No. of cylinder		6
Depth, D (M)		21.35	Power (kW)	MCR	13,560 × 105.0
Draft, d (M)		13.60		NCR	12,204 × 101.4
Displacement (Ton)		121,427.50	Firing order		1-5-3-4-2-6
Block Coeff., CB		0.82	Forces and Moments Due to Unbalance	Order	Force (N) Moment (KN-M)
Dead weight (Ton)		102,064.20		2 nd _v	1,360
Light weight (Ton)		19,363.30	Guide force and Moment	2 nd _x	204
2th moment of inertia of midship (M ⁴)	I _V	432.45		3 rd _x	542
	I _H	1,371.51		4 th _x	401
Effective shear area of midship (M ²)	A _V	1.84		6 th _h	873
	A _H	2.59		-	-
Section area (M ²)		6.08	Sketch of screw aperture		
PROPELLER					
Type		FIXED PITCH			
No. of blades		4			
Pitch ratio		0.67			
Exp. area ratio		0.52			
Skew (Degree)		20.00			
Diameter, (M)		7.20			
Rudder type		-			
Remarks					

Table 2 Conditions during noise measurement

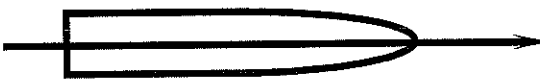
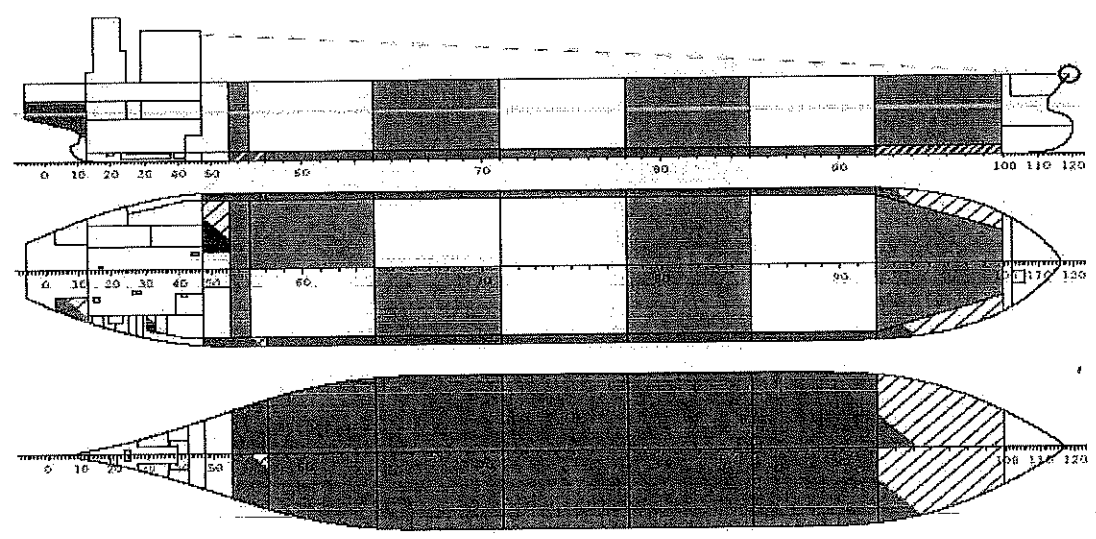
Test condition			Date	2010. 09. 02.
			Place	Off PUSAN
Sea condition (Beaufort #)		3	Heading angle in degree (P01°)	
Height of swell (M)		1.00		
Weather		Blue sky		
Depth of water (M)		105.00		
Wind	Velocity (M/SEC)	3.93		
	Direction	P02°		
Draft	Fore (M)	13.66		
	Aft (M)	13.67		
	Mean (M)	13.64		
Displacement (Ton)		120,847.20		
Propeller immersion from shaft center to water surface (%)		184.00		
Remarks				
				

Table 3 Noise levels at measured spaces

Space		Noise level dB(A)		Remarks
Deck	Name	Contract Spec.	Measurement	
Nav.	Wheel house	65	58	
Nav.	Radio space	65	60	
Nav.	Pilot	60	52	
Nav.	Bridge wing (St'bd)	70	70	
Nav.	Bridge wing (Port)	70	68	
C	2/Eng. bed room	60	56	
C	2/Eng. day room	65	53	
C	Owner	60	57	
C	C/Eng. bed room	60	52	
C	C/Eng. day room	65	52	
C	C/Eng. Office	65	52	
C	3/Eng.	60	51	
C	2/Off.	60	53	
C	Captain office	65	54	
C	Captain day room	65	53	
C	Captain bed room	60	52	
C	C/Off. day room	65	57	
C	C/Off. bed room	60	57	
C	3/Off.	60	54	
C	Conference room	65	60	
C	4/Eng.	60	55	
C	Electrician	60	55	
B	No.1 oiler	60	54	
B	Crew 1	60	55	
B	Crew 2	60	53	
B	Bosun	60	53	
B	Chief cook	60	53	
B	Crew 3	60	52	
B	Crew 4	60	54	
B	Crew 5	60	53	

Table 4 Noise levels at measured spaces

Space		Noise level dB(A)		Remarks
Deck	Name	Contract Spec.	Measurement	
B	Cadet 3/4	60	55	
B	Officer's laundry	75	63	
B	Crew 11	60	54	
B	Crew 8/9	60	60	
B	Crew 6/7	60	56	
B	Crew 10	60	56	
B	Gymnasium	65	54	
B	Cadet 1/2	60	52	
A	Crew's mess room	65	60	
A	Galley	75	70	
A	Crew's recreation room	65	58	
A	Cargo control room	65	62	
A	Ship's office	65	61	
A	Officer's recreation room	65	57	
A	Duty mess room	65	59	
A	Officer's mess room	65	55	
Upper	Crew's laundry	75	59	
Upper	Hospital	60	57	
Upper	Catering	60	60	
Upper	Crew 13	60	58	
Upper	Crew 12	60	54	
Upper	Tally office	65	58	
Upper	Suez crew room	65	55	
2 nd	Engine control room	75	66	
2 nd	Workshop	85	72	
3 rd	Main engine platform on 3 rd deck(P)	110	93	
	Main engine platform on 3 rd deck(S)	110	94	
Floor	Main engine platform on floor(P)	110	90	
	Main engine platform on floor(S)	110	90	

DISTRIBUTION		Q'TY	BIBLIOGRAPHY			PAGE	1/26					
Basic R&D Team			ALT. NO.	DATE	DESCRIPTION	REMARK						
General Design Part.		1	0	2010.09.06	BASIC R&D TEAM							
					VIBRATION & NOISE PART							
TOTAL		1										
APP. BY	J.U.Park	DATE: 2010. 09. 06.						SCALE: NONE		SHIP. NO.		
APP. BY	M.S.Kim	<u>TITLE : 114K CRUDE OIL TANKER</u> THE REPORT OF VIBRATION MEASUREMENT M.V. "EMARAT DUBAI ANGEL"						N-208				
CHK. BY	J.H.Kwon					DWG. NO.						
DWN. BY	S.H.Lee					TRIMV00VS003R						
TEL. NO	7355					ALT NO.	0					
 HANJIN HEAVY INDUSTRIES & CONSTRUCTION CO., LTD.												

C O N T E N T S

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Table 2 Conditions during vibration measurement

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1. Preface

The vibration measurement on board N-208 (M.V. "EMARAT DUBAI ANGEL") was carried out during endurance test (106.0 rpm) at sea trial (2010. 09. 03.)

The particulars of test ship are tabulated in Table 1.

2. Vibration measurement

2.1 Instruments

The vibration measurement was performed by portable vibration meter.

* Data collector : REALWAVE POCKET ANALYZER PA101(SV)

* Accelerometer : 3148E(DYTRAN)

2.2 Test conditions

The actual test condition during measurement and sea trial condition are tabulated in Table 2.

2.3 Measured points and results

The measured points and vibration levels are listed in Table 3 ~ 4.

The sketch and vibration components of measured points are shown in Fig. 1.1 ~ 1.5 and Fig. 2.1 ~ 2.14.

Table 1 Particulars of test ship

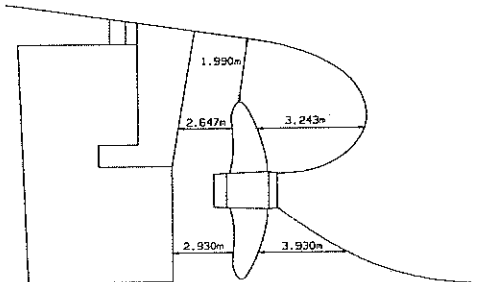
HULL			MAIN ENGINE		
Kind and type		TANKER	Kind and type		MAN B&W 6S60MC-C
Class		LR	Bore (MM)		600
Length, LBP (M)		239.00	Stroke (MM)		2,400
Breadth, B (M)		44.00	No. of cylinder		6
Depth, D (M)		21.35	Power (kW)	MCR	13,560 × 105.0
Draft, D (M)		13.60		NCR	12,204 × 101.4
Displacement (Ton)		121,427.50	Firing order		1-5-3-4-2-6
Block Coeff., CB		0.82	Forces and Moments Due to Unbalance	Order	Force (N) Moment (KN-M)
Dead weight (Ton)		102,064.20			
Light weight (Ton)		19,363.30	Guide force Moments	2 nd _v	1,360
2nd moment of inertia of midship (M ⁴)	I _V	432.45		2 nd _x	204
	I _H	1,371.51		3 rd _x	542
Effective shear area of midship (M ²)	A _V	1.84		4 th _x	401
	A _H	2.59		6 th _h	873
Section area (M ²)		6.08	-	-	
PROPELLER			Sketch of screw aperture		
Type		FIXED PITCH			
No. of blades		4			
Pitch ratio		0.67			
Exp. Area ratio		0.52			
Skew (Degree)		20.00			
Diameter, (M)		7.20			
Rudder type		-			
Remarks					

Table 2 Conditions during vibration measurement

Test condition			Date	2010. 09. 02.		
			Place	Off PUSAN		
Sea condition (Beaufort #)		3	Heading angle in degree (P01 °)			
Height of swell (M)		1.00				
Weather		Blue sky				
Depth of water (M)		105.00				
Wind	Velocity (M/SEC)	3.93				
	Direction	P02°				
Draft	Fore (M)	13.66				
	Aft (M)	13.67				
	Mean (M)	13.64				
Displacement (Ton)		120,847.20				
Propeller immersion from shaft center to water surface (%)		184.00				

Remarks

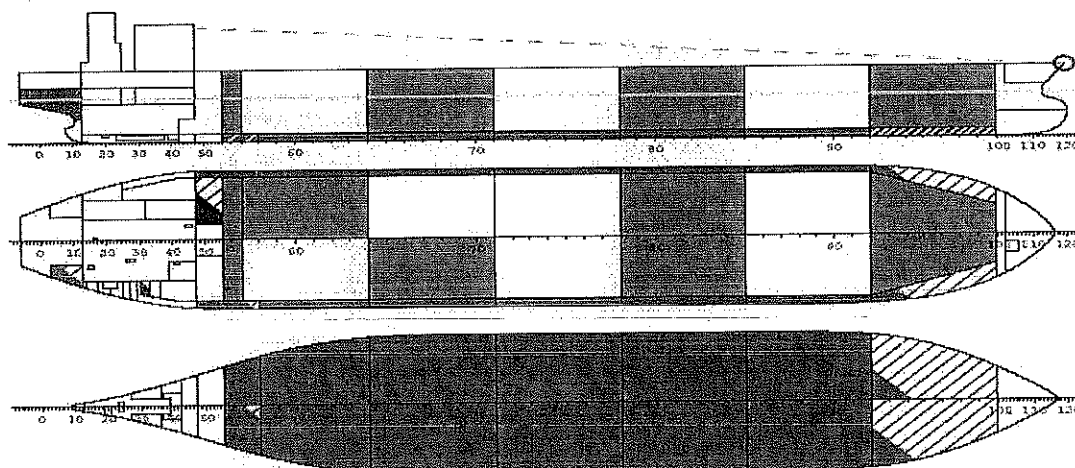


Table 3 The measured vibration levels at each location and direction

No.	Locations	Dir.	Level (mm/sec, RMS)	Freq. (Hz)
1	Platform of the radar mast	L	4.06	14.13
		T	1.30	21.12
2	Wheel house on Nav. bridge deck	V	1.96	14.13
		L	1.87	10.63
		T	1.00	10.63
3	Bridge wing(st'bd) on Nav. bridge deck	V	1.39	10.63
		L	3.00	14.13
		T	1.20	14.13
4	C/Eng. day room on C deck	V	0.41	14.13
5	Captain day room on C deck	V	0.94	10.63
6	No.1 oiler on B deck	V	0.59	10.63
7	Crew (5) on B deck	V	1.12	10.63
8	Crew's recreation room on A deck	V	0.59	10.63
9	Officer's recreation room on A deck	V	0.91	10.63
10	Officer's mess room on A deck	V	0.99	10.63
11	Hospital on upper deck	V	0.59	10.63
12	Tally office on upper deck	V	1.05	10.63
13	Stern on upper deck	V	1.02	14.13
14	Steering gear room on 2 nd deck	V	1.44	28.25
15	Support of the lower platform of the incinerator in funnel	L	6.27	7.13
		T	20.86	10.63
16	F.W. expansion tank on 2 nd deck in E/R	L	0.38	14.13
		T	2.68	10.63
17	Deck plate at workshop on 2 nd deck in E/R	V	0.37	14.13
18	Engine control room on 2 nd deck in E/R	V	0.89	14.13
19	Transformer on 2 nd deck in E/R	T	2.37	11.00
20	Gauge board of the No.2 auxiliary boiler on 2 nd deck in E/R	T	3.10	43.50
21	Gauge board of the No.1 auxiliary boiler on 2 nd deck in E/R	T	4.36	14.13
22	No.2 air conditioner on 2 nd deck in E/R	T	5.64	14.13
23	Support structure of No.1 & No.2 air conditioner on 2 nd deck in E/R	T	10.24	18.75
24	No.1 & No.2 M/E cylinder oil serve tank on 2 nd deck in E/R	T	2.34	21.35

Table 4 The measured vibration levels at each location and direction

No.	Locations	Dir.	Level (mm/sec, RMS)	Freq. (Hz)
25	Fore top of the main engine on 3 rd deck in E/R	V	1.94	10.63
		L	2.23	14.13
		T	8.38	10.63
26	No.2 M/E auxiliary blower motor on 3 rd deck in E/R	V	5.18	10.88
		L	3.08	45.50
		T	2.59	58.13
27	Inlet pipe of turbo charger on 3 rd deck in E/R	L	20.58	26.50
28	Upper platform grating of the M/E on 3 rd deck in E/R	V	12.03	10.63
29	F.O transfer pump on floor in E/R	L	19.65	21.88
		T	12.48	21.88

Note: V : Vertical Direction

L : Longitudinal Direction

T : Transverse Direction

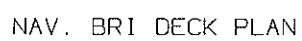
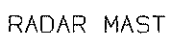


Fig.1.2 The sketch of measured points

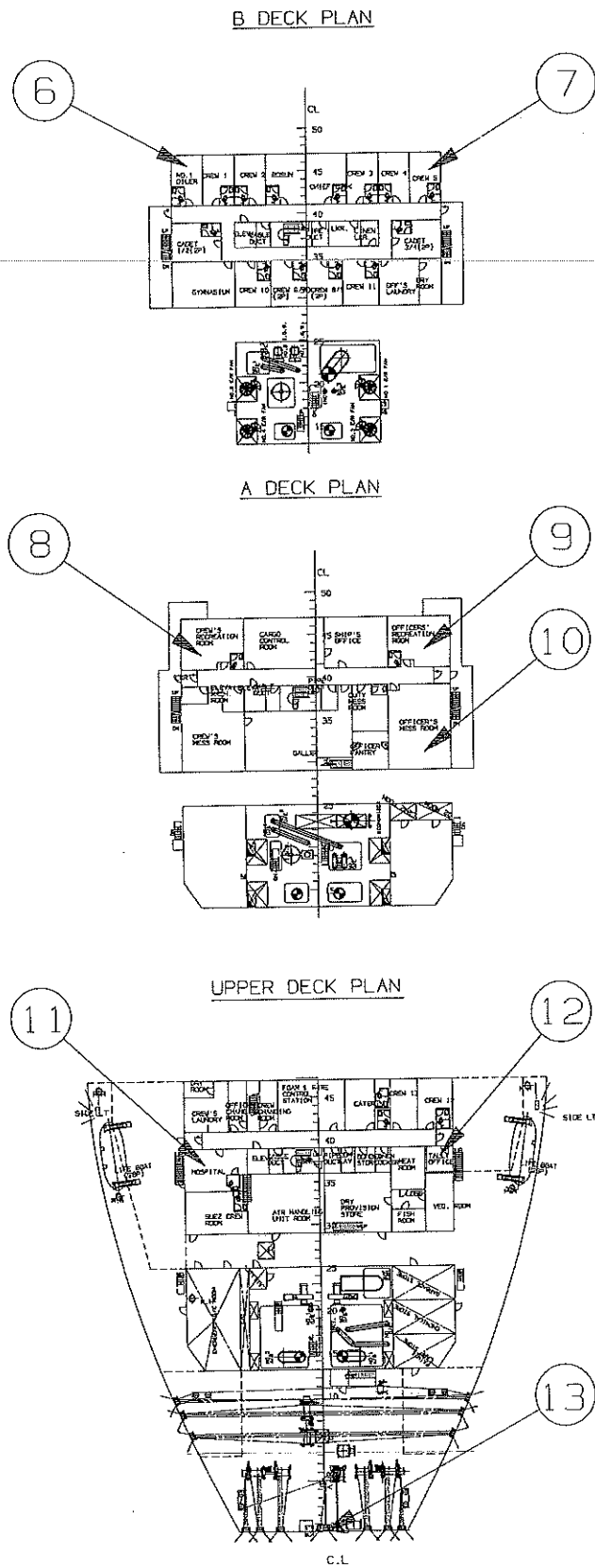
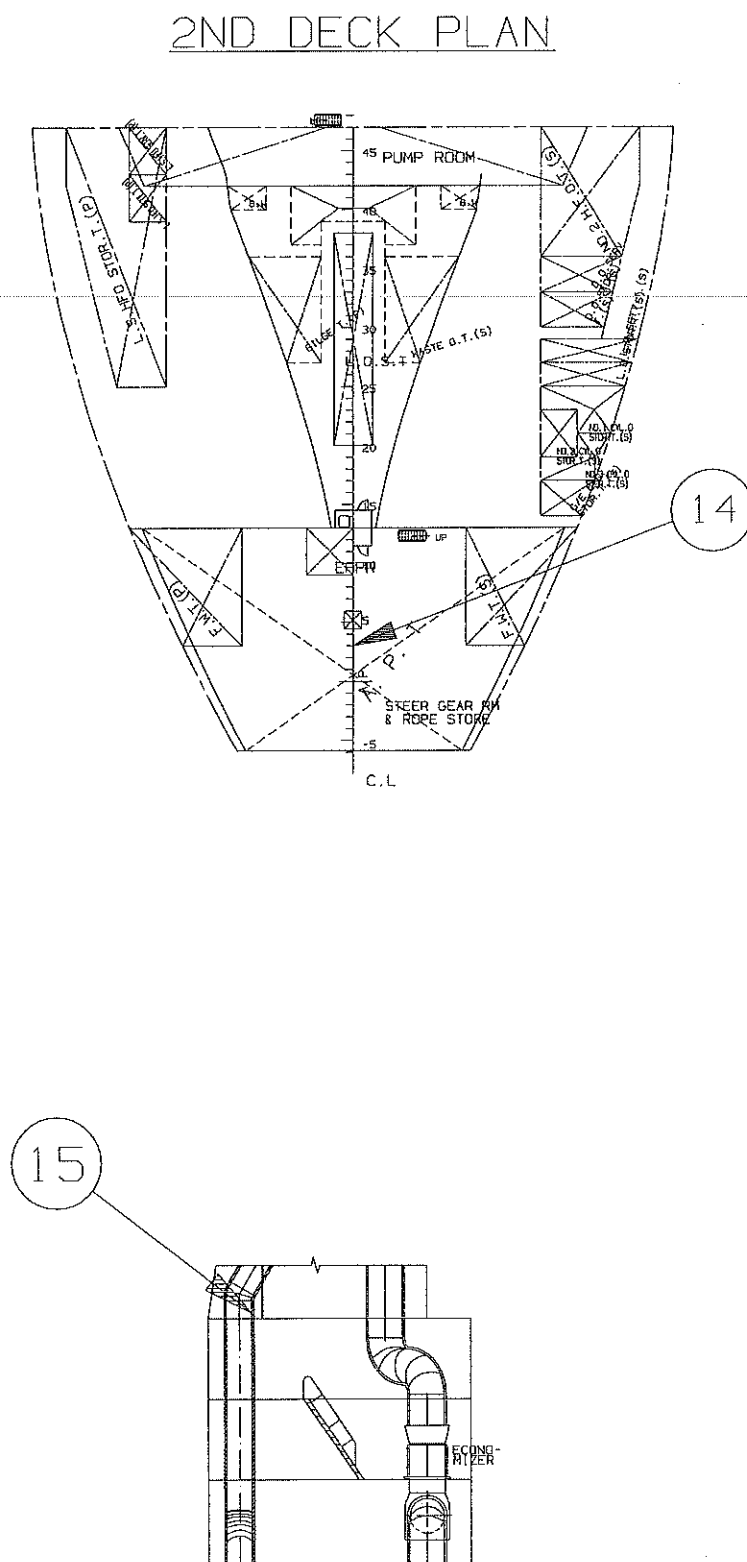


Fig.1.3 The sketch of measured points



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FLOOR PLAN



Fig.2.1 The measured vibration components at each point

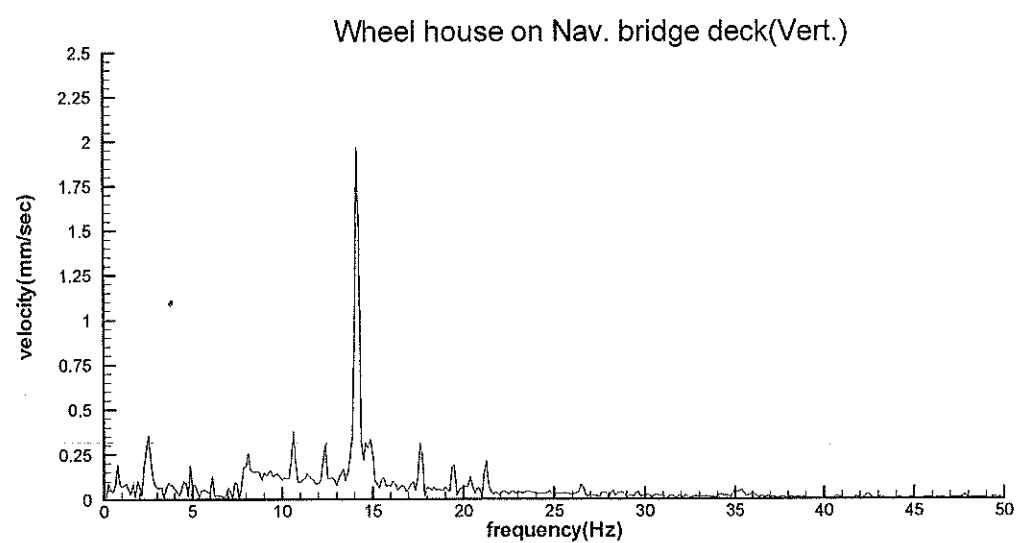
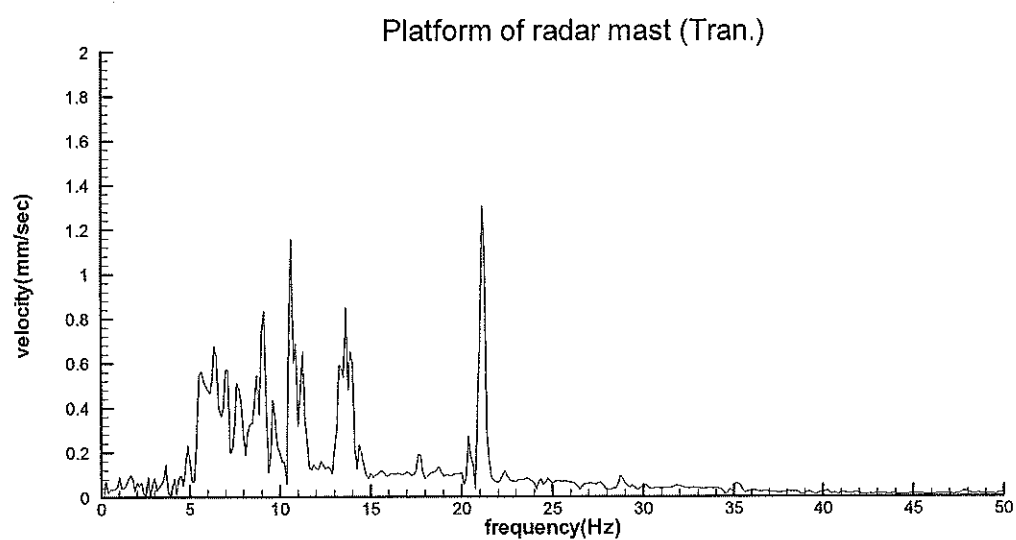
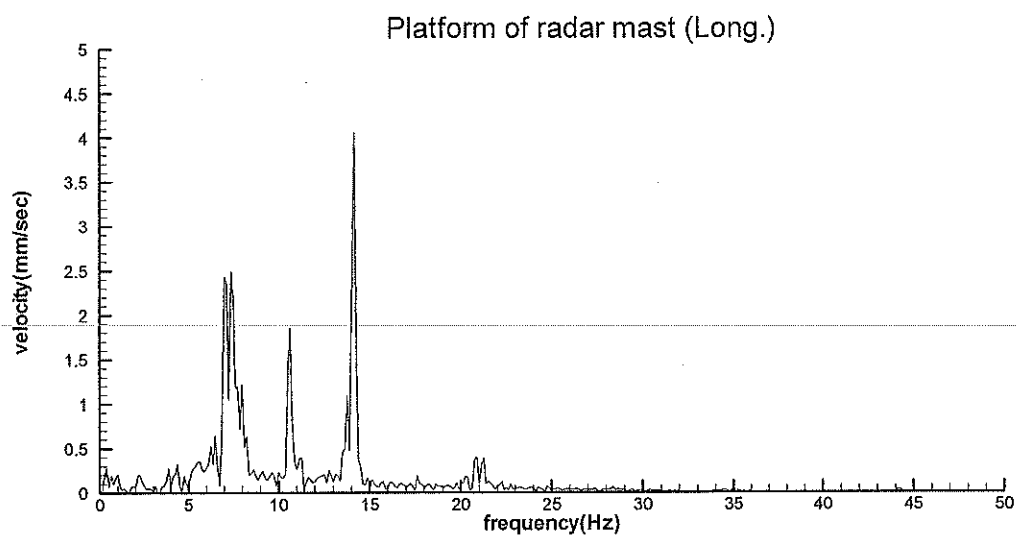


Fig.2.2 The measured vibration components at each point

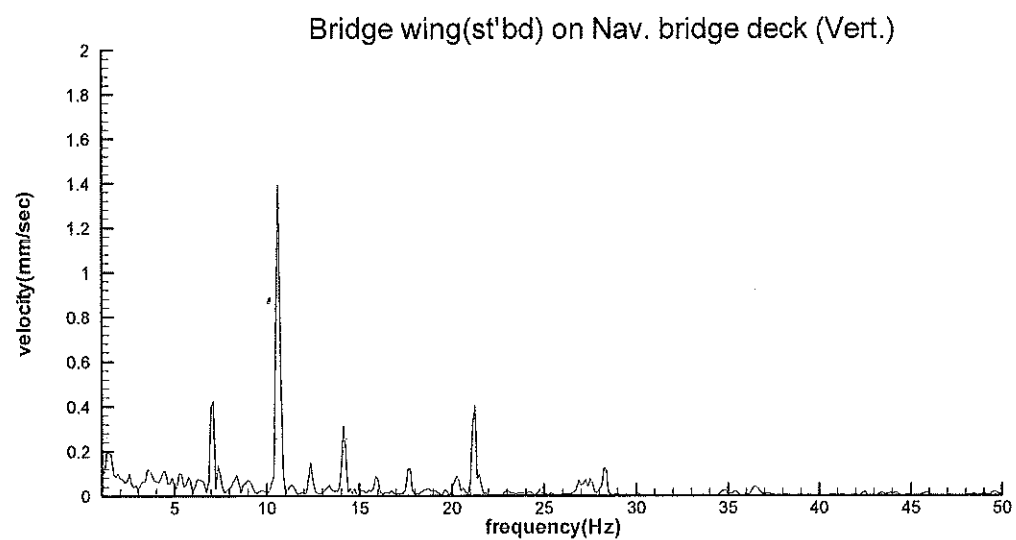
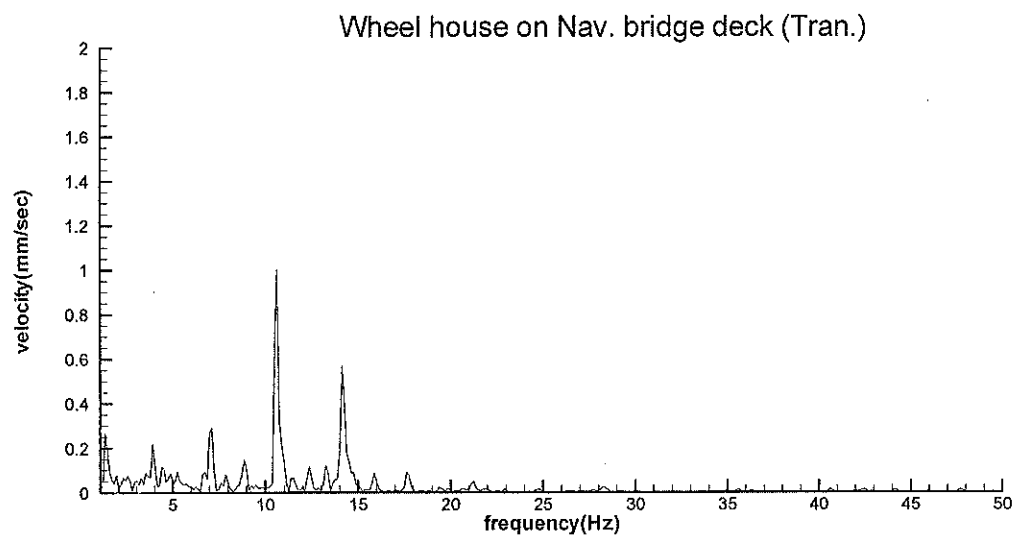
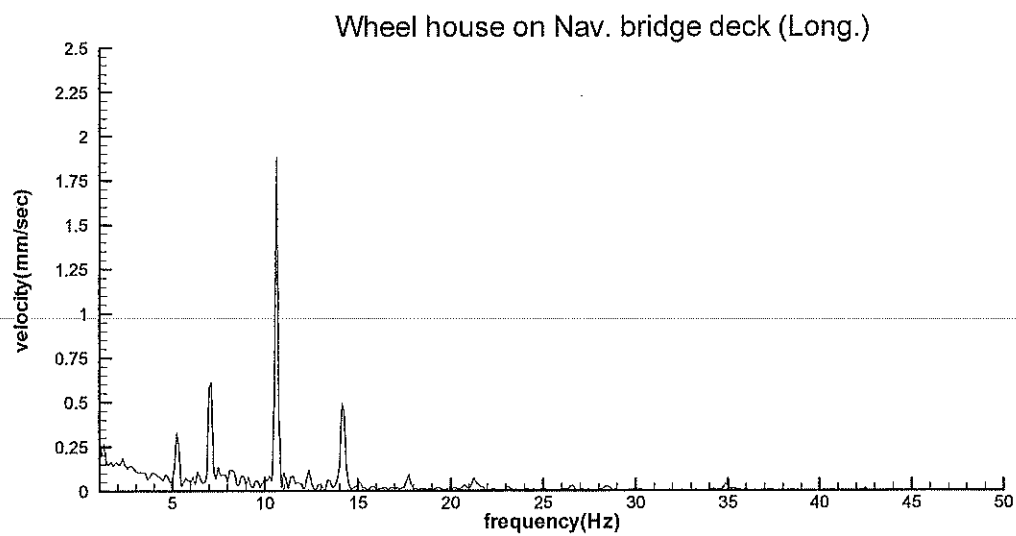


Fig.2.3 The measured vibration components at each point

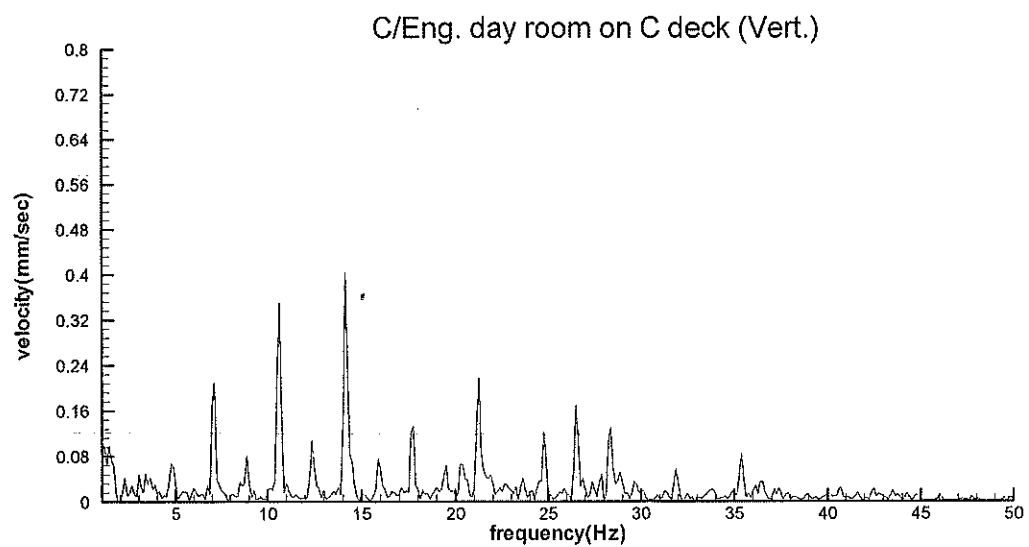
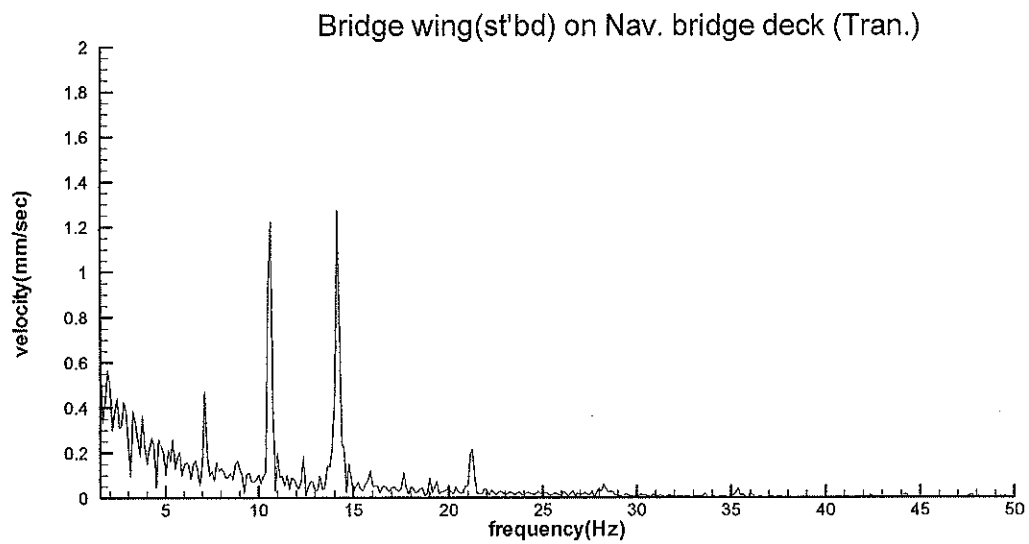
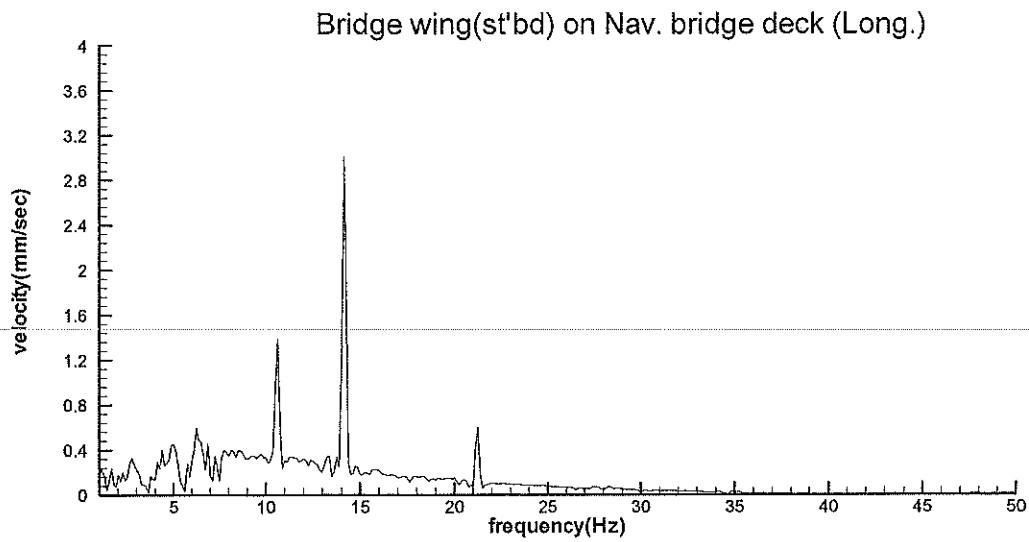


Fig.2.4 The measured vibration components at each point

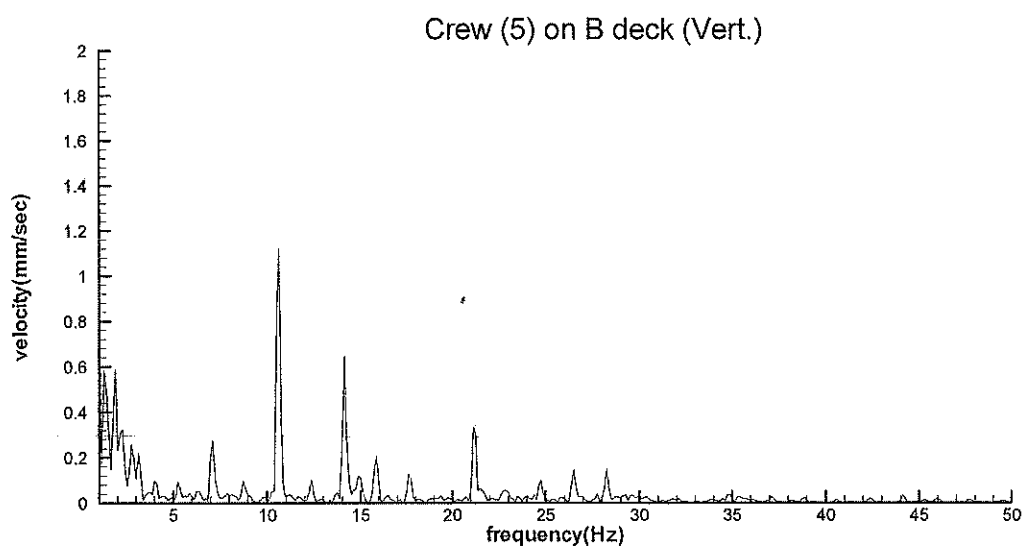
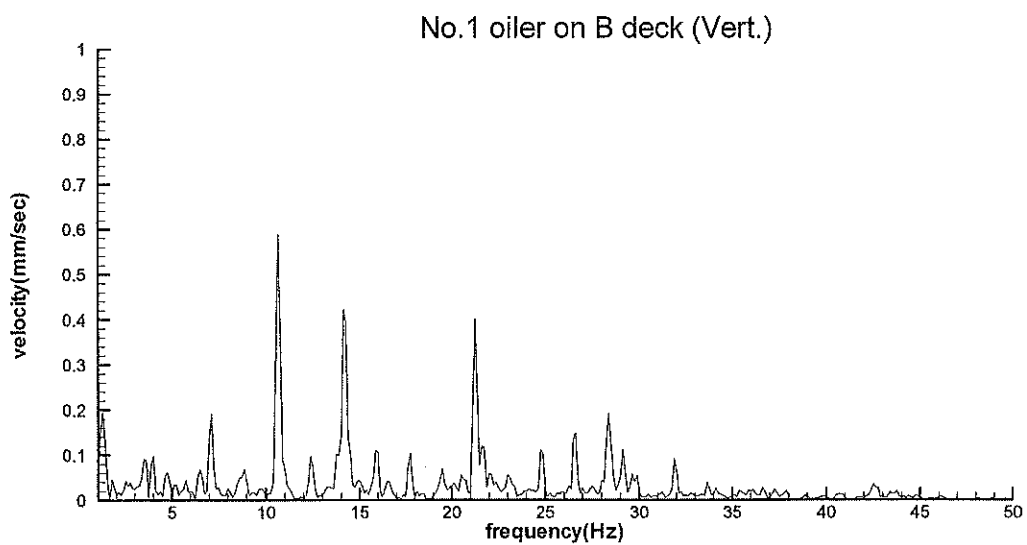
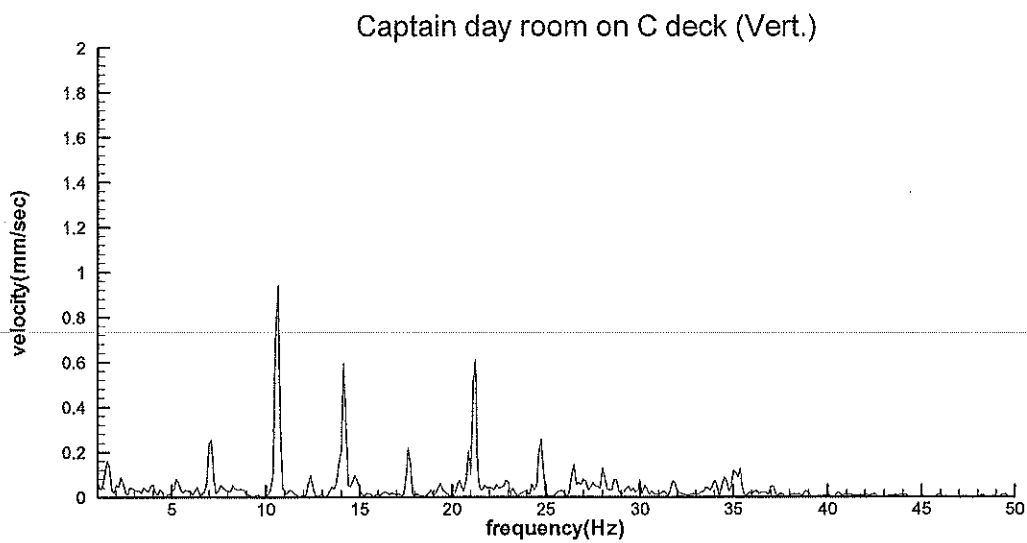


Fig.2.5 The measured vibration components at each point

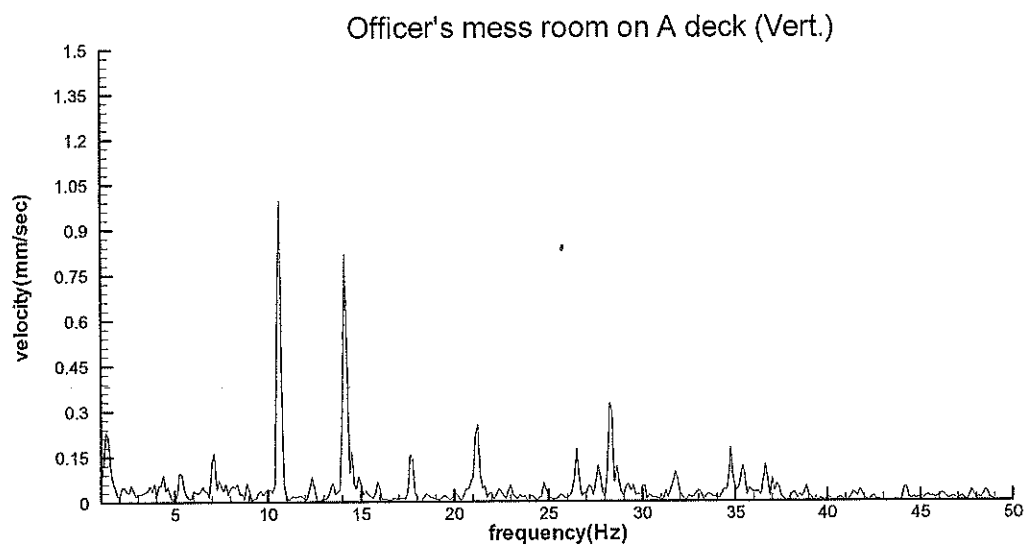
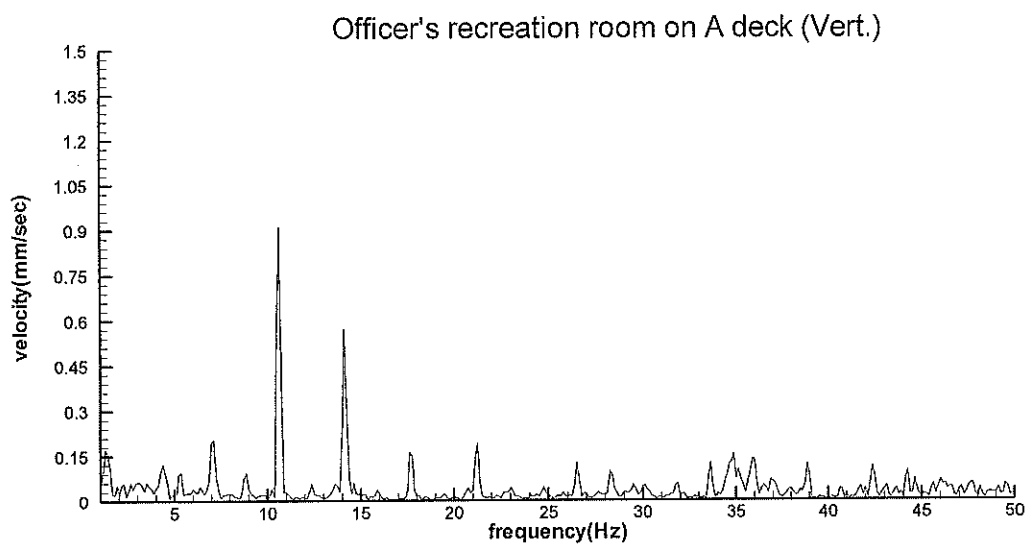
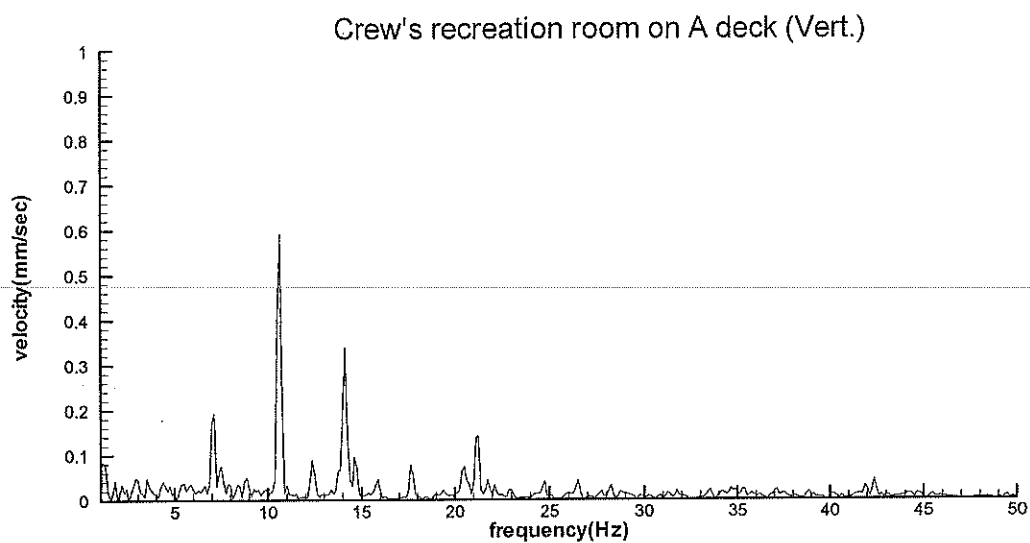


Fig.2.6 The measured vibration components at each point

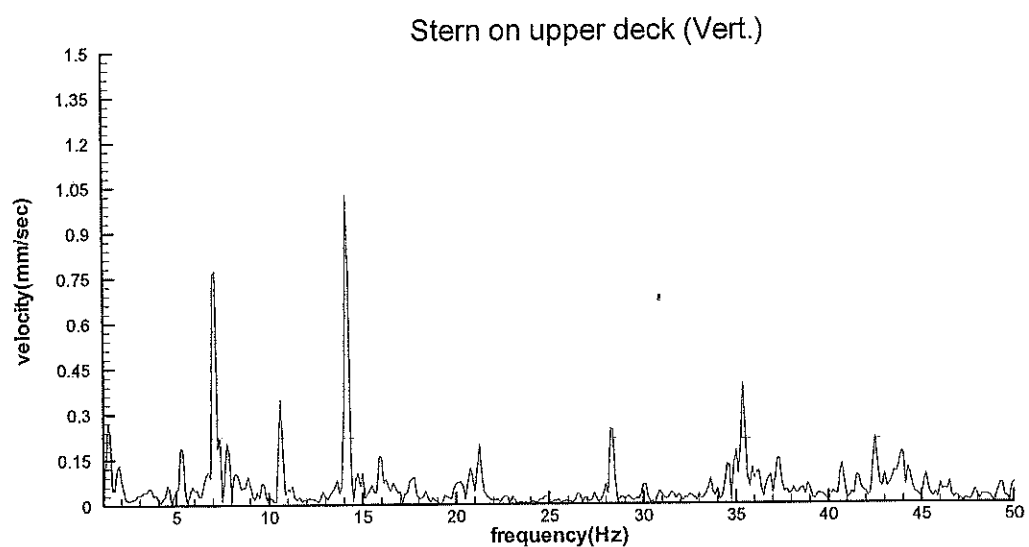
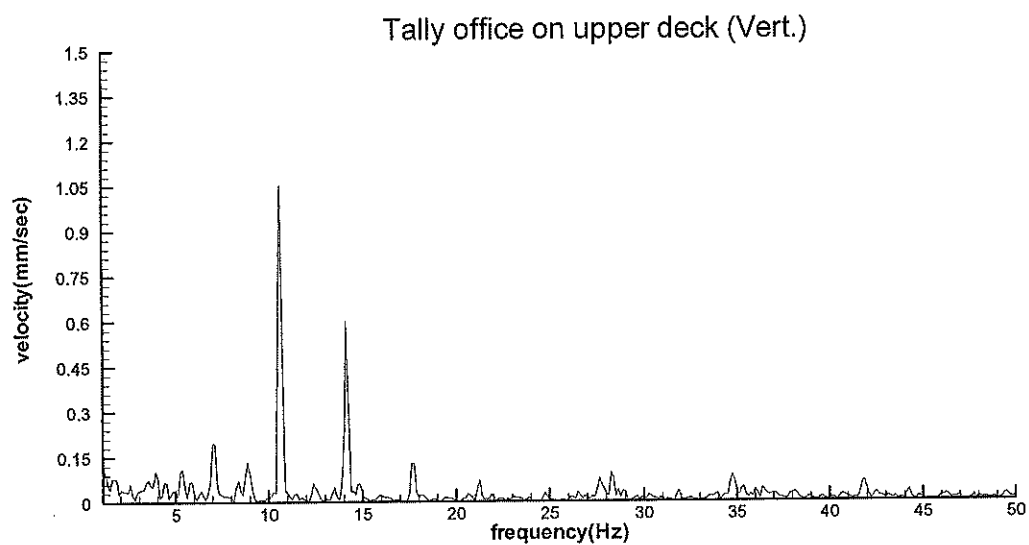
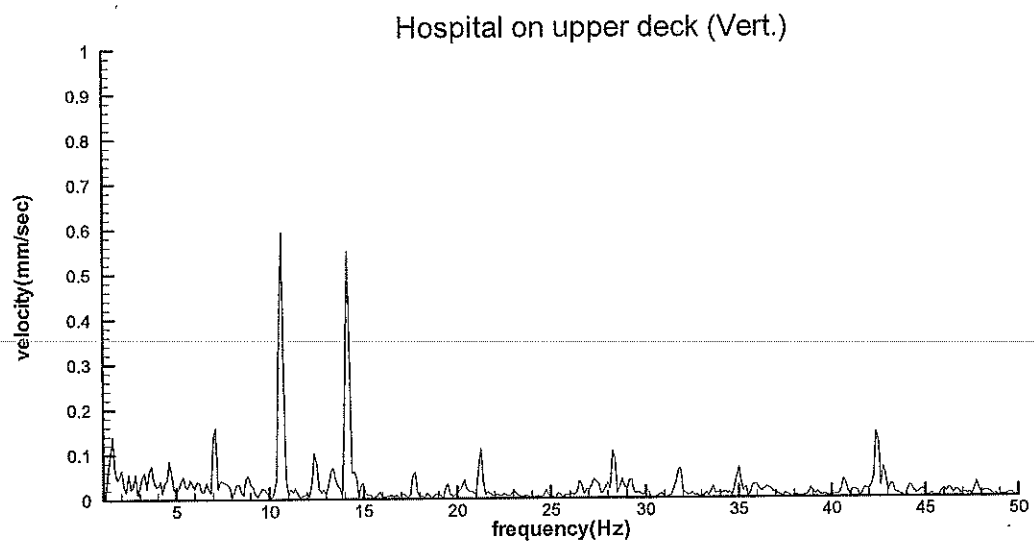


Fig.2.7 The measured vibration components at each point

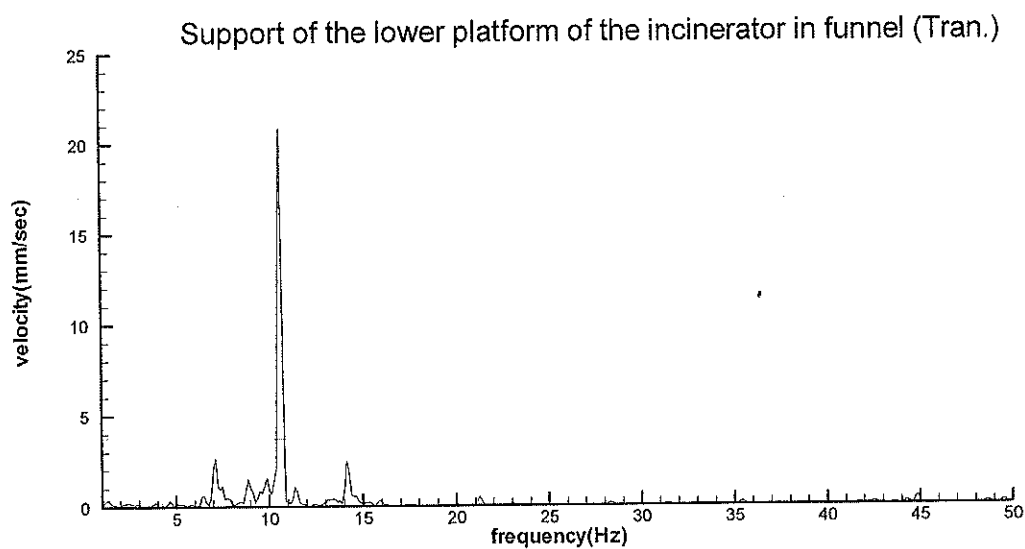
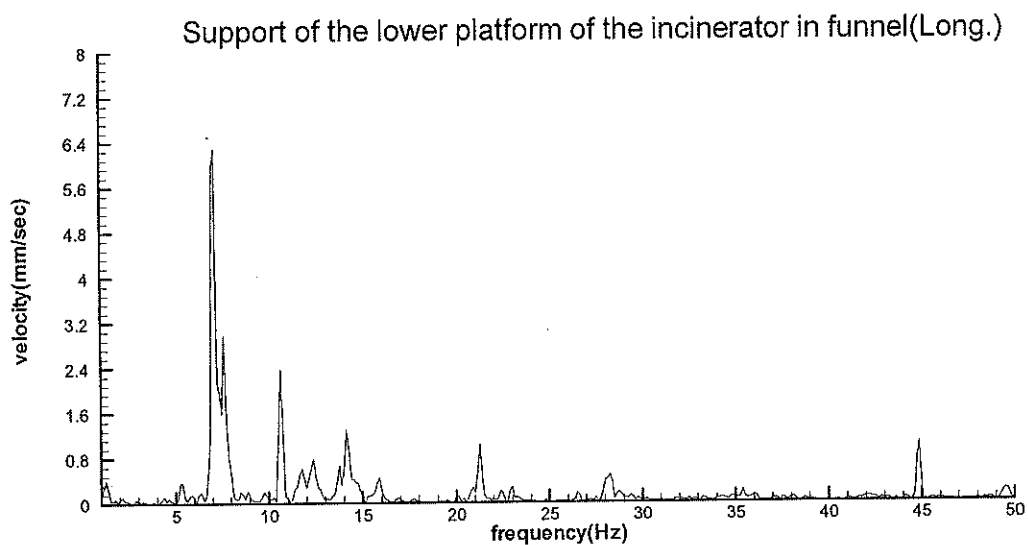
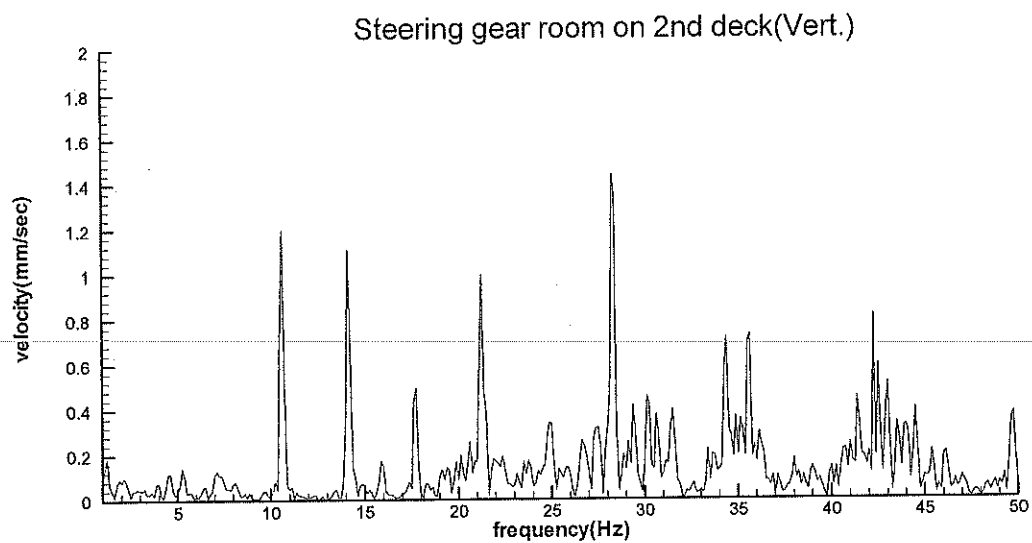


Fig.2.8 The measured vibration components at each point

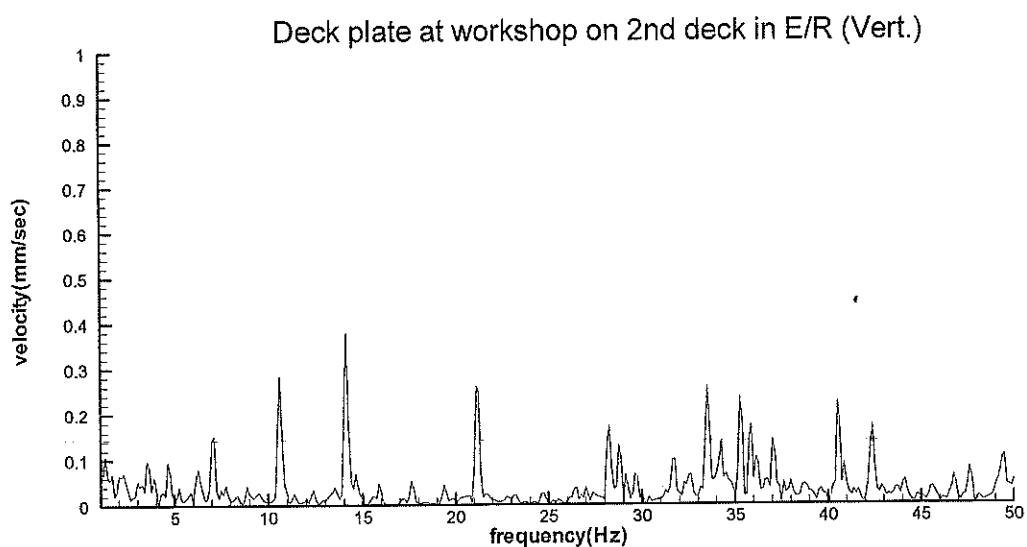
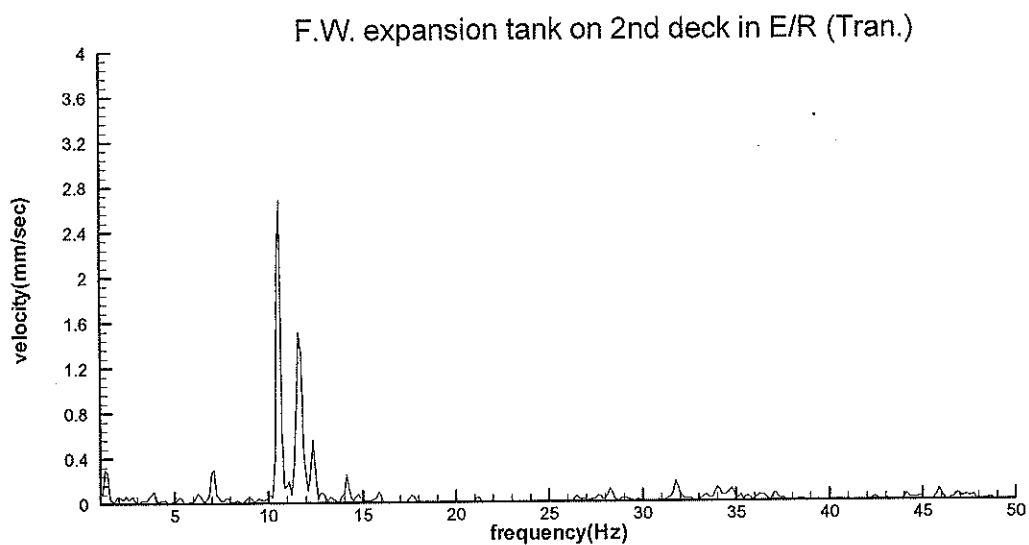
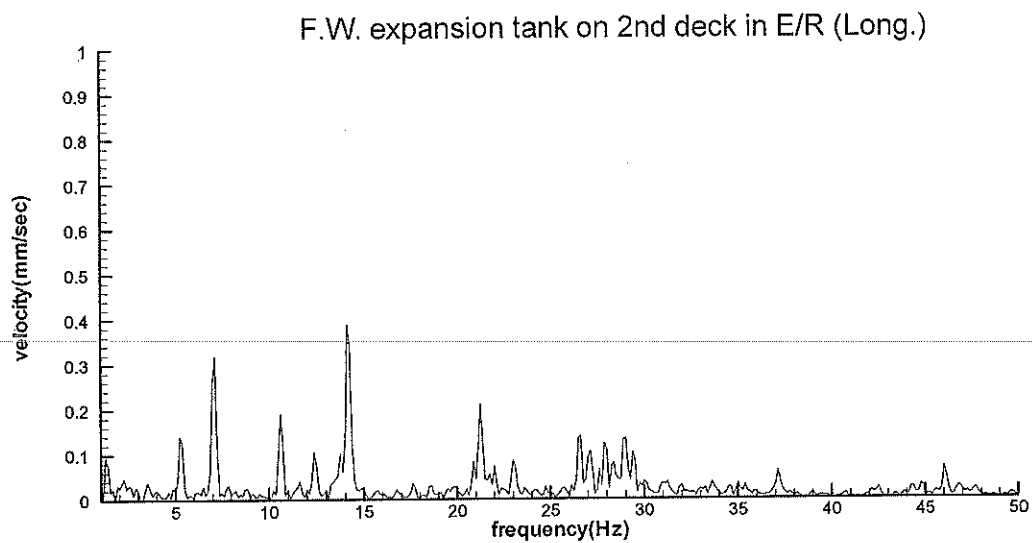


Fig.2.9 The measured vibration components at each point

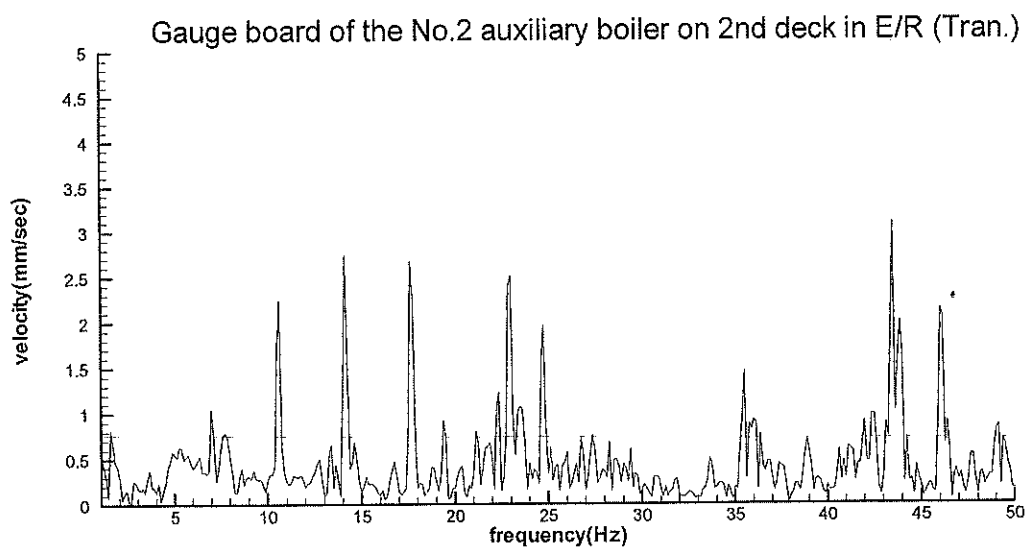
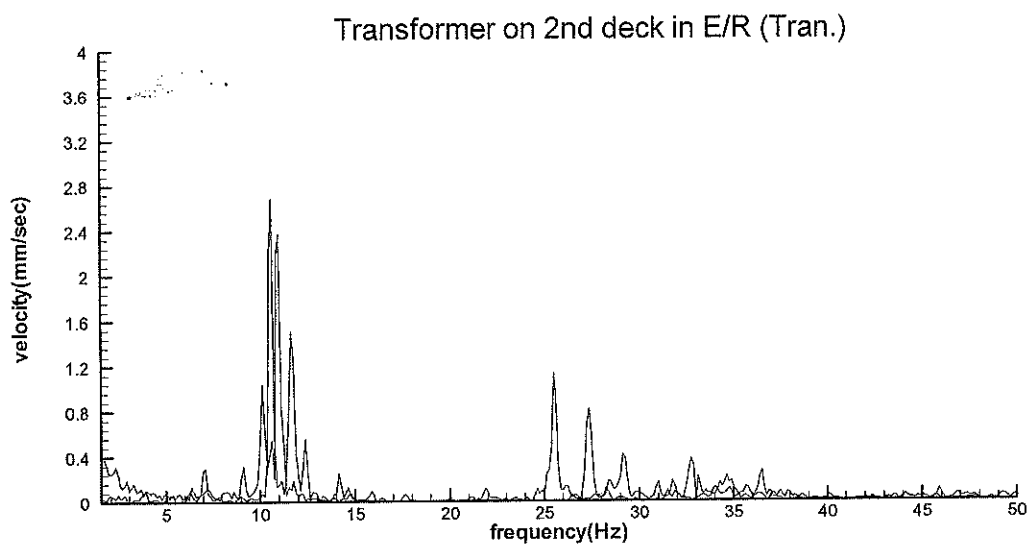
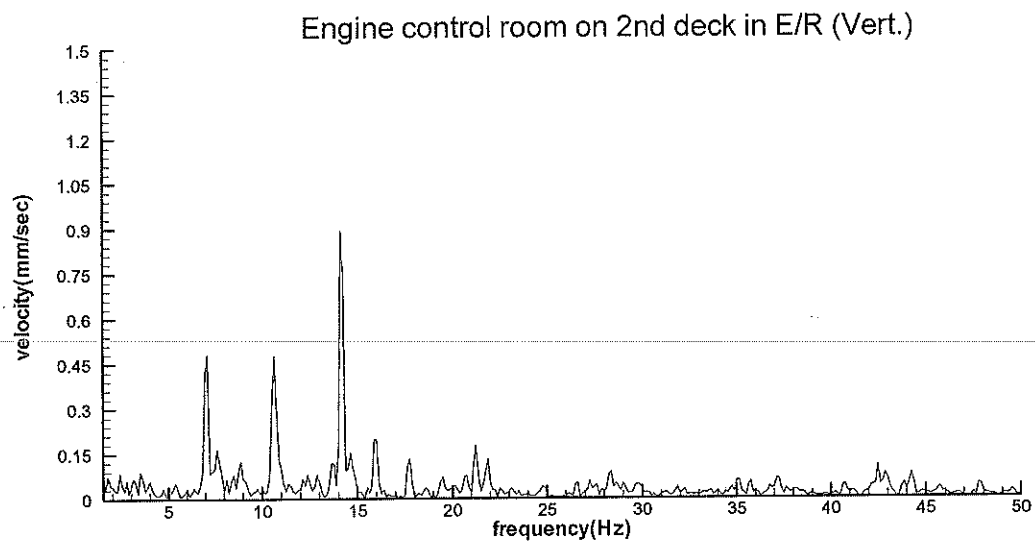


Fig.2.10 The measured vibration components at each point

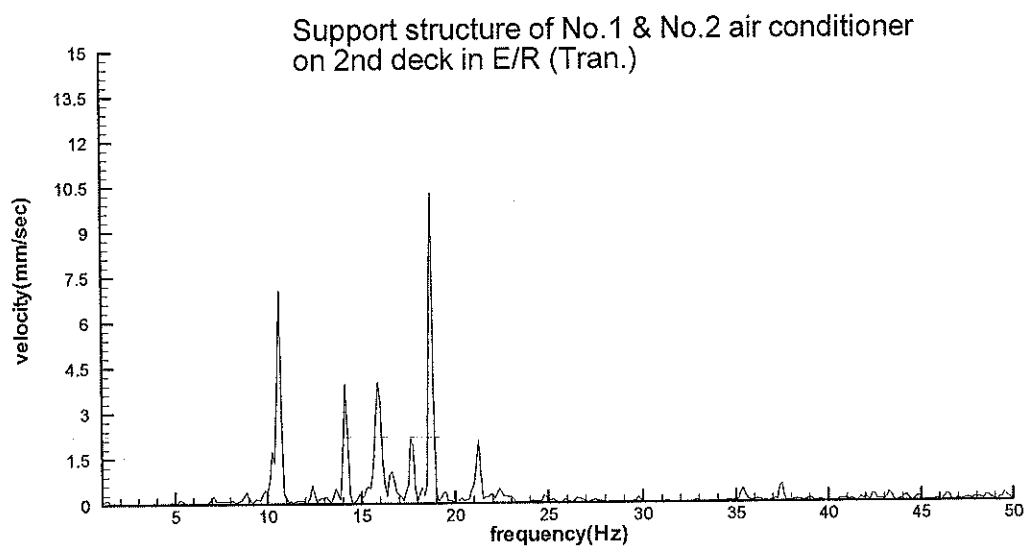
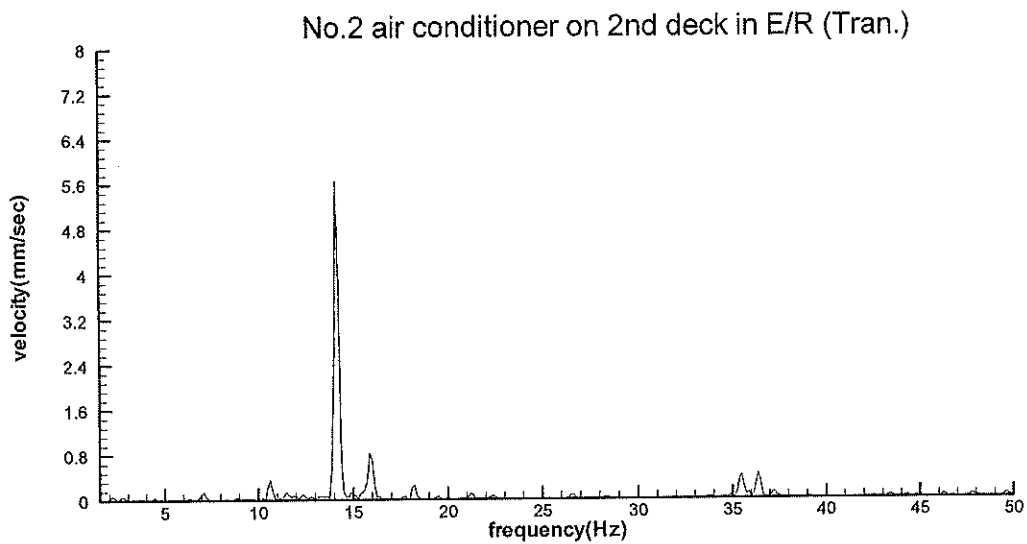
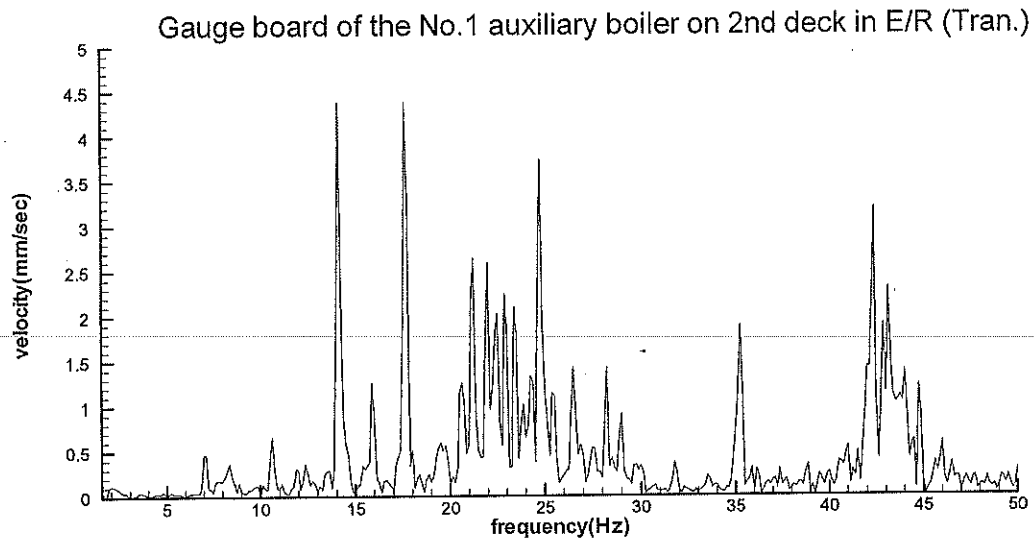


Fig.2.11 The measured vibration components at each point

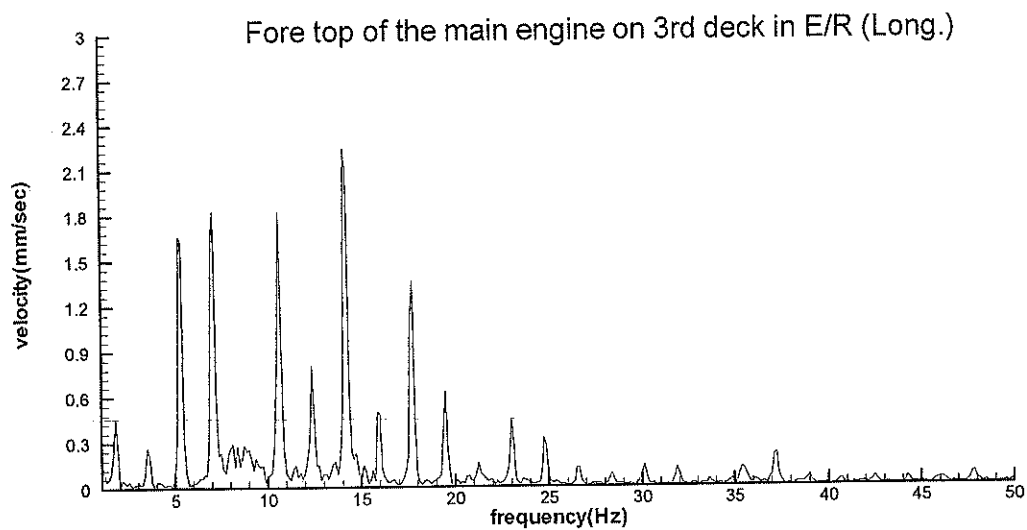
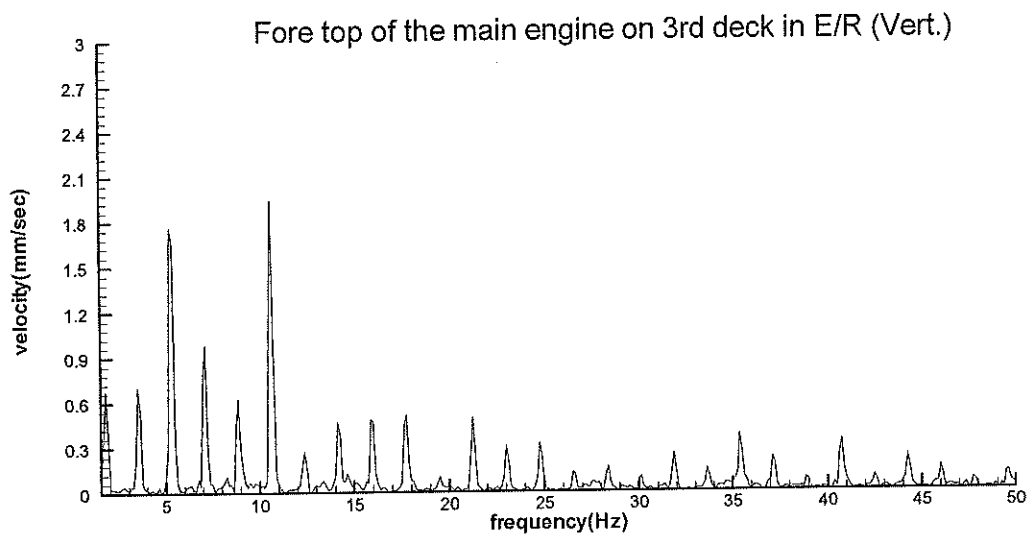
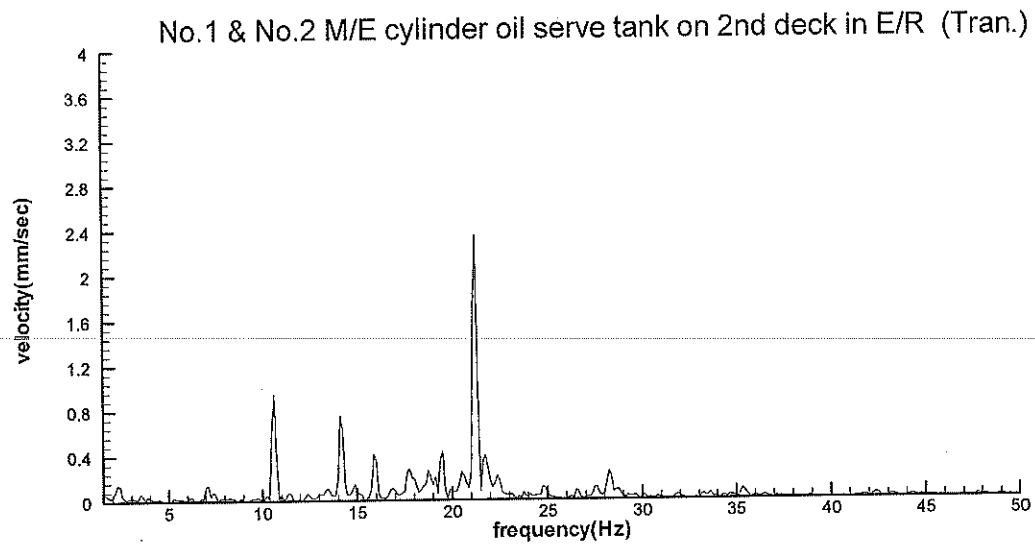


Fig.2.12 The measured vibration components at each point

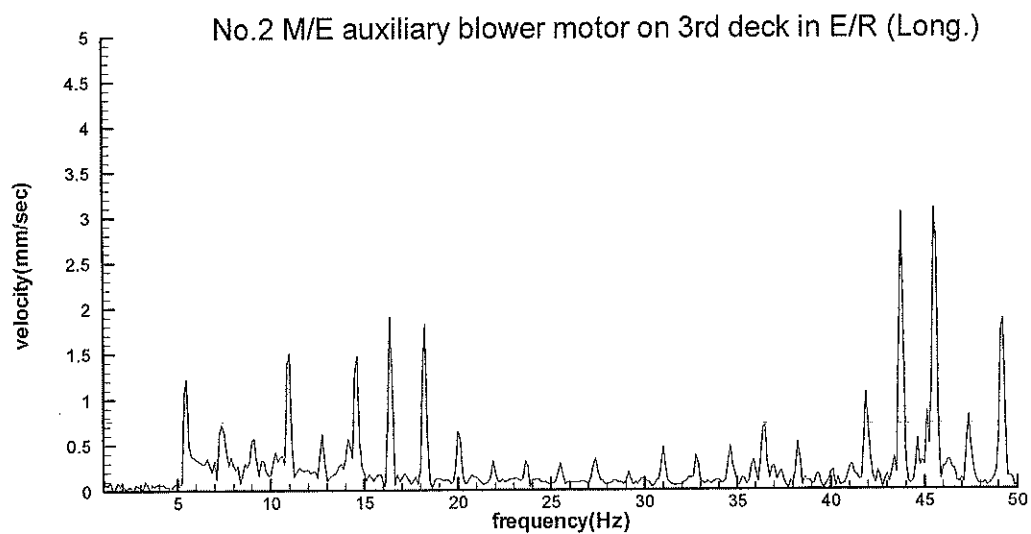
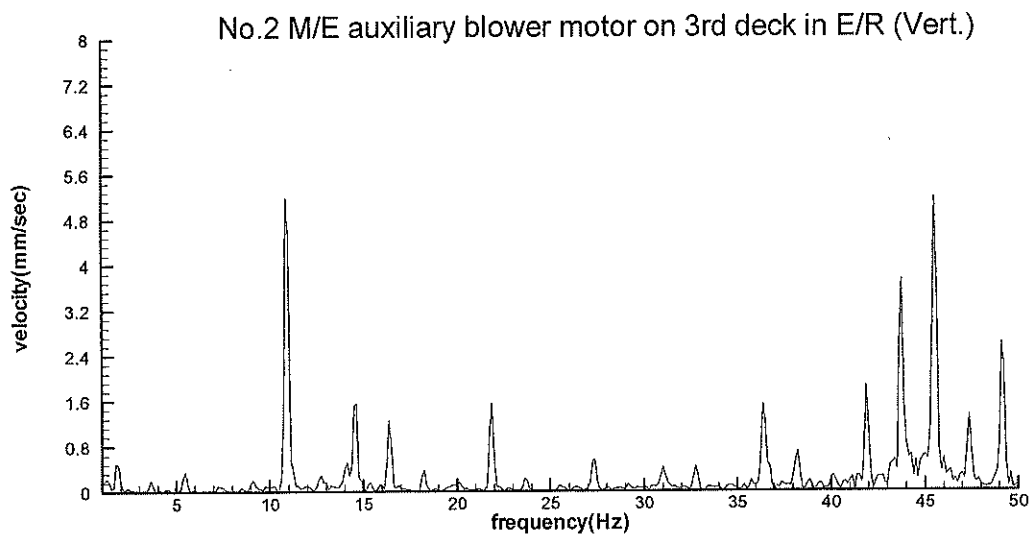
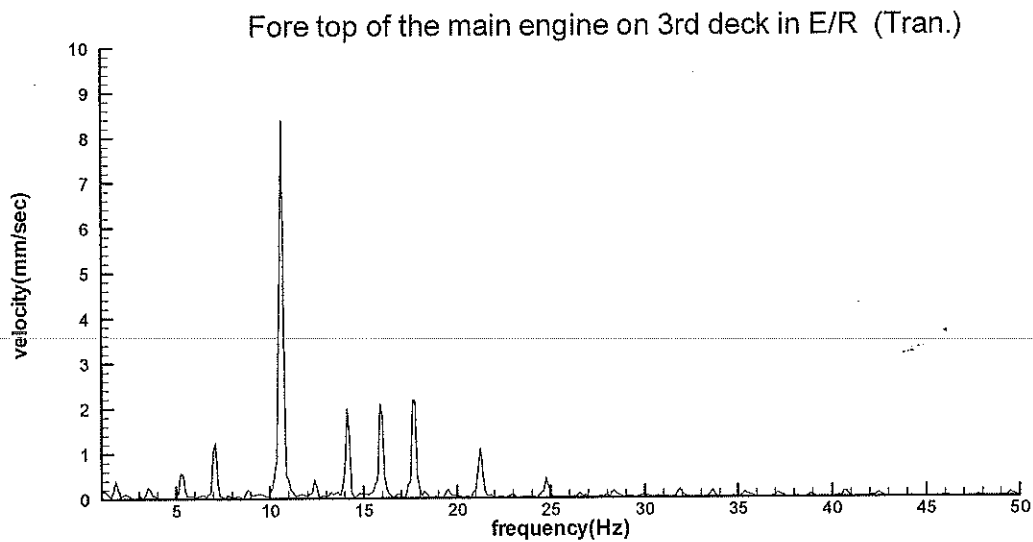


Fig.2.13 The measured vibration components at each point

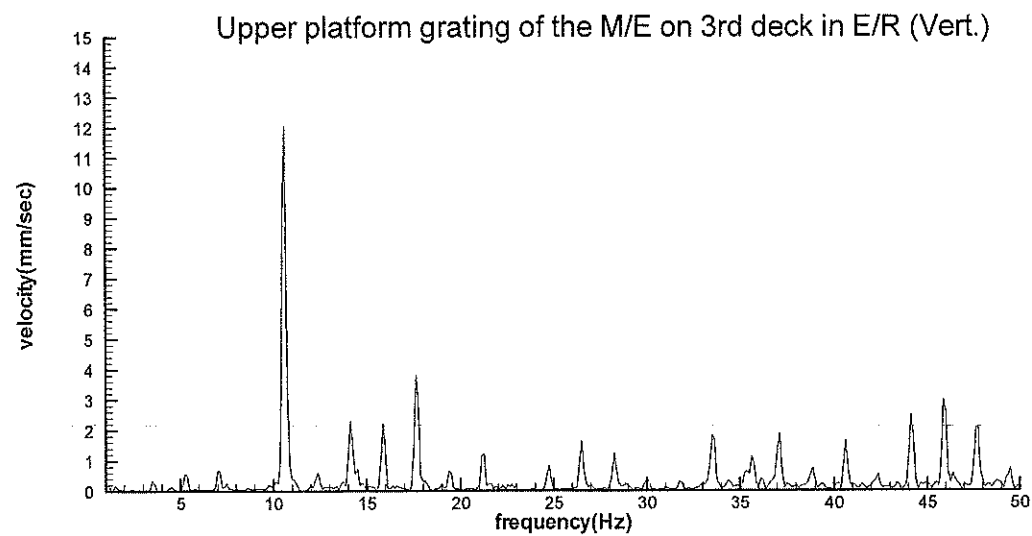
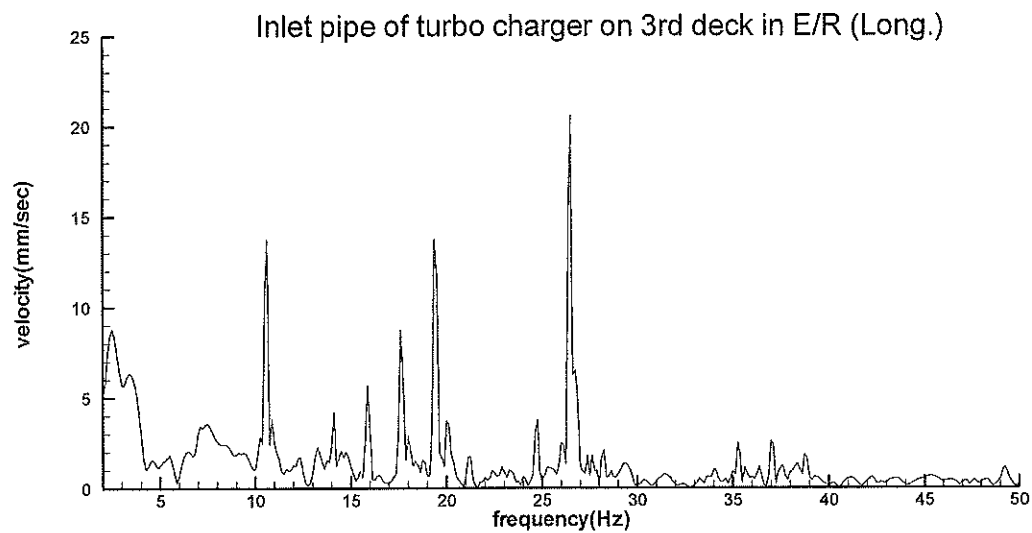
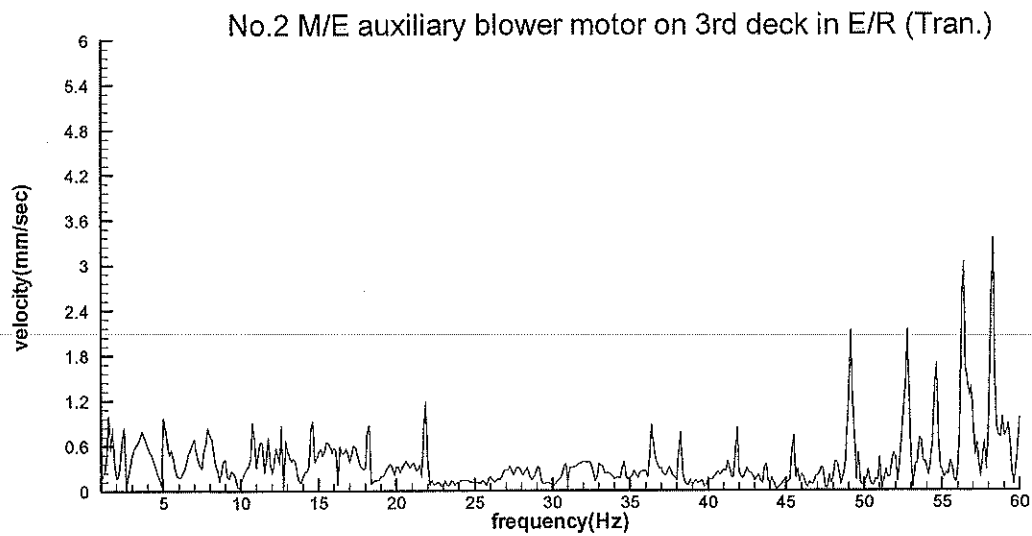


Fig.2.14 The measured vibration components at each point

